Algebra I

Fall 2025

1 Groups

1.1 Basics

1.1.1 Groups and Subgroups

Definition 1. Let G be a set with a map $\cdot : G \times G \to G, (g_1, g_2) \mapsto g_1 \cdot g_2$

- G is a **semigroup** if $\cdot$ is associative, i.e.

$$\begin{array}{ccc} G \times G \times G & \xrightarrow{\mathrm{id}\times\cdot} & G \times G \\ {\scriptstyle \cdot\times\mathrm{id}}\downarrow & & \downarrow{\scriptstyle \cdot} \\ G \times G & \xrightarrow[\cdot]{} & G \end{array}$$

  commutes

- G is a **monoid** if G is a semigroup and has a neutral element (identity element) $e \in G$ such that $eg = g = ge \forall g \in G$

- G is a **group** if G is a monoid and every element has an inverse i.e. for all $g \in G \exists h \in H$ such that $hg = e = gh$

Example 2. 1. G monoid $\implies$ e unique

2. G group $\implies$ inverses are unique (write $g^{-}1$ for inverse of $g \in G$)

Definition 3. We say G is **abelian** if $\cdot$ is commutative.

Definition 4. If G is a group, we say $U \subseteq G$ is a **subgroup** if U is a group with respect to $\cdot|_{U\times U} \to U \subseteq G$. (Notation: $U \leq G$, $|G|$ = order of G)

Example 5.

- $(\mathbb{Z}, +), (\mathbb{C}, +), (\mathbb{R}, +), (\mathbb{Q}, +)$
- D_n dihedral group of order $2n$
- $Gl_n(F), F$ field
- A_n, S_n

Proposition 6. *Given a group G and index set I, if $U_i \leq G$ for $i \in I$,*

$$\bigcap_{i \in I} U_i \leq G$$

Proof. **Exercise**. □

Definition 7. G group, $M \subseteq G$ subset:

$$\langle M \rangle := \bigcap_{M \subseteq U \leq G} U \leq G$$

is the **subgroup generated by M.**

G is **cyclic** if $G = \langle \{g\} \rangle$ for some $g \in G$ (write $G = \langle g \rangle$)

$|g| := |\langle g \rangle|$ is the **order** of $g \in G$

Note 8. $\mathbb{Z}$ is cyclic, $\mathbb{Z}/n\mathbb{Z}$ is cyclic for $n \in \mathbb{Z}$

Proposition 9. *Let $U \leq G$. $g_1 \ g_2 \iff g_1^{-1} g_2 \in U (\iff g_2 \in g_1 U)$ defines an equivalence relation on G with classes $\{gU | g \in G\}$.*

Proof. **Exercise**. □

Definition 10. Let G be a group, $U \leq G$ a subgroup.

gU is the (left) **coset** of u in G

$G/U =$ set of (left) cosets

$[G : U] = |G/U| =$ **index** of U in G

Theorem 11 (Index Formula)**.** $U \leq H \leq G \implies [G : U] = [G : H][H : U]$

Corollary 12 (Lagrange's Theorem)**.** *Let $H \leq G$.*

1. $[G : H] \cdot |H| = |G|$
2. *If $G < \infty$, $|H|$ divides $|G|$*

Corollary 13. *If $|G| \leq \infty$, $g^{|G|} = e$ for all $g \in G$.*

1.1.2 Normal Subgroups

Proposition 14. *G group, $U \leq G$, $g \in G$. Then $gUg^{-1} \leq G$.*

Definition 15. G group, $U, H \leq G$.

U and H are **conjugate** if $U = gHg^{-1}$ for some $g \in G$

$N \leq G$ is **normal** if $gNg^{-1} = N$ for all $g \in G$ (Notation: $N \trianglelefteq G$)

G is **simple** if $\{e\}, G$ are the only normal subgroups of G

Example 16. 1. $\mathrm{SL}_n(\mathbb{C}) = \{A \in \mathrm{GL}_n(\mathbb{C}) \mid \det(A) = 1\}$ special linear group

2. The only abelian simple groups are the cyclic groups of prime order $(\mathbb{Z}/p\mathbb{Z})$

Proposition 17. *1. Intersections of normal subgroups are normal.*

2. The subgroup generated by any collection of normal subgroups of G is normal in G.

1.1.3 The subgroup lattice

Definition 18. If M is a set with relation "$\leq$" that is reflexive, transitive, and antisymmetric, M is called a **poset (partially ordered set)**.

A poset M is called a **lattice** if any two elements $m_1, m_2 \in M$ satisfy:

1. There exists a least upper bound $u \in M$ such that $m_1 \leq u$, $m_2 \leq u$, and if $m_1 \leq p$ and $m_2 \leq p$, then $u \leq p$. (Write $u = m_1 \vee m_2$)

2. There exists a greatest lower bound $l \in M$ such that $l \leq m_1$, $l \leq m_2$, and if $r \leq m_1$ and $r \leq m_2$ then $r \leq l$. (Write $l = m_1 \wedge m_2$)

Definition 19. A lattice is called **modular** if whenever $m_1 \geq m_2$, one has

$$m_1 \wedge (m_2 \vee m_3) = m_2 \vee (m_1 \wedge m_3)$$

for all $m_1, m_2, m_3 \in M$.

Theorem 20. *Let G be a group.*

1. *The subgroups of G, ordered by $\leq$, form a lattice under $U_1 \wedge U_2 = U_1 \cap U_2$ and $U_1 \vee U_2 = \langle U_1, U_2 \rangle$.*

2. *The normal subgroups of G form a modular lattice.*

Example 21. Let S be a set. The power set $\mathcal{P}(S)$ with intersection and union forms a lattice, which is modular.

1.1.4 The group of rotational symmetries of the dodecahedron is simple

Let $D =$ the rotational symmetry group of the icosahedron

An icosahedron has 20 vertices, 30 edges, and 12 faces.

Any element of D takes vertices to vertices; so 20 possibilities for fixed vertex A and 3 possibilities for adjacent vertex $B \implies D$ has $|D| = 60$. Elements of D:

- Elements of order 1: the identity (1)
- Elements of order 2: there are 30 edges, 15 parallel pairs of edges, so 15 180° rotations (reflections) around axes through edge midpoints (15)

- Elements of order 3: there are 20 vertices, 10 opposite pairs, and 2 rotations about each (20)
- Elements of order 5: there are 12 faces, 6 pairs of opposite faces, each having 4 non-identity rotations, gives 24 rotations about center of a face (24)

This adds up to all 60 elements of D.

All subgroups of order 2, 3, or 5 are conjugate. Proof: **exercise**

Elements of order 2 (or 3, or 5) generate D. Proof: **exercise**

1.2 Homomorphisms

1.2.1 Quotients

Proposition 1. *If G is a group and $N \trianglelefteq G$, then G/N is naturally endowed with a group structure. If $|G| < \infty$, $|G/N| = \frac{|G|}{|N|}$.*

Definition 2. Given G a group and $N \trianglelefteq G$, the quotient G/N is called the **quotient group** of G by N.

Definition 3. For $g, h \in G$, the **commutator** of g and h is

$$[g, h] = ghg^{-1}h^{-1}.$$

The subgroup generated by all commutators,

$$G' := [G, G] := \langle [g, h] \mid g, h \in G \rangle,$$

is called the **commutator subgroup** or **derived subgroup** of G.

Definition 4. The **center** of G is

$$Z(G) = \{g \in G \mid gh = hg \text{ for all } h \in G\}.$$

Example 5. For $G = \mathrm{GL}_2(\mathbb{C})$, what is G'?

The matrix $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \notin G'$.

Every element of G' must have determinant 1. So $G' \leq SL_2(\mathbb{C})$. It turns out that $G' = SL_2(\mathbb{C})$. (More generally $GL_n(\mathbb{C})' = SL_n(\mathbb{C})$).

Example 6. Groups of order < 8:

1. If $|G| = 4$,

(a) $\exists g \in G$ with $|g| = 4 \implies \mathbb{Z}/4\mathbb{Z}$

(b) For all $g \in G, |g| \leq 2 \implies G = \{e, g_1, g_2, g_1g_2\}$. $(g_1g_2)^2 = e \implies G$ abelian.

2. If $|G| = 6$,

(a) $\exists g \in G$ with $|g| = 6 \implies \mathbb{Z}/6\mathbb{Z}$

(b) $|g| \leq 2$ for all $g \in G$ can't happen so there is an element g of order 2:

(c) If this $|g| = 3$, $N = \{e, g, g^2\} \leq G$. By counting, $N \trianglelefteq G \implies G = h \sqcup hN$, some $h \in G$. If $h^2 \in hN, h \in N \Rightarrow\Leftarrow$. So $h^2 \in N$.

- If $h^2 = e$, $hgh^-1 \in N$ $\{e\}$ so $hgh^-1 = g$ or $g^2 \implies D_3$
- If $h^2 \neq e$, $h^2 = g$ so $|h| = 6 \implies \mathbb{Z}/6\mathbb{Z}$

Proposition 7. *Let G be a group and $N \leq G$. TFAE:*

1. *$N \trianglelefteq G$ and G/N is abelian*

2. *$G' \leq N$*

In particular, $G' \trianglelefteq G$ and G/G' is abelian

Proof. (1) $\implies$ (2) :
If G/N is abelian and $N \trianglelefteq G$,, then for $g, h \in G$, $ghN = hgN$, so $ghg^{-1}h^{-1} \in N$ for all g, h, meaning $G' \leq N$.
(2) $\implies$ (1) :
If $G' \subseteq N$, then for $g \in G, n \in N, gng^-1 = gng^{-1}n^{-1}n = [g, n]n \in N$ so $N \trianglelefteq G$. And $gNg^{-1} = N \implies gN = Ng \implies G/N$ is abelian. □

Note 8. Taking the quotient by the commutator subgroup "abelianizes" a group; $G/[G, G]$ is the largest abelian quotient of G.

1.2.2 Homomorphism Theorems

Definition 9. G, H groups, $\varphi : G \to H$ map.

φ is a **group homomorphism** if $\varphi(gh) = \varphi(g)\varphi(h)$ for all $g, h \in G$

φ is a **group isomorphism** if φ is a bijective homomorpihism

Two groups G and H are **isomorphic** if $\exists$ an isomorphism $\varphi : G \to H$ (Notation: $G \cong H$)

φ is an **automorphism** if $\varphi : G \to G$ is an isomorphism

$\mathrm{Aut}(G) = \{\text{automorphisms } \phi : G \to G\}$

$\operatorname{Ker} \varphi = \{g \in G |\ \varphi(g) = e_H\}$ is the **kernel** of φ

Example 10. • Conjugation by a fixed element $g \in G$,

$$c_g : G \to G, \quad x \mapsto gxg^{-1},$$

is an automorphism.

- $\exp : (\mathbb{R}, +) \to (\mathbb{R}_0, \cdot)$ homomorphism
- $\det : GL_n(K) \to (K \setminus \{0\}, \circ)$ homomorphism

Proposition 11. *Let G be a group and $N \trianglelefteq G$. The canonical map $\pi : G \to G/N, g \mapsto gN$ is a homomorphism with $\ker(\pi) = N$ (called the canonical homomorphism to the quotient).*

Theorem 12. *Let $\varphi : G \to H$ be a group homomorphism. Then:*

1. $\ker(\varphi) \trianglelefteq G$,
2. $\varphi(G) \leq H$,
3. $G/\ker(\varphi) \cong \varphi(G)$.

Corollary 13. *Given G, H groups, $\varphi : G \to H$ surjective homomorphism. If $N \trianglelefteq H$, then $M := \varphi^{-1}(N) \trianglelefteq G$ and $G/M \cong H/N$.*

Proof. Let $g, h \in M$. $\varphi(gh^{-1}) = \varphi(g)\varphi(h^{-1}) = \varphi(g)\varphi(h)^{-1} \in N \implies gh^{-1} \in \varphi^{-1}(N) = M \implies M \leq G$.

If $m \in M, g \in G, \varphi(mgm^{-1}) = \varphi(m)\varphi(g)\varphi(m^{-1}) \implies mgm^{-1} \in \varphi^{-1}(N) = M$ so $M \trianglelefteq G$.

Define $\psi : G \to H \to H/N$ by $\psi(g) = \varphi(\pi(g))$. This is surjective (**why?**) with $\operatorname{Ker}\psi = M$. So $G/M \cong G/\ker\psi \cong$ (by Theorem 2) $\psi(G) = H/N$. □

1.2.3 Isomorphism Theorems

Theorem 14 (First Isomorphism Theorem)**.** *Let G be a group, $N \trianglelefteq G, M \trianglelefteq G$ such that $N \leq M$. Then*

$$\frac{(G/N)}{M/N} \cong G/M.$$

Proof. **Exercise** □

Theorem 15 (Second Isomorphism Theorem)**.** *Let G be a group, $N \trianglelefteq G$, and $U \leq G$. Then*

$$U/(U \cap N) \cong NU/N.$$

Proof. Check: $\langle N, U\rangle = NU$ since N is normal. $N \trianglelefteq NU$ and $N \cap U \trianglelefteq U$. Then define $\phi : U \to G/N$ by $u \mapsto uN$ and use the homomorphism theorem. □

Example 16. $S_3 \leq S_4, V_4 = \langle (12)(34), (13)(24), (14)(23)\rangle \trianglelefteq S_4$.

Check: $S_3V_4 = S_4$. The theorem implies:

$$(S_3V_4)/V_4 = S_4/V_4 \cong S_3/(S_3 \cap V_4) \cong S_3.$$

Definition 17. If V, V' are lattices, $\Phi : V \to V'$ is a **lattice homomorphism** if

- Φ preserves the partial order
- $\Phi(u \vee v) = \Phi(u) \vee \Phi(v)$
- $\Phi(u \wedge v) = \Phi(u) \wedge \Phi(v)$

If Φ is bijective, Φ is a **lattice isomorphism**

Theorem 18 (Correspondence Theorem)**.** *G, H groups, $\varphi : G \to H$ homomorphism. Then*

$$V := \{U \leq G \mid \operatorname{Ker} \varphi \leq U\}, \quad V' := \{W \leq H \mid W \leq \varphi(G)\}$$

are lattices and $\Phi : V \to V', U \mapsto \varphi(U)$ is a lattice isomorphism.

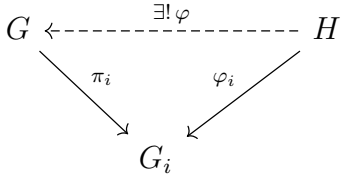
Proof. **Exercise.** □

1.3 Direct Products and Abelian Groups

1.3.1 Direct products

Definition 1. Let I be a nonempty index set, $\{G_i\}_{i \in I}$ a collection of groups. A group G together with homomorphisms $\pi_i : G \to G_i$ is called a **direct product** of the G_i if for every group H with homomorphisms $\varphi_i : H \to G_i$, there exists a unique homomorphism $\varphi : H \to G$ with $\pi_i \circ \varphi = \varphi_i$ for all $i \in I$.

Universal mapping property (UMP):



π_i is called the **canonical projection onto the i-th factor**

Notation: $G = \prod_{i \in I} G_i$. If $I = \{1, \ldots, n\}$ write $G = G_1 \times \cdots \times G_n$.

Example 2. $V_4 = \{e, g_1, g_2, g_1 g_2\}$ (Klein 4).

Claim: $V_4 \cong (\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$.

Say the first $(\mathbb{Z}/2\mathbb{Z})$ is $G_1 = \langle a \rangle$, and the second $(\mathbb{Z}/2\mathbb{Z})$ is $G_2 = \langle b \rangle$. Define:

$$\pi_1 : g_1 \mapsto a, g_2 \mapsto e \pi_2 : g_1 \mapsto e, g_2 \mapsto b$$

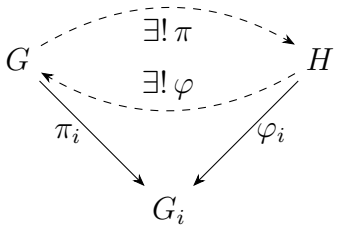
H arbitrary with $\varphi_1 : H \to G_1, \varphi_2 : H \to G_2$. $h \in H \implies \varphi_1(h) \in \{e, a\}$ and $\varphi_2(h) \in \{e, b\}$. Define $\varphi(h)$ according to these cases:

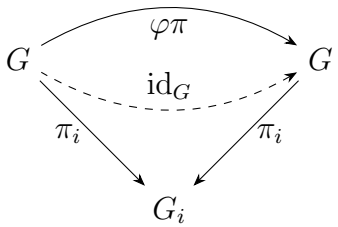
$$\varphi(h) = \begin{cases} e & \text{if } \varphi_1(h) = e, \varphi_2(h) = e, \\ g_1 & \text{if } \varphi_1(h) = a, \varphi_2(h) = e, \\ g_2 & \text{if } \varphi_1(h) = e, \varphi_2(h) = b, \\ g_1 g_2 & \text{if } \varphi_1(h) = a, \varphi_2(h) = b \end{cases}$$

Then $\pi_1 \circ \varphi = \varphi_i$ by construction and φ is unique with this property.

Theorem 3. *If $I \neq \varnothing$ is an index set and G_i is a group for all $i \in I$, $\prod_{i \in I} G_i$ exists and is unique up to isomorphism.*

Proof. Let G, H be direct products with projections π_i, φ_i. Since G is a direct product, there is a unique homomorphism $\varphi : H \to G$ such that $\pi_i \circ \varphi = \varphi_i$. Since H is a direct product, there is a unique homomorphism $\pi : G \to H$ such that $\varphi_i \circ \pi = \pi_i$.





So $\varphi\pi : G \to G$ is a homomorphism such that $\pi_i(\varphi\pi) = \pi_i$. But id_G makes the diagram commute, and G being a direct product makes the working map commute, so $\varphi\pi = \mathrm{id}_G$ by UMP. So φ, π are isomorphisms (inverses) i.e. $G \cong H$. So we have uniqueness.

For existence:
Let $G := \{(g_i) | i \in I, g_i \in G_i\}$ (cartesian product as sets). G becomes a group with component-wise multiplication.

Let $\pi_i : G \to G_i, (g_j)_{j \in I} \mapsto g_i$. Need to verify UMP:
Let H, φ_i be as in definition. Define $\varphi : H \to G$ via $\varphi(h) = (\varphi_i(h))_{i \in I}$. Check: this works. $\square$

Proposition 4. *Let G be a group and $U_1, \ldots, U_n \trianglelefteq G$. If (a) $G = U_1 \cdots U_n$ and (b) $U_i \cap U_1 U_2 \ldots U_{i-1} U_{i+1} \ldots U_n = \{e\}$ for all $i = 1, \ldots, n$, then*

1. *$G \cong \prod_{i=1}^n U_i$*
2. *Every element $g \in G$ has a unique expression of the form $g = g_1 \ldots g_n, g_i \in G_i$. This is compatible with the product.*

Proof. The product map $U_1 \times \cdots \times U_n \to G$, $(g_1, \ldots, g_n) \mapsto g_1 \cdots g_n$, is a homomorphism: For $g_i \in U_i, g_j \in U_j, i \neq j$, we have $g_i^{-1} g_j^{-1} g_i g_j \in U_i \cap U_j = \{e\}$ (by U_i's being normal) so $g_i g_j = g_j g_i \implies \varphi$ is a homomorphism.

(b) $\implies$ φ has trivial kernel $\implies$ φ is an isomorphism.

Part (2): **Exercise.** □

Corollary 5 (Chinese Remainder Theorem for Groups)**.** *Let G be a finite group and $N_1, \ldots, N_k \trianglelefteq G$ with $|N_i| = m_i$ for $i = 1, \ldots, n$, m_i's pairwise coprime, and $|G| = m_1 \ldots m_n$. Then*

$$G \cong N_1 \times \cdots \times N_k.$$

1.3.2 Cyclic Groups

Recall: G is cyclic if $G = \langle G \rangle$, some $g \in G$.

Note 6. G cyclic $\implies$ G abelian. G cyclic of prime order $\implies$ G simple (has no proper normal subgroups).

Theorem 7. *Every subgroup of a cyclic group is cyclic.*

Proof. **Exercise** □

Example 8. All subgroups of $\mathbb{Z}$ are of the form $n\mathbb{Z} (n \in \mathbb{N})$.

Theorem 9. *If G is a group, $a \in G, |a| < \infty, m \in \mathbb{N}$, then*

$$|a^m| = \frac{|a|}{\gcd(m, |a|)}.$$

Proof. **Exercise** □

Recall: For $n \in \mathbb{N} \setminus \{0\}$, define

$$\varphi(n) := |\{a \in \mathbb{N} \mid 1 \leq a \leq n, \gcd(a, n) = 1\}| = \text{Euler } \varphi\text{-function}$$

Example 10. $\varphi(6) = |\{1, 5\}| = 2$
$\varphi(p) = p - 1$ (p prime)

Corollary 11. *A cyclic group of order n has $\varphi(n)$ generators.*

Proof. Say $G = \langle g \rangle = \{e, g, g^2, \ldots, g^{n-1}\}$. By Theorem 2, $|g^m| = n \iff \gcd(m, n) = 1 \iff g^m \in \{a \in \mathbb{N} \mid 1 \leq a \leq m, \gcd(m, n) = 1\}$. □

Theorem 12. *If $G = \langle g \rangle$ is a finite cyclic group of order n, then for each divisor $d \mid n$ there exists a unique subgroup $H \leq G$ such that $|H| = d$.*

Proof. Say $G = \langle g \rangle, |g| = |G| = n, n = d \cdot m$. Theorem 2 $\implies$ $|g^m| = \frac{n}{\gcd(m,n)} = d$ i.e. $|\langle g^m \rangle| = d$ so H exists.

For uniqueness: Say $U \leq G, |U| = d$. Theorem 1 $\implies$ U is cyclic $\implies$ $U = \langle g^k \rangle$ some k. $d = |g^k| - \frac{dm}{\gcd(n,k)} \implies \gcd(k, n) = m \implies m \mid k$. Write $k = m \cdot k'$. Then $g^k = g^{mk'} = (g^m)^{k'} \in \langle g^m \rangle \implies U \leq \langle g^m \rangle$. But $|U| = |\langle g^m \rangle|$ so $U = \langle g^m \rangle$. □

1.3.3 Structure Theorem for Finitely Generated Abelian Groups

Definition 13. A group G is **finitely generated** if $G = \langle g_1, \ldots, g_n \rangle$ for some $g_i \in G$.

Example 14. $\mathbb{Q}^\times = \langle -1, p \text{ primes} \rangle$ not finitely generated.

Theorem 15 (Structure Theorem 1: Elementary Divisors)**.** *If G is a finitely generated abelian group, there exists $r \in \mathbb{N}, e_i \in \mathbb{N}_0, i = 1, \ldots, r$ such that*

$$e_1 \mid e_2 \mid \cdots \mid e_r \text{ and } G \cong \prod_{i=1}^{r} \mathbb{Z}/e_i\mathbb{Z}$$

(G is a direct product of cyclic groups)

Note 16. $e_i = 0$ is possible

Definition 17. The e_i's in Theorem 1 are alled the **structure constants** of G (these uniquely determine G up to isomorphism).

Corollary 18 (Structure Theorem 2: Primary Divisors)**.** *Let G be a finitely generated abelian group. There exist primes $p_1 \ldots, p_s$ (not necessarily distinct) and $k_1, \ldots, k_s \in \mathbb{N}\backslash\{0\}, m \in \mathbb{N}_0$ such that*

$$G \cong \mathbb{Z}^m \times \mathbb{Z}/p_1^{k_1}\mathbb{Z} \times \cdots \times \mathbb{Z}/p_s^{k_s}\mathbb{Z}.$$

Proof. **Exercise.** □

Example 19. Groups of order 8 up to isomorphism:
If abelian,

$$\mathbb{Z}/8\mathbb{Z} \quad \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}, \quad \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$$

If not abelian, has to have an element g with order 4 (if they're all 2, it's abelian). So let $N := \langle g \rangle \trianglelefteq G$ (has index 2). Take $h \in G \setminus N$. Write $G = N \sqcup hN$ (coset decomposition).

$h^{-1}gh \in N$, cannot be g or e (because then G would be abelian) so is g^2 or g^3.

If $h^{-1}gh = g^2, (h^{-1}gh)^2 = g^4 = e \implies h^{-1}g^2h = e \implies g^2 = e \to\leftarrow$. So must have $h^{-1}gh = g^3$. Then $G = \{e, g, g^2, g^3, h, hg, hg^2, hg^3\}$.

Cases:

- If $|h| = 2$, $G = \langle g, h | g^4 = e = h^2, h^{-1}gh = g^3 \rangle \implies G \cong D_4$
- $|h| = 4 \implies h^2 = g^2 \implies G \cong Q_8 = \langle \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \rangle \leq GL_2(\mathbb{C})$.

Note 20. If $H \leq G$ has index 2, then H is automatically normal, since G acts by left multiplication on cosets and the complement is a single other coset.

1.4 Permutation Groups

1.4.1 Actions of groups on sets

Example 1. Let G = rotational symmetry group of tetrahedron: G maps vertices to vertices so $G \leq S_4$.
Let $g = (234) \in G$ (rotation about vertex 1)
Let $h = (12)(34) \in G$ (reflection across opposite edge midpoints)
$\langle g, h \rangle = A_4 \implies A_4 \leq G$.
$(23) \notin G \implies G \neq S_4 \implies G = A_4$.

Definition 2. Let G be a group, M be a nonempty set. We say G **acts on** M if there is a map $G \times M \to M$, $(g, x) \mapsto g \cdot x$ satisfying:

$$e \cdot x = x, \qquad (gh) \cdot x = g \cdot (h \cdot x)$$

for all $g, h \in G, x \in M$.

We call $G \times M \to M$ a **(left) action of** G **on** M

For $x \in M$, the **stabilizer** of x in G is $G_x = \{g \in G \mid g \cdot x = x\}$

The **orbit** of x under G is $G.x = \{g \cdot x \mid g \in G\}$.

We say that G **acts transitively** on M if $G \cdot x = M$ for some $x \in M$ (i.e. there is a single orbit)

$\pi : G \to \{\text{self maps of } M\}$, $g \mapsto \pi_g := (x \mapsto g \cdot x)$ is the **permutation representation** of G on M

The action is **faithful** if $g \cdot x = x$ for all x implies $g = e$ (i.e. if π is injective)

Theorem 3. *If G acts on M, then the associated permutation representation is a homomorphism $\pi : G \to \mathrm{Sym}(M)$ with $\ker \pi = \cap_{x \in M} G_x$.*

Note 4. Hence there is a bijection between group actions of G on M and homomorphisms $G \to \mathrm{Sym}(M)$.

Corollary 5 (Cayley's Theorem)**.** *Every group has a faithful permutation representation.*

Proof. Let G act on itself by left translation. Then the resulting homomorphism $G \to \mathrm{Sym}(G)$ is injective. □

Definition 6. The representation of G by left translation is called the **regular representation**.

Corollary 7. *If G is a finite group, $G \leq S_n$ for some n.*

1.4.2 The class equation

Lemma 8. *If G is a group, M a nonempty set with a G-action, the relation $x \sim y \iff x \in G \cdot y$ defines an equivalence relation on M.*

Proof. **Exercise.** □

Theorem 9. *Suppose G acts on $M \neq \varnothing$.*

1. *$x \in M \implies G_x \leq G$ and $[G : G_x] = |G.x|$.*
2. *Let V be system of representatives of the orbits of G on M. Then*

$$|M| = \sum_{x \in V} |G.x| = \sum_{x \in V} [G : G_x]$$

Proof. 1) $g, g \in G_x \implies gh^{-1} \in G_x \implies G_x \leq G$.
Define $G \cdot x \to \{gG_x \mid g \in G\}, g \cdot x \mapsto gG_x$. This is a bijection, so $[G : G_x] = |G \cdot x|$. □

Example 10. Symmetry group of dodecahedron:

Faces: $|S| = 12$. 1 orbit of size 12, stabilizer has size 5 $\implies |D| = 60$.

Edges: $|S| = ?$. Stabilizer has order 2 so there are $60/2 = 30$ edges.

Definition 11. If G acts on M we say $x \in M$ is a **fixed point** if $G_x = G$.

$\mathrm{Fix}_G(M) = \{\text{fixed points of } G\} = \{x \in M \mid G_x = G\}$.

Corollary 12.

$$|M| = |\operatorname{Fix}_G(M)| + \sum_{x \in M, 1 < [G:G_x]} [G : G_x]$$

Theorem 13 (Fixed Point Theorem)**.** *Suppose p is prime, G is a finite group of order p^r, and G acts on $M \neq \varnothing$ where $|M| < \infty$. Then*

$$|M| \equiv |\operatorname{Fix}_G(M)| \pmod{p}$$

In particular, if $p \nmid |M|$, there is at least 1 fixed point.

Definition 14. A group G acts on itself by **conjugation**:

$$g \cdot x = gxg^{-1}.$$

The **orbit** of x under this action is the **conjugacy class** of x:

$$[x]_G := G \cdot x = [x]_G = \{gxg^{-1} \mid g \in G\}.$$

The **centralizer** of x is

$$C_G(x) := G_x = \{g \in G \mid gx = xg\}.$$

For a subset $H \subseteq G$, define the **centralizer of** H is $C_G(H) = \{g \in G \mid gh = hg \text{ for all } h \in H\}$

Note 15. Conjugacy classes are the orbits of this action, and the stabilizer of x is $C_G(x)$. Hence by the orbit–stabilizer theorem:

$$|[g]_G| = [G : C_G(g)].$$

Corollary 16. *1.* $|[g]_G| = [G : C_G(g)]$

In particular, the size of $[g]_G$ *divides* $|G|$ *if* G *is finite*

2. (Class Equation): If C *is a system of representatives of conjugacy classes,*

$$\sum_{g \in C} [G : C_G(g)] = |G|$$

Example 17. For the icosahedral group D of order 60, there are conjugacy classes corresponding to:

- the identity,
- 15 elements of order 2,
- 20 elements of order 3,
- 24 elements of order 5.

The class equation is $60 = 1 + 15 + 20 + 12 + 12$ (need all the terms to divide 60).

A normal subgroup contains its entire conjugacy class. $K \trianglelefteq D \implies |K|$ is a sum of elements of $\{1, 15, 20, 12, 12\}$ and $|K|$ divides $60 \implies |K| = 60 \implies K = D$.

Example 18. Classify all finite abelian groups of order 2025:
$2025 = 3^4 \cdot 5^2$. The possibilities are (written as products of cyclic groups):

1. $\mathbb{Z}/81\mathbb{Z} \times \mathbb{Z}/25\mathbb{Z}$
2. $\mathbb{Z}/81\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}$
3. $\mathbb{Z}/27\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/25\mathbb{Z}$
4. $\mathbb{Z}/27\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}$
5. $\mathbb{Z}/9\mathbb{Z} \times \mathbb{Z}/9\mathbb{Z} \times \mathbb{Z}/25\mathbb{Z}$
6. $\mathbb{Z}/9\mathbb{Z} \times \mathbb{Z}/9\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}$
7. $\mathbb{Z}/9\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/25\mathbb{Z}$
8. $\mathbb{Z}/9\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}$
9. $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/25\mathbb{Z}$
10. $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}$

Definition 19. The **center** of G is $Z(G) = \{g \in G \mid gx = xg \text{ for all } x \in G\} = \bigcap_{x \in G} C_G(x)$.

Proposition 20. *1. $Z(G)$ is an abelian normal subgroup of G*

2. If $G/Z(G)$ is cyclic, G is abelian

Proof. 1) from definition

2) $G/Z(G)$ cyclic order $d \in \mathbb{Z} \cup \{\infty\} \implies$ if $g \in G$ is the preimage of the generator, $G/Z(G) = \{g^i Z(G) \mid i \in \mathbb{Z}\}$.

$g_1, g_2 \in G \implies g_1 = g^i z_1$ and $g_2 = g^j z_2$ some $z_1, z_2 \in Z(G)$. So $g_1 g_2 = g^i z_1 g^j z_2 = g_2 g_1$.

□

Definition 21. Let $M = \{U \mid U \leq G\}$ set of subgroups of G. We have the action $G \times M \to M, (g, U) \mapsto gUg^{-1}$.

$G \cdot U = \{H \leq G \mid \exists g \in G \text{ such that } H = gUg^{-1}\} =: [U]_g$ is the class of subgroups **conjugate** to U

$G_U = \{g \in G \mid gUg^{-1} = U\} =: N_G(U)$ **normalizer of U in G**

Note 22. $U \leq G \implies N_G(U) \leq G$ and $U \trianglelefteq N_G(U)$.

$U \trianglelefteq G \iff N_G(U) = G$.

Corollary 23. $U \leq G \implies |[U]_G| = [G : N_G(U)]$.

Theorem 24. *If G is a finite p-group (i.e. $|G| = p^r$ for a prime p), if $\{e\} \neq N \trianglelefteq G$, then $N \cap Z(G) \neq \{e\}$. In particular, $Z(G)$ is nontrivial.*

Proof. G acts on N by conjugation.

$\mathrm{Fix}_G(N) = \{n \in N \mid gng^{-1} = n \forall g \in G\}$.

By the fixed point theorem, $|\mathrm{Fix}_G(N)| \equiv |N| \equiv 0 \pmod{p}$. So p divides $|\mathrm{Fix}_G(N)|$. Now, $|\mathrm{Fix}_G(N)| \neq 0$ since $\mathrm{Fix}_G(N) = Z(G) \cap N$ which contains 1. So $|Z(G) \cap N| \geq p$. □

Corollary 25. $|G| = p^2 \implies G$ *is abelian.*

Proof. If $|Z(G)| = p$, $|G/Z(G)| = p \implies G/Z(G)$ is cyclic, hence G is abelian.

If $|Z(G)| = p^2, Z(G) = G \implies G$ abelian. □

1.4.3 The Groups S_n and A_n

Definition 26. Any permutation $\sigma \in S_n$ can be written as a product of disjoint cycles $\sigma = (a_1 \dots a_{r_1}) \dots (a_s \dots a_{r_a})$.

The underordered tuple $(r_1, \dots, r_s)$ is called the **cycle type.**

Proposition 27. *Let $\tau, \sigma \in S_n$.*
τ, σ are conjugate $\iff$ τ, σ have the same cycle type.

Note 28. Recall:

- S_n is generated by transpositions
- For $\sigma \in S_n$, the parity of the number of transpositions needed to express it is unique, so we an define $\text{sign}(\sigma) = (-1)^{\#\text{transpositions}}$
- $A_n = \{\sigma \in S_n \mid \text{sign}(\sigma) = \pm 1$ is the **alternating group**. A_n has index 2 and $A_n \trianglelefteq S_n$.

Proposition 29. *1. When $n \geq 3$, A_n is generated by the 3-cycles.*

2. When $n \geq 5$, A_n is generated by double transpositions.

Proof. $S_n = \langle (ij) \mid 1 \leq i, j \leq n, i \neq k \rangle$.

1. Let $n \geq 3$. $A_n = \langle (ij)(kl) \mid 1 \leq i, j \leq n, i \neq j, 1 \leq k, l \leq n, k \neq l \rangle$.

   Let $(ij)(kl)$ be such a product. If $n \geq 3$, then $j = k \implies (ij)(kl) = (ijl)$. And $j = l \implies (ij)(kl) = (ijl)$.

   On the other hand, if $j \notin \{k, l\}, (ij)(kl) = (ij)(jk)(jk)(kl) = (ijk)(jkl)$.

2. Let $n \geq 5$. If (ijk) is a 3-cycle, then $\exists l, m \notin \{i, j, k\}$. So $(ijk) = (ij)(lm)(lm)(jk)$.

□

Proposition 30. *For $n \geq 5$, A_n is the only proper normal subgroup of S_n.*

Proof. Suppose $N \trianglelefteq S_n$ is proper and nontrivial. Then there exists some $\sigma \in N, i \in \{1, \ldots, n\}$, so that $\sigma(i) \neq i$.

Let $j \notin \{i, \sigma(i)\}$. Let $\tau := (ij)$. Then by Proposition 1, $\sigma^{-1}\tau^{-1}\sigma$ is a transposition. Let $\rho := \sigma^{-1}\tau^{-1}\sigma\tau$. Then $\rho(j) = \sigma^{-1}\tau^{-1}\sigma\tau(j) = \sigma^{-1}tau^{-1}\sigma(i) = \sigma^{-1}\sigma(i) = i \neq j$. Thus $\rho \neq e$ i.e. ρ is a product of two transpositions such that ρ is a 3-cycle or ρ is a double transposition, and $\rho \in A_n$.

$N \trianglelefteq S_n \implies \rho \in N$ $(\rho = \sigma^{-1}(\tau^{-1}\sigma\tau))$. By Proposition 1, N either contains all 3-cycles or all double transpositions.

So $A_n \leq N$, and since $N \neq S_n$, we have $N = A_n$. □

Theorem 31. *When $n \geq 5$, A_n is a simple nonabelian group.*

Proof. Assume $N \trianglelefteq\!\!\!\!\neq A_n$ is a proper normal subgroup and that N is nontrivial.

We know that N is not a normal subgroup of S_n, by Proposition 3. So there exists some $\tau \in S_n \setminus A_n$ such that $\tau N \tau^{-1} \neq N$. Without loss of generality, assume τ is a transposition.

So $\tau N\tau \trianglelefteq \tau A_n\tau = A_n$.

Let $H := \langle \tau N\tau, N\rangle = \tau N\tau N$ (since they are both normal).

Observe that $H \trianglelefteq A_n$ but $\langle A_n, \tau\rangle = S_n$ (since A_n is a maximal subgroup), so $H \trianglelefteq S_n$, thus by Proposition 3, $H = A_n$.

Now $N \cap \tau N\tau \trianglelefteq S_n \implies N \cap \tau N\tau = \{e\} \implies A_n = N \times \tau N\tau \implies |A_n| = |N|^2$. But $\frac{n!}{2}$ is not a square. $\rightarrow\leftarrow$ □

1.5 Composition Series and Solvability

1.5.1 Subnormal series and the Jordan-Hölder Theorem

Definition 1. A finite chain of subgroups

$$G = N_0 \trianglerighteq N_1 \trianglerighteq \cdots \trianglerighteq N_r = \{e\}$$

is called a **subnormal series** of length r.

Two subnormal series

$$G = N_0 \trianglerighteq N_1 \trianglerighteq \cdots \trianglerighteq N_r = \{e\}, \quad G = M_0 \trianglerighteq M_1 \trianglerighteq \cdots \trianglerighteq M_r = \{e\}$$

of the same length are **equivalent** if

$$N_i/N_{i+1} \cong M_{\sigma(i)}/M_{\sigma(i+1)}.$$

for some $\sigma \in S_r$ for all $i = 0, \ldots, r-1$.

A subnormal series is a **composition series** if all the quotients N_i/N_{i+1} are simple. The quotients are then called **composition factors**.

Example 2.
1. $\mathbb{Z}$ does not admit a composition series because we can always find more groups to stick in.
2. $S_5 \trianglerighteq A_5 \trianglerighteq \{e\}$ is a composition series of S_5. The composition factors are $S_5/A_5 \cong \mathbb{Z}/2\mathbb{Z}$ and $A_5/\{e\} \cong A_5$.

Definition 3. A normal subgroup $N \trianglelefteq G$ is a **maximal normal subgroup** if N is maximal in the normal subgroup lattice.

Note 4. N is a maximal normal subgroup $\iff$ G/N is simple (otherwise $S \trianglelefteq G/N$ gives $\pi^{-1}(S)$ a larger normal subgroup).

Lemma 5. *Let G be a group with a composition series. Let $M \trianglelefteq G$ be a maximal normal subgroup. The G has a composition series starting with $G \trianglerighteq M \trianglerighteq \ldots$ which is equivalent to the given one.*

Proof. Let $G \gneq N_1 \gneq \cdots \gneq N_r = \{e\}$ be a composition series. We'll proceed by induction on r.

If $r = 1$ then $G \gneq \{e\}$ is a composition series with $\{e\}$ the unique maximal subgroup.

If $r > 1$, without loss of generality assume $N_1 \neq M$. The $\langle N_1, M\rangle = N_1M = G$ since they are maximal.

Let $H_2 := N_1 \cap M$. $H_2 \trianglelefteq G$. By the isomorphism theorems,

$$G/N_1 \cong MN_1/N_1 \cong M/M \cap N_1 = M/H_2$$

and

$$G/M \cong MN_1/M \cong N_1/M \cap N_1 = N_1/H_2$$

So $M/H_2, N_1/H_2$, are simple (because N_1, M are maximal simple. So H_2 is a maximal normal subgroup in M and N_1.

By induction, N_1 has composition series $N_1 \gneq H_2 \gneq \dots$ equivalent to $N_1 \gneq N_2 \gneq \dots$

We have that $M \gneq H_2 \gneq \dots$ is a composition series of $M \implies$ we get a composition series which is equivalent to the given one, by the isomorphism statements above. □

Theorem 6 (Jordan–Hölder Theorem)**.** *Every two composition series over a group are equivalent. (In particular, all composition series have the same length).*

Proof. If $r = 0$ the statement is vacuously true.

Now say G has a composition series of length $t \leq r$. Let $G \gneq N_1 \gneq \cdots \gneq N_r = \{e\}$ be a composition series. Then all composition series of that group are equivalent.

$r = 1 : \checkmark$
$r > 1 : G \gneq M_1 \gneq \cdots \gneq M_s = \{e\} \implies$ by Lemma 1 G has a composition series $G \gneq M_1 \gneq \dots$ which is equivalent to $G \gneq N_1 \cdots \gneq N_r$. Then we get a composition series of M_1 of length $r-1$. By induction, this is equivalent to $M_1 \gneq M_2 \gneq \dots$. So both composition series of G are equivalent to $G \gneq N_1 \cdots \gneq N_r$. □

1.5.2 Solvable Groups

Definition 7. Let G be a group. Define the **i-th commutator subgroup of G (derived series)**:

$$G^{(0)} = G, \quad G^{(1)} = [G, G], \quad G^{(i+1)} = [G^{(i)}, G^{(i)}].$$

G is called **solvable** if this series terminates at the identity after finitely many steps, i.e. $G^{(r)} = \{e\}$ for some r.

Example 8. 1. Every abelian group is solvable (since $G' = \{e\}$).

2. Non-abelian simple groups are not solvable, since then $G' = G$.

Theorem 9. *1. Subgroups of solvable groups are solvable.*

2. Quotients of solvable groups are solvable.

3. Direct products of solvable groups are solvable.

Proof. **Exercise.** □

Proposition 10. *Every finite abelian group has a composition series with cyclic composition factors of prime order.*

Theorem 11. *For a finite group G, the following are equivalent:*

1. G is solvable.

2. G has a subnormal series with abelian factors.

3. G has a composition series whose factors are cyclic of prime order.

Proof. (1) $\Rightarrow$ (2): G solvable $\implies G \gneq G^{(1)} \gneq G^{(2)} \gneq \cdots \gneq G^{(r)} = \{e\}$ is a subnormal series with abelian quotients.

(2) $\Rightarrow$ (3): refine subnormal series using Proposition 2 to get composition series as desired.

(3) $\Rightarrow$ (1): Say $G = N_0 \gneq N_1 \gneq \cdots \gneq N_s = \{e\}$ is a composition series with cyclic factors. $N_{i+1} \geq N_i' \geq (G^{(i)})' = G^{(i+1)} \implies \{e\} = N_s \geq G^{(s)} \implies G^{(s)} = \{e\} \implies G$ is solvable. □

Theorem 12 (Feit-Thompson Theorem)**.** *Groups of odd order are solvable.*

1.5.3 Nilpotent Groups

Definition 13. A **normal series** of G is a chain of finite length $G = N_0 \gneq N_1 \cdots \gneq N_r = \{e\}$ where $N_i \trianglelefteq G$ for all i.

A **central series** for a group G is a normal series for which $N_i/N_{i+1} \leq Z(G/N_{i+1})$ for all $i = 0, \ldots, r-1$. A group admitting a central series is called **nilpotent**.

Theorem 14. *1. Every subgroup of a nilpotent group is nilpotent.*

2. Every quotient of a nilpotent group is nilpotent.

3. Nilpotent groups are solvable.

Proof. **Exercise** □

Proposition 15. *Define inductively*

$$G_{(0)} := G, \quad G_{(i+1)} = [G_{(i)}, G] \text{ (the subgroup generated by } [g,h] \text{ where } g \in G_{(i)}, h \in G).$$

Then $G_{(i+1)} \leq G_{(i)}$ and $G_{(i)} \trianglelefteq G$ for all i.

Proof. $i = 0 : G_{(1)} = [G, G] = G' \trianglelefteq G = G_{(0)}$ ✓

$G_{(i+1)} = [G_{(i)}, G] \leq [G_{(i-1)}, G] = G_{(i)}$.

For $g \in G, gG_{(i+1)}g^{-1} = [gG_{(i)}g^{-1}, G]$ (by induction) $= [G_{(i)}, G] = G_{(i+1)}$. □

Corollary 16. $G_{(i)}/G_{(i+1)} \leq Z(G/G_{(i+1)})$.

Proof. Take $g \in G_{(i)}, h \in G$. Then $[g, h] \in [G_{(i)}, G] = G_{(i+1)}$. □

Definition 17. The series in Proposition 2 is called the **lower central series** of G.

Note 18. This is not always a central series—only when it ends in $\{e\}$.

Definition 19. The **upper central series** of a group G is defined as

$$Z_0(G) = \{e\} \trianglelefteq Z_1(G) \trianglelefteq \dots, \qquad Z_{i+1}(G)/Z_i(G) = Z(G/Z_i(G)).$$

Note 20. This is not always a central series—only when it ends in G.

Theorem 21. *If $G = N_0 \trianglerighteq \cdots \trianglerighteq N_r = \{e\}$ is a central series for G, then*

1. *$N_i \geq G_{(i)}$ for all $i = 0, \dots, r$*
2. *$N_{r-j} \leq Z_j(G)$ for all $j = 0, \dots, r$*

Proof. **Exercise** □

Theorem 22. *For a group G, the following are equivalent:*

1. *G is nilpotent.*
2. *The upper central series terminates at G.*
3. *The lower central series terminates at $\{e\}$.*

Proof. (1) ⇒ (2): given a central series $G = N_0 \trianglerighteq \cdots \trianglerighteq N_r = \{e\}$, by Theorem 4, $G = N_0 \leq Z_r(G) \leq G \implies Z_r(G) = G \implies$ the upper central series ends at G.

(2) ⇒ (3): Suppose $Z_s(G) = G$ for some s. Then the upper central series is a central series. By Theorem 4, $Z_{s-i}(G) \geq G_{(i)}$ for all $i = 0, \dots, s$. When $s = i$, $\{e\} = Z_0(G) \geq G_{(s)} \implies G_{(s)} = \{e\} \implies$ the lower central series ends at $\{e\}$.

(3) ⇒ (1): clear. □

Example 23. S_3 is nonabelian

$S_3 \trianglerighteq A_3 \trianglerighteq \{e\}$

The factors are of cyclic prime order and this is a composition series, so S_3 is solvable

$Z_0(S_3) = \{e\}$

$Z(S_3/Z_0(S_3)) = \{e\} \implies Z_i(S_3) = \{e\}$ for all i. So S_3 is not nilpotent.

Example 24. $Q_8 = \{\pm 1, \pm I, \pm J, \pm K\}$ where
$I = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$,and $K = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} = IJ$.

$Z(Q_8) = \{\pm 1\}$.

$|Q_8/Z(Q_8)| = 4 \implies$ abelian $\implies Z(Q_8/Z(Q_8)) = Q_8 \implies Z_1(Q_8) = Q_8 \implies Q_8$ is nilpotent. Hence Q_8 is also solvable.

1.6 Sylow Theory

1.6.1 Sylow Theorems

Lemma 1. *If G is a finite abelian group of order m and p is a prime dividing m, then G has a subgroup of order p.*

Proof. Structure theorem, reduce to cyclic case. □

Definition 2. Let p be a prime. A finite group G of order $|G| = p^r$ for $r \in \mathbb{N} \setminus \{0\}$ is called a **p-group.**

A p-group is a **p-Sylow subgroup** of the finite group G if $H \leq G$ and p does not divide $[G : H]$.

Theorem 3 (First Sylow Theorem)**.** *If G is a finite group and $p \mid |G|$ where p is prime, then G has a p-Sylow subgroup.*

Proof. Proceed by induction on $m = |G|$.

If $m = 1$ or m is prime, done.

Otherwise, suppose $H \subsetneq G$ is a subgroup such that p does not divide $[G : H]$. Then every p-Sylow subgroup of H is a p-Sylow subgroup of G, and we are done by induction. Hence, we may assume p divides the index of every proper subgroup of G.

Then by the class equation,

$$|G| = Z(G) + \sum_{[G:C_G(g)]>1} [G : C_G(g)].$$

Then since $|G|$ and $\sum_{[G:C_G(g)]>1}[G : C_G(g)]$ are divisible by p, $|Z(G)| \neq 1$. So p divides the order of the center $Z(G)$, thus by Lemma 1 there is $H \leq Z(G)$ with $|H| = p$. $H \leq Z(G) \implies H \trianglelefteq G$.

Quotienting by H and applying the inductive hypothesis gives a Sylow subgroup of G:
If $p^n \mid |G|$ and $p^{n+1} \nmid |G|$ then $p^{n-1} \mid |G/H|$ and $p^n \nmid |G/H|$. Then by induction G/H has a p-Sylow subgroup $P \leq G/H$ with $|P| = p^{n-1}$. Then if $\pi : G \to G/H$ is the quotient map, letting $K := \pi^{-1}(P) \leq G$, we have $H \leq K$ and $K/H \cong P$ so $|K| = p^n$. □

Example 4. Let $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ be the finite field with p elements, and $G = \mathrm{GL}_n(\mathbb{F}_p)$, the group of n by n matrices with nonzero determinant with entries in $\mathbb{F}_p$. Find a p-Sylow subgroup.

$|G| = (p^n-1)(p^n-p)\dots(p^n-p^{n-1})$. The highest power of p dividing $|G|$ is $1+2+\cdots+(n-1)$.

Want: $H \leq G$ such that $|H| = p^{1+2+\cdots+(n-1)}$.

Let

$$H = \left\{ \begin{pmatrix} 1 & * & \cdots & * \\ 0 & 1 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} \right\} = U_n(\mathbb{F}_p) \leq GL_n(\mathbb{F}_p)$$

H is a Sylow p-subgroup of order $p^{n(n-1)/2}$.

Theorem 5 (Second Sylow Theorem)**.** *Let $|G| < \infty$ and $p \mid |G|$ be a prime.*

1. *Every p-subgroup of G is contained in a p-Sylow subgroup.*
2. *Any two p-Sylow subgroups are conjugate in G.*
3. *If $n_p =$ the number of p-Sylow subgroups of G, then $n_p \equiv 1 \pmod{p}$ and $n_p \mid \frac{|G|}{p^r}$ where p^r is the highest power of p dividing $|G|$.*

Example 6.

1. Consider S_3, with $|S_3| = 6 = 2 \cdot 3$.
   - $n_3 \equiv 1 \pmod{3}$ and $n_2 \mid 2$ so $n_3 = 1$.
   - $n_2 \equiv 1 \pmod{2}$ and $n_2 \mid 3$ so $n_2 = 1$ or 3
   - The transpositions each generate a subgroup of order 2, so $n_2 = 3$
2. There are no simple groups of order $40 = 2^3 \cdot 5$. Indeed, $n_5 \equiv 1 \pmod{5}$ and $n_5 \mid 8$, so $n_5 = 1$. Thus the 5-Sylow subgroup is normal.

Proof. (Of Theorem 3):

1. Let $P \leq G$ be a p-Sylow subgroup. Let $S = \{\text{conjugates of } P \text{ in } G\}$. Everything in S is a p-Sylow subgroup.

   $P \leq N_G(P) \leq G$. Now $[G : N_G(P)]$ is not divisible by p since $p \nmid [G : P]$. We can use the orbit equation:

   $$|S| = [G : N_G(P)] \implies p \nmid |S|$$

   H also acts on P by conjugation, and H is a p-group. We can use the fixed point theorem: $\exists \tilde{p} \in S$ such that $N_G(\tilde{p}) \geq H$. Without loss of generality, assume $H \leq N_G(P) \implies HP \leq N_G(P)$ and $P \trianglelefteq HP$. So

   $$|H| = \frac{|H||P|}{|H \cap P|}$$

   is a p-power and divides $|G|$ (since $HP \leq G$), hence $|HP| = |P|$ and thus $HP = P \implies HP \leq P$.

2. Apply the proof of (1) to $H = Q$ p-Sylow

3. Let $Q \leq G$ be p-Sylow. As above, if $Q \leq N_G(P)$ then $Q = P$. Hence Q acts on S with exactly one fixed point (itself), thus $n_p \equiv |S| \equiv 1 \pmod{p}$.

□

Example 7. Every simple group of order 60 is isomorphic to A_5.

Let $|G| = 60 = 2^2 \cdot 3 \cdot 5$.

- $n_5 \equiv 1 \pmod{5}$ and $n_5 \mid 12$, so $n_5 = 1, 6$.
- $n_3 \equiv 1 \pmod{3}$ and $n_3 \mid 20$, so $n_3 = 1, 10$.
- $n_2 \equiv 1 \pmod{2}$ and $n_2 \mid 15$, so $n_2 = 3, 5, 15$.

G simple implies $\ker \pi$ is trivial (where π embeds G into the permutation group), so: If $n_2 = 3$, then $\pi : G \to S_3$ is injective $\to\leftarrow$.
If $n_2 = 5$, then $\pi : G \to S_5$ is injective $\implies G \cong A_5$.
If $n_2 = 15$ and $n_5 \equiv \pmod{5}$ and $n_5 \neq 1$, then $n_5 = 6$. In this case:
The 5-Sylows intersect trivially by having prime order, so the 5-Sylows contain 24 nontrivial elements and thus the 2-Sylows contain ≤ 35 nontrivial elements. So there are P, Q 2-Sylows so that $|P \cap Q| = 2$. Let $M := N_G(P \cap Q)$. Observe: $P \leq M, G \leq M$ since P, Q are abelian. So $6 \leq |M|$ and $4 \mid |M| \implies |M| \in \{12, 20, 60\}$.

If $|M| = 60, N_G(P \cap Q) = G \implies P \cap Q \trianglelefteq G \to\leftarrow$.
If $|M| = 20, \pi : G \to S_3$ is injective $\to\leftarrow$.
If $|M| = 12, \pi : G \to S_5$ is injective $\implies G \cong A_5$.

Example 8. Let $|G| = 42 = 2 \cdot 3 \cdot 7$.
$n_2 \in \{1, 3, 7, 21\}, n_3 \in \{1, 7\}$, and $n_7 \in \{1\}$. For $p = 2, 3, 7$, we have $n_p(p-1)$ nontrivial elements of order p. So we get 6 elements of order 7, ≤ 21 elements of order 2, and ≤ 14 elements of order 3. $6 + 21 + 14 + 1 = 28 + 14 = 42$ *does not* imply $n_3 = 7$ and $n_2 = 21$, because there exist elements of other orders.

Example 9. Let

$$U = \left\{ \begin{pmatrix} 1 & * & \cdots & * \\ 0 & 1 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} \right\} \leq GL_n(\mathbb{F}_p) = G$$

Recall that $|U| = 1 \cdot p \cdot \cdots \cdot p^{n-1}$ and $|G| = (p^n - 1)(p^n - p) \dots (p^n - p^{n-1})$.

$n_p \equiv 1 \pmod{p}$ and n_p divides $(p^n - 1)(p^{n-1} - 1) \dots (p - 1)$. Now $n_p = [G : N_G(U)] = \frac{|G|}{|N_G(U)|}$.

In fact, $N_G(U) = \{g \in G \ gUg^{-1} = U\} = \{APA^{-1} \in U \forall P \in U\} = \{\text{upper triangular matrices in } GL_n(\mathbb{F}_p)\}$.

So $|N_G(U)| = (p-1)^n$ hence $n_p = (1 + p)(1 + p + p^2) \dots$

1.6.2 Normalizers of p-Sylow Subgroups

Let $\mathrm{Syl}_p(G) :=$ set of p-Sylows of G.

Proposition 10. *Let $P \in \mathrm{Syl}_p(G)$. Then:*

1. *P is the unique p-Sylow subgroup in $N_G(P)$.*
2. *$N_G(P)$ is the largest subgroup with that property.*

Theorem 11. *If $N \trianglelefteq G$ and $P \in \mathrm{Syl}_p(N)$, then*

1. $G = N \cdot N_G(P)$
2. $P \trianglelefteq N \implies P \trianglelefteq G$

Proof. 1. Let $g \in G$. **Frattini argument:** $gPg^{-1} \leq gNg^{-1} = N$. So $|gPg^{-1}| = |P| \implies gPg^{-1}$ is a p-Sylow subgroup of N. That is, there exists $n \in N$ such that $nPn^{-1} = gPg^{-1} (\iff n^{-1}gPg^{-1}n = P) \implies n^{-1}g \in N_G(P) \implies g \in N \cdot N_G(P)$.

2. $P \trianglelefteq N, N \leq N_G(P) \implies G = N \cdot N_G(P) = N_G(P) \implies P \trianglelefteq G$.

□

Example 12. Let $G = GL_2(\mathbb{F}_3), N = SL_2(\mathbb{F}_3)$.

(Recall: $\det : GL_n(\mathbb{F}_3) \to \mathbb{F}_3^\times$ has $\ker(\det) = SL_n(\mathbb{F}_3)$ so $|SL_n(\mathbb{F}_3)| = \frac{|GL_n(\mathbb{F}_3)|}{|\mathbb{F}_3^\times|}$.)

$|G| = (3^2 - 1)(3^2 - 1) = 48$
$|N| = 4 \cdot 6 = 24$

Check: N has a unique 2-Sylow (Q_8) which is normal in G by Theorem 2. This is not a 2-Sylow of G but is contained in all of them (there are three).

Corollary 13. *Let $P \in \mathrm{Syl}_p(G)$. If $N_G(P) \leq H \leq G$, then $N_G(H) = H$ and $[G : H] \equiv 1 \pmod{p}$. In particular, $N_G(N_G(P)) = N_G(P)$ (normalizers of p-Sylow subgroups are self-normalizing).*

Proof. Theorem 2 $\implies N_G(H) = G \cdot N_{N_G(H)}(P) \leq H \cdot N_G(P) = H$.
Moreover, $N_G(P) = N_H(P)$ since $N_G(P) \leq H \implies n_p(G) = [G : N_G(P)] \equiv 1 \pmod{p}$ and $n_p(H) = [H : N_H(P)] \equiv 1 \pmod{p}$. So $[G : H] \equiv 1 \pmod{p}$. □

1.6.3 p-Groups

Theorem 14. *Every p-group is nilpotent.*

Sketch of proof. Induction on $|G| = m$.

If $m = 1$, ✓.

For $m > 1$, by the fixed-point theorem, $Z(G) = \{e\} \implies G/Z(G)$ is a p-group and nilpotent by induction. Then $G/Z(G)$ has a central series

$$G/Z(G) = N_0/Z(G) \supsetneq \cdots \supsetneq N_r/Z(G) = \{e\} \implies G = N_0 \supsetneq \cdots \supsetneq N_r = \{e\}$$

By the correspondence theorem, this is a normal series.

$$N_i/N_{i+1} \cong \frac{N_i/Z(G)}{N_{i+1}/Z(G)} \leq Z(\frac{G/Z(G)}{N_{i+1}/Z(G)}) \cong Z(G/N_{i+1}),$$

so this is a central series. Hence G is nilpotent. □

Corollary 15. *Every nontrivial p-group has a composition series with cyclic order p factors.*

Corollary 16. *If p is a prime so that $p \mid |G|$, $|G| = p^r m$, and $p \nmid m$, then for all $s \in \{0, \ldots, r\}$ there exists $H \leq G$ so that $|H| = p^s$.*

Proof. $r = 0$ ✓

$r > 0 \implies$ there exists a p-Sylow $H \leq G$ with $|H| = p^r$. By Corollary 2, there exists a central series $H = H_0 \supsetneq \cdots \supsetneq H_r = \{e\}$ with $|H_i/H_{i+1}| = p, |H_i| = p^{r-i}$. □

Corollary 17 (Cauchy's Theorem)**.** *If $|G| < \infty, p \mid |G|$ then there exists $H \leq G$ with $|H| = p$.*

Definition 18. Let G be a finite group. Then the **Frattini subgroup of** G is

$$\Phi(G) := \bigcap_{M \leq G \text{ maximal}} M.$$

Note 19. $\Phi(G) \trianglelefteq G$. In fact, any automorphism of G maps $\Phi(G)$ to itself (i.e. $\Phi(G)$ is a characteristic subgroup).

Definition 20. If G is a group and $g \in G$, we say g is a **non-generator** if $\langle x, g\rangle = G \implies \langle x\rangle = G$.

Proposition 21. *Let G be a finite group. Then $\Phi(G) = \{g \in G \mid g \text{ is a non-generator}\}$. In particular, if $H \leq G$ and $H\Phi(G) = G$, then $H = G$.*

Definition 22. An abelian p-group G is **elementary abelian** if every nontrivial element has order p. ($G \cong (\mathbb{Z}/p\mathbb{Z})^r$).

Lemma 23. *If G is a p-group, every proper maximal subgroup $M \subsetneq G$ is normal of index p.*

Proof. **Exercise** □

Proposition 24. *Let G be a p-group. Then $G/\Phi(G)$ is elementary abelian. If $H \trianglelefteq G$ such that G/H is elementary abelian, then $\Phi(G) \leq H$.*

Proof. By Lemma 6, every maximal subgroup $M \leq G$ is normal in G of index p. Then the quotients G/M are isomorphic to $\mathbb{Z}/p\mathbb{Z}$ so $G' \leq M$ for all proper maximal $M \leq G$ (since G/M is abelian), hence $G' \leq \Phi(G) \implies G/\Phi(G)$ is abelian.

M maximal $\implies |G/M| = p, g^p \in M$ for all proper maximal $M \leq G, \forall g \in G$. Then $g^p \in \Phi(G)$ for all $g \in G \implies$ every nontrivial element in $G/\Phi(G)$ has order $p \implies G/\Phi(G)$ is elementary abelian.

Suppose $H \trianglelefteq G, G/H$ is elementary abelian, and $\Phi(G) \leq H$. Say $|G/H| = p^n$ i.e. $G/H \cong \langle g_1 H\rangle \times \cdots \times \langle g_n H\rangle$ for cosets $g_i H$ of order p. Define $U_i := \langle g_j H | j \neq i\rangle$ and let $H_i \leq G$ such tat $H_i/H = U_i$.

Note: U_i is maximal in G/H by having index p. And $\bigcap U_i = \{e\}$ (e.g. $U_{i_1} \cap U_{i_2} = \langle g_j H \mid j \neq i_1, i_2\rangle$ so intersecting them all leaves no $g_j H$'s). Then $H_i \leq G$ is maximal and $\bigcap H_i = H$ hence $\Phi(G) \leq H$. □

Corollary 25. *Let $G^p := \langle g^p \mid g \in G\rangle$. Then $\Phi(G) = G'G^p$.*

Proof. From the proof of Proposition 7, $g \in G \implies g^p \in \Phi(G) \implies G^p \leq \Phi(G)$, and $G' \leq \Phi(G)$, so $G'G^p \leq \Phi(G)$.

Conversely, $G/G'G^p$ is abelian and $g^p \in G'G^p$ so $G/G'G^p$ is elementary abelian. Then $\Phi(G) \leq G'G^p$ by Proposition 7. □

Theorem 26 (Burnside's Basis Theorem)**.** *Let G be a finite p-group. If $\{g_1\Phi(G), \ldots, g_d\Phi(G)\}$ generates $G/\Phi(G)$ then G is generated by $g_1, \ldots, g_d$. (this shows a correspondence between minimal generating sets of G and bases of $G/\Phi(G)$).*

1.6.4 Groups of Order *pq*

So far:

- $|G| = p^2 \implies G$ abelian
- $|G| = p^r \implies G$ solvable
- $|G| = pq \implies$??? (we already showed that S_3 is not nilpotent)

Lemma 27. *Let $|G| < \infty$ and let p be the smallest prime dividing $|G|$. Then any subgroup of index p in G is normal.*

Proof. Recall: prime index implies maximal (if $M \leq G$ of index p and $M \leq K$ then $p = [G : M] = [G : K][K : M]$ so either $[G : K] = 1$ or $[K : M] = 1$).

Let $H \leq G$ be maximal. Let $N := N_G(H)$. So $N = H$ or $N = G$.

Assume (for contradiction) that $N = H$. Then we have the map $\pi : G \to S_p$ by acting on conjugates of H. Let

$$K := \ker \pi = \{g \in G \mid g \in N_G(hHh^{-1}) \forall h \in G\} =$$
$$\{g \in G \mid g \in hN_G(H)h^{-1} \forall h \in G\} = \bigcap_{h \in G} hHh^{-1}$$

.
$|\pi(G)| = [G : K] = [G : H][H : K] = p \cdot [H : K]$.
H is not normal in G so $[H : K] > 1$.
Now $|\pi(G)| \mid |S_p| \implies [H : K] \mid (p-1)!$. But also $[H : K] \mid |G| \to\leftarrow$. ($p$ is the smallest prime dividing $|G|$) □

Corollary 28. *If $|G| = pq$ with $p < q$ primes, then G has a normal q-Sylow subgroup.*

Theorem 29. *If $|G| = pq$ with p, q primes, then G is solvable.*

Theorem 30. *If $|G| = pq$ for primes $p < q$, if $q \not\equiv 1 \pmod{p}$, then G is cyclic.*

Proof. By Corollary 2, there exists a unique $Q \leq G$ with $|Q| = q$. $n_p = 1 + kp$ for some k, some $n_p \in \{1, q\}$. Then $q \not\equiv 1 \pmod{p} \implies n_p = 1 \implies G \cong P \times G \cong \mathbb{Z}/(pq)\mathbb{Z}$. □

Theorem 31 (Burnside)**.** *If $|G| = p^a q^b$, where p, q are primes, then G is solvable.*

1.6.5 More on nilpotent groups

Lemma 32. *If G is nilpotent, and U is a proper subgroup of G, then U is a proper subgroup of $N_G(U)$.*

Proof. Say we have a central series of G: $N_0 := G \trianglerighteq \cdots \trianglerighteq N_r = \{e\}$. Take $0 \leq i$ so that $U \geq N_{i+1}$ and U is a proper subgroup of N_i. Take $u \in U, n \in N_i$: $[n_i, u] \in N_{i+1} \subset U$, since N_i is central modulo N_{i+1}. SO $N_i \leq N_G(U)$ but $N_i \not\leq U$. □

Theorem 33. *The following are equivalent:*

1. *G is nilpotent.*
2. *All Sylow subgroups of G are normal.*
3. *G is the direct product of its Sylow subgroups.*
4. *Elements of coprime order commute in G.*

Proof. (1) ⇒ (2): Let p be a prime dividing $|G|$ and let $P \in \mathrm{Syl}_p(G)$. Then by Corollary 2.3, $N_G(N_G(P)) = N_G(P)$. By Lemma 1, $N_G(P) = G$.

(2) ⇒ (3): Corollary 3.1.3

(3) ⇒ (4): Say we have g, h such that $\gcd(|g|, |h|) = 1$. Then if $G \cong P_1 \times \ldots P_r$ where P_i is a p_i-Sylow, sorted so that $g = g_1 \ldots g_s, g_i \in P_i$ and $h = h_{s+1} \ldots h_r, h_i \in P_i$, then $gh = hg$.

(4) ⇒ (1): Let $p_1, \ldots, p_r$ denote prime divisors of $|G|$, and let $P_1, \ldots, P_r$ denote the corresponding p_i-Sylow subgroups. Define $\varphi : P_1 \times \ldots P_r \to G, (g_1, \ldots, g_r) \mapsto g_1 \ldots g_r$. φ is a homomorphism since elements of coprime order commute.

$P_1 \ldots P_r = \langle P_1, \ldots, P_r \rangle$ since φ is a homomorphism. $g \in \ker \varphi \implies g = (g_1, \ldots, g_r)$ so that $g_1 \ldots g_r = e \implies g_1 = \cdots = g_r = e \implies \varphi$ is injective.

$|P_1 \times \cdots \times P_r| = |G|$ so φ is an isomorphism. Each P_i is nilpotent (by being a p-group), and the direct product of nilpotents is nilpotent. □

1.6.6 Sylow subgroups of symmetric groups

Example 34. For S_8, determine size of 2-Sylow subgroup of S_8.

Answer: 2^7.

In general: what is the highest power p^M dividing $n!$? Factors of $n!$ divisible by p are $p, 2p, \ldots, kp$ where $k = \lfloor \frac{n}{p} \rfloor$. So n is divisible by p^k and p-powers in $k!$ so $\lfloor \frac{k}{p} \rfloor = \lfloor \frac{n}{p^2} \rfloor$ so inductively

$$M = \sum_{i=1}^{\infty} \left\lfloor \frac{n}{p^i} \right\rfloor.$$

Write

$$n = a_0 p^0 + a_1 p^1 + a_2 p^2 + \cdots + a_r p^r, a_i \in \{0, \ldots, p-1\}. \quad \text{(p-adic expansion)}$$

Then

$$\begin{aligned} M &= (a_1 + a_2 p + \cdots + a_r p^{r-1}) + (a_2 + a_3 p + \cdots + a_r p^{r-2}) + \cdots + a_r \\ &= a_1 + a_2(1+p) + a_3(1+p+p^2) + \cdots + a_r(1 + \cdots + p^{r-1}) \end{aligned}$$

Decompose $\{1, \ldots, n\}$ into a_r blocks of p^r letters, a_{r-1} blocks of p^{r-1} letters, ..., a_0 blocks of 1 letter.

Suppose we have p-Sylow subgrops P_i of S_{p_i} for $i = 0, \ldots, r$. Then $(P_0)^{a_0} \times \cdots \times (P_r)^{a_r} \leq S_n$ since $|P_i| = p^{1+\cdots+p^{i-1}}$.

So: need to find p-Sylows of $S_{p^i}, i \in \mathbb{N}$. Proceed by induction:

When $i = 1$, $P_1 = \langle (1 \ldots p) \rangle$ p-Sylow of S_p.

When $i = 2$, S_{p^2} contains p subset of order p and so we get $Q_1 \times \cdots \times Q_p$ where $Q_1 = \langle (1 \ldots p) \rangle = \langle c_1 \rangle$, $Q_2 = \langle (c_{p+1} \ldots 2p) \rangle = \langle c_2 \rangle$, ...

We get $|Q_1 \times \ldots Q_p| = p^p$ and all factors are isomorphic to P_1. Now $b = (1, p+1, 2p+1, \ldots)(2, p+2, \ldots) \ldots (p, 2p, \ldots)$ has order p.

Let $b^{-1} c_i b = c_{i+1}$ (take the indices modulo p).

Let $P := \langle b, c_1, \ldots, c_p \rangle$. P has order p^{1+p} so is a p-Sylow of S_{p^2}.

For the induction step, say $i = r+1$ and that P_r is a p-Sylow of S_{p^r}. Then $Q_1 \times \cdots \times Q_p \leq S_{p^{r+1}}, Q_i \leq P_r \forall i$.

Define $c := (1, p^r + 1, \ldots)(2, p^r + 2, \ldots) \ldots (p^r, 2p^r, \ldots)$. Then $\langle c, Q_1, \ldots, Q_p \rangle$ has order $p^{1+\cdots+p^{r-1}} \cdot p$ so is p-Sylow.

Example 35. The 2-Sylow subgroup of S_8 has order $2^7 = 128$. To find one: In $S_2 : \langle (12) \rangle$.

In $S_4 : \langle (12), (34), (12)(34) \rangle = \langle (12), (13)(24) \rangle$.

In $S_8 : \langle (12), (13)(24), (56), (57)(68), (15)(26)(37)(48) \rangle$ gives a 2-Sylow subgroup of S_8.

Example 36. Classify all groups of order 12.

$12 = 3 \cdot 2^2$. We know $n_3 \equiv 1 \pmod 3, n_3 \mid 4 \implies n_3 \in \{1, 4\}$.

Case 1: If $n_3 = 1$, we have a normal 3-Sylow $N \trianglelefteq G$ with $|N| = 3$. Let H be a 2-Sylow. $H \cap N = \{e\} \implies NH = G$. H acts on N by conjugation.

Case 2: If $n_3 = 4$, the 3-Sylows cover 8 nontrivial elements. This leaves 4 elements, so only room for one 2-Sylow subgrouup $M \trianglelefteq G$. If Q is a 3-Sylow, then $MQ = G$ as above and Q acts on M by conjugation.

1.7 Group Extensions and Group Cohomology

1.7.1 Semidirect Product

Recall: $\mathrm{Aut}(G) = \{\varphi : G \to G \mid \varphi \text{ bijective homomorphism}\}$. We have $\mathrm{id} \in \mathrm{Aut}(G), \mathrm{id}(g) = g$ for all $g \in G$.

Before: actions of groups G on sets X corresponded to homomorphisms $G \to S(X)$. Now: actions of groups G on other groups H via automorphisms will correspond to homomorphisms $G \to \mathrm{Aut}(H)$.

Example 1.

1. $G = \mathbb{Z}/3\mathbb{Z} = \langle g \rangle$. If $\varphi \in \mathrm{Aut}(G), \varphi \neq \mathrm{id}$ then $\varphi(g) = g^2$. So $\varphi^2(g) = \varphi(\varphi(g)) = \varphi(g^2) = \varphi(g)^2 = g^4 = g$. So $\varphi^2 = \mathrm{id} \implies \mathrm{Aut}(G) \cong \mathbb{Z}/2\mathbb{Z}$.

2. $G = \mathbb{Z}/4\mathbb{Z} = \langle g \rangle$.

   If $\varphi \in \mathrm{Aut}(G), \varphi \neq \mathrm{id}$ and $\varphi(g) = g^2$ then since $|g| = 4$ and$|g^2| = 2$ we get a contradiction.

   So say $\varphi(g) = g^3$. Then $\varphi^2(g) = \varphi(\varphi(g)) = \varphi(g^3) = \varphi(g)^3 = g^9 = g$.

   One can check that $\varphi^2 = \mathrm{id}$ and that $\mathrm{Aut}(G) \cong \mathbb{Z}/2\mathbb{Z}$.

3. $G = V_4 \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} = \{e, g, h, gh\}$. V_4 is elementary abelian so we can think of it as $\mathbb{F}_2^2$. Then $\mathrm{Aut}(G) \cong GL_2(\mathbb{F}_2)$ and has size 6.

   G has 3 groups of order 2. Take the element that permutes them and take a transposition. $\mathrm{Aut}(G) = \langle \omega, \tau \rangle, \omega : g \mapsto h \mapsto gh$ (3-cycle) and $\tau : g \mapsto h \mapsto g$ (transposition). $\mathrm{Aut}(G) \cong D_3 \cong S_3$.

Definition 2. Let G and H be groups. An **action of G on H** by automorphisms is a homomorphism

$$\varphi : G \to \mathrm{Aut}(H).$$

We then write $g \cdot h = \varphi(g)(h)$ for $g \in G$, $h \in H$ and say G acts on H via φ.

Example 3. Conjugation is an action of G on itself and on its normal subgroups.

Proposition 4. *Let G act on a group H via $\varphi : G \to \mathrm{Aut}(H)$. Then:*

1. *If $U \subseteq H$ such that $G.U \leq U$, then φ induces an action of G on U.*
2. *If $N \trianglelefteq H$ and $G.N \subseteq N$, then φ induces an action of G on H/N.*
3. *If $N \trianglelefteq H$ so that $N \subseteq \mathrm{Ker}\,\varphi$, then φ induces an action of G/N on H.*

Proof. **Exercise.** □

Theorem 5. *Given groups N and H, suppose H acts on N via $\varphi : H \to \mathrm{Aut}(N)$. Then*

$$G := \{(n, h) \mid n \in N, h \in H\}$$

is a group via

$$(n_1, h_1) \cdot (n_2, h_2) := (n_1\, \varphi(h_1)(n_2),\ h_1 h_2).$$

Moreover,

1. *$H^* = \{(e, h) \mid h \in H\} \leq G$ and $H^* \cong H$.*
2. *$N^* = \{(n, e) \mid n \in N\} \trianglelefteq G$ and $N^* \cong N$.*
3. *$H^* \cap N^* = \{e\}$, $N^* H^* = G$, and $G/N^* \cong H^* \cong H$. (outer semidirect product)*

Proof. **Exercise** □

Example 6. Let $H = \mathbb{Z}/2\mathbb{Z} = \langle \tau \rangle$, $N = \mathbb{Z}/3\mathbb{Z} = \langle \sigma \rangle$, $\mathrm{Aut}\, N = \{\mathrm{id}, \sigma \mapsto \sigma^2\}$.

H acts on N via $\varphi(\tau)(\sigma) = \sigma^2$. Observe that $\varphi(\tau^2)(\sigma) = (\varphi(\tau)(\sigma))^2 = \sigma^2 \cdot \sigma^2 = \sigma$. So $G = \{(n, h)\}$ as in Theorem 2 $\implies$ G is nonabelian and order 6 $\implies$ $G \cong S_3$.

Definition 7. (External Semidirect Product)
If H, N, G, φ are as in Theorem 2, G is called the **semidirect product of N and H via φ** (Notation: $G = N \rtimes H$ or $G = N \rtimes_\varphi H$)

Proposition 8 (Internal Semidirect Product)**.** *If G is a group, $N \trianglelefteq G$, $H \leq G$, if $N \cap H = \{e\}$ and $NH = G$, then $G = N \rtimes_\varphi H$ where $\varphi : H \to \mathrm{Aut}\, N, \varphi(h)(n) = hnh^{-1}$.*

Proof. If $G = NH$, then if $g \in G$, $g = nh$ for some $n \in N, h \in H$. And this is unique because $N \cap H = \{e\}$. Now $g_1 = n_1 h_1$ and $g_2 = n_2 h_2$ $\implies$ $g_1 g_2 = n_1 h_1 n_2 h_2 = (n_1 h_1 n_2 h_1^{-1})(h_1 h_2) \in NH$, and the right hand side is the unique way of representing the element as a prodict in NH.

The $N \rtimes_\varphi H \to G, (n, h) \mapsto n \cdot h$ is an isomorphism. □

Definition 9. In the situation of Proposition 3, H is called a **complement to** N.

Example 10. Let

$$G = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \right\} \in GL_2(K)$$

for a field K. Then let

$$D = \left\{ \begin{pmatrix} a & 0 \\ 0 & c \end{pmatrix} \right\} \leq G, \quad U = \left\{ \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \right\} \trianglelefteq G$$

Then $G = UD, D \cap U = \{e\}$, so $G \cong U \rtimes D$.

Example 11. Groups of order 12:

Say $|G| = 12$.

- Case 1: $N \trianglelefteq G, |G| = 12, N$ is a 3-Sylow, $N = \langle g \rangle$. Let H be a 2-Sylow, $|H| = 4$. Then H is complement to N. By Proposition 3, there exists some action $\varphi : H \to \operatorname{Aut} N$ such that $G \cong N \rtimes_\varphi H$. We know that $\operatorname{Aut} N \cong \mathbb{Z}/2\mathbb{Z}$ with generator $\tau : g \mapsto g^2$.
  - Case 1a: $H \cong \mathbb{Z}/4\mathbb{Z} = \langle h \rangle$. $\operatorname{Hom}(H, \operatorname{Aut} N) = \operatorname{Hom}(\mathbb{Z}/4\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) = \{\varphi_o, \varphi_1\}$ where $\varphi_0(h) = 1$ and $\varphi_1(h) = \tau$.

    Either $G \cong \mathbb{Z}/3\mathbb{Z} \rtimes_{\varphi_0} \mathbb{Z}/4\mathbb{Z} \cong \mathbb{Z}/12\mathbb{Z}$, or
    $G \cong \mathbb{Z}/3\mathbb{Z} \rtimes_{\varphi_1} \mathbb{Z}/4\mathbb{Z} \cong G_{12}$ (new to us).
  - Case 1b: If $H = V_4 = \langle h_1, h_2 \rangle$. Then $\operatorname{Hom}(H, \operatorname{Aut} N) = \operatorname{Hom}(V_4, \mathbb{Z}/2\mathbb{Z}) = \{\psi_0, \psi_1, \psi_2, \psi_3\}$ where
    $\psi_0(h_1) = \psi_0(h_2) = 1$, $\psi_1(h_1) = \tau, \psi_1(h_2) = 1$, $\psi_2(h_1) = 1, \psi_2(h_2) = \tau$, and $\psi_3(h_1) = \tau, \psi_3(h_3) = \tau$.

    Then either $G \cong \mathbb{Z}/3\mathbb{Z} \rtimes_{\psi_o} V_4 = \mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ or
    $G \cong \mathbb{Z}/3\mathbb{Z} \rtimes_{\psi_1} V_4 \implies G = \langle h_1, h_2, g \rangle, h_1^2 = h_2^2 = g^3 = e, h_1 g_1 h_1^{-1} = g^2, h_2 g h_2^{-1} = g \implies |h_2 g| = 6 \implies G = D_6$.

    Check: ψ_2, ψ_3 give the same group.
- Case 2: G has normal 2-Sylow $M \trianglelefteq G$. If $Q = \langle g \rangle$ is the 3-Sylow, Q is complement to M.
  - Case 2a: If $M \cong \mathbb{Z}/4\mathbb{Z}$, then $\operatorname{Aut} m \cong \mathbb{Z}/2\mathbb{Z}$ and $\operatorname{Hom}(\mathbb{Z}/3\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) = \{e\} \implies G \cong \mathbb{Z}/12\mathbb{Z}$.
  - Case 2b: If $M \cong V_4 \cong \langle h_1, h_2 \rangle$, Then $\operatorname{Aut} M \cong S_3 = \langle \omega, \tau \rangle$, where $\omega : h_1 \mapsto h_2 \mapsto h_1 h_2$ and $\tau : h_1 \mapsto h_2 \mapsto h_1$.

    $\operatorname{Hom}(\mathbb{Z}/3\mathbb{Z}, S_3) = \{\psi_0, \psi_1, \psi_2\}$ where $\psi_0 : g \mapsto \mathrm{id} \implies G \cong \mathbb{Z}/6\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$
    $\psi_1 : g \mapsto \omega \implies G = \langle h_1, h_2, g \rangle, h_1^2 = h_2^2 = g^3 = e, g h_1 g^{-1} = h_2, g h^2 g^{-1} = h_1 h_2 \implies G \cong A_4$ (Check: ψ_2 gives A_4 also)

  Conclude: 3 isomorphism classes of nonabelian groups of order 12: G_{12}, D_6, A_4.

Example 12. Let $N = D \cap SL_2(K) = \left\{ \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} \right\} \leq SL_2(K)$,

and let $H = \left\langle \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right\rangle, |H| = 4$.

$G := NH, N \trianglelefteq G$ but N has no complement in G (since $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}^2 = -\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \in N$. (complement would have to be order 2.)

Example 13. $Q_8 = \{\pm 1, \pm I, \pm J, \pm K\}$. Letting $N = \langle I \rangle, |N| = 4$. So $N \trianglelefteq Q_8$.

$|Q_8/N| = 2$. If there were a complement, it would have to be order 2. So there is no complement to N in Q_8, since the only element of order 2 is $-1 \in N$ and this won't help generate the whole group.

1.7.2 Schur–Zassenhaus Theorem (reduction to the case of an abelian normal subgroup)

Theorem 14 (Schur–Zassenhaus)**.** *Let G be a finite group, $N \trianglelefteq G$ and $|N|$, $[G : N]$ be coprime. Then N has a complement in G.*

Proof. In this section, we only show that if G, N is a counterexample to the theorem with $|G|$ minimal, then N must be abelian.

Let p be prime, $p \mid |N|$, and $P \in \mathrm{Syl}_p(N)$. Then by the Frattini argument, $G = N \cdot N_G(P)$.

Step 1: Show $P \trianglelefteq N$.
Define $N_1 := N_G(P) \cap N, G_1 := N_G(P)$. Then $N_1 \trianglelefteq G_1$ and

$$G_1/N_1 = N_G(P)/(N_G(P) \cap N) \cong \frac{N \cdot N_G(P)}{N} = G/N.$$

Then $\gcd(\frac{|N_1|}{|G_1/N_1|}) = 1$ ($|N_1| \mid |N|$ and $|G_1/N_1| = |G/N|$).

Suppose $G_1 \neq G$. Then N_1 has a complement H_1 in G_1 by hypothesis (by minimality of G). So $|H_1| = [G_1 : N_1] = [G : N]$ and $H_1 \cap N \leq H_1 \cap G_1 \cap N = H_1 \cap N_1 = \{e\}$ so H_1 is complement to N in $G \rightarrow\leftarrow$. So $G_1 = G$, meaning $G = N_G(P)$ hence $P \trianglelefteq G \implies P \trianglelefteq N$. So we now know that N is nilpotent.

Step 2: Show N is abelian. Since N is nilpotent, the lower central series is a central series, so we must have $N' \leq N$. Let $G_2 := G/N'$, and let $N_1 := N/N'$. Now cine $G/N \cong \frac{G/N'}{N/N'}$, we have $[G_2 : N_2] = [G : N]$ so $|N_2|, [G_2 : N_2]$ are coprime.

If $N' \neq \{e\}$, then N_2 has a complement H_2 in G_2 (by minimality of G). Let $H \leq G$ such that $H/N' = H_2$ (note that $H \neq G$). $H_2N_2 = G_2 \implies HNN' = H \implies HN = G$. Bu t$H \cap N \neq \{e\}$ since $N' \leq H$, so H is not a complement.

$[H : N'] = |H_2| = [G_2 : N_2] = [G : N] \implies |N'|, [H : N']$ are coprime, so N' has a complement K in H, by minimality. That is, $KN' = H, K \cap N' = \{e\}$ so $KN = KN'N = HN = G$. So $K \cdot |N| = [H : N'] \cdot |N| = [G : N] \cdot |N| = |G|$. So K is complement to N in G, hence N is abelian ($N' = \{e\}$). □

1.7.3 Group Extensions and Factor Sets

Definition 15. A **short exact sequence** of groups is a diagram of the form

$$1 \longrightarrow N \xrightarrow{\iota} G \xrightarrow{\pi} H \longrightarrow 1$$

where ι is an injective homomorphism, π is a surjective homomorphism, N, G, H are groups, and $\ker(\pi) = \mathrm{im}(\iota)$.

Definition 16. A **group extension of H by N** is a short exact sequence

$$1 \to N \to G \to H \to 1$$

Definition 17. Two group extensions of H by N

$$1 \to N \xrightarrow{\iota_1} G_1 \xrightarrow{\pi_1} H \to 1, \quad 1 \to N \xrightarrow{\iota_2} G_2 \xrightarrow{\pi_2} H \to 1$$

are **equivalent** if there exists a homomorphism $\psi : G_1 \to G_2$ making the following diagram commute:

$$\begin{array}{ccc} N & \xrightarrow{\iota_1} & G_1 \\ {\scriptstyle \iota_2}\downarrow & \swarrow_{\psi} & \downarrow{\scriptstyle \pi_1} \\ G_2 & \xrightarrow[\pi_2]{} & H \end{array}$$

Note 18.
1. If $\varphi : G_1 \to G_2$ is as in the above definition, then φ is an isomorphism. Using this, one can show this is an equivalence relation on the set of extensions of H by N.
2. Some books call this an extension of N by H.
3. Notation: $\mathrm{Ext}(H, N) = \{\text{extensions of } H \text{ by } N\}/\sim$.

Definition 19. We say a group extension

$$1 \to N \to G \xrightarrow{\pi} H \to 1$$

splits if there exists a homomorphism $s : H \to G$ such that $\pi \circ s = \mathrm{id}_H$. Such a map s is called a **section to** π or a **splitting**.

Proposition 20. *Let $1 \to N \to G \to H \to 1$ be a group extension and $s : H \to G$ a homomorphism. Then the following are equivalent:*

1. *The extension splits via s*
2. *$s(H)$ is a complement to N in G*

Lemma 21. *If $1 \to A \to G \xrightarrow{\pi} H \to 1$ is a group extension with A abelian, then H acts on A via $\varphi(h)(a) := \tilde{h}a\tilde{h}^{-1}$ (where $\tilde{h}$ is any element so that $\pi(\tilde{h}) = h$).*

Proof. G acts on A by conjugation. A acts on itself trivially by being abelian. So $G/A \cong H$ acts in the expected way. □

Note 22.
1. If A is not abelian this is false.
2. If $1 \to A \to G \to H \to 1$ is an extension and A is abelian, then the two properties in Proposition 1 are equivalent to: (*) The extension is equivalent to $1 \to A \to A \rtimes H \to H \to 1$ where the action of H on A is given in Lemma 2.

Example 23. Calculate all extensions of the form

$$1 \longrightarrow \mathbb{Z}/3\mathbb{Z} \longrightarrow G \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 1.$$

Say $\mathbb{Z}/3\mathbb{Z} = \langle g \rangle$ and $\mathbb{Z}/2\mathbb{Z} = \langle h \rangle$. We have $\mathrm{Aut}(\mathbb{Z}/3\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z} = \langle g \mapsto g^2 = g^{-1} \rangle$. Therefore possible actions are:

1. the trivial action. Get: $1 \to \mathbb{Z}/3\mathbb{Z} \to \mathbb{Z}/6\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 1$ (split extension, $\mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$)
2. the nontrivial action. Get: $1 \to \mathbb{Z}/3\mathbb{Z} \to S_3 \to \mathbb{Z}/2\mathbb{Z} \to 1$ (split extension, $\mathbb{Z}/3\mathbb{Z} \rtimes \mathbb{Z}/2\mathbb{Z}$)

Thus, since $\gcd(3, 2) = 1$, by Schur–Zassenhaus all extensions split and these two describe all splitting extensions.

From now on: H group, A abelian group.

Let $1 \to A \to G \to H \to 1$ be an extension, and let $s : H \to G$ be a set-theoretic section which is normalized ($s(e_H) = e_G$, $\pi \circ s = \mathrm{id}$). In general, s is not always a homomorphism Then define

$$f : H \times H \to A$$
$$(g, h) \mapsto s(g)s(h)s(gh)^{-1}$$

We note that $\pi(s(g)s(h)s(gh)^{-1}) = \pi s(g)\pi s(h)\pi s(gh)^{-1} = gh(gh)^{-1} = e \implies$
$s(g)s(h)s(gh)^{-1} \in \ker \pi = A$.

f is also normalized ($f(g, e) = e, f(e, h) = e \forall g, h \in H$.)

Proposition 24. *Let $1 \to A \to G \to H$ be a group extension, $s : H \to G$ a set-theoretic section, and associate to s a function $f : H \times H \to A$ as above. Then the extension is up to equivalence uniquely determined by f and the action of H on A.*

Proof. Let $G_f := A \times H$ (as a set). Then $\varphi_f : G_f \to G, (n, h) \mapsto n \cdot s(h)$ is a bijection.

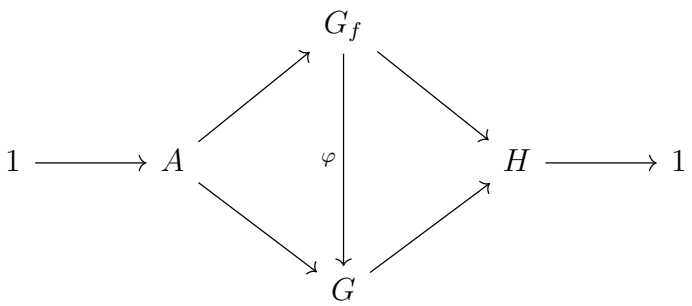
Define a group structure on G_f as follows:

$$(n_1, h_1)(n_2, h_2) = (\varphi(n_1)(h_1h_2)f(h_1, h_2),\ h_1h_2)$$

In G we have

$$\varphi(n_1)(h_1n_2)f(h_1, h_2)s(h_1h_2) \in A \cdot s(H)$$

so φ is a group homomorphism and in fact an isomorphism. Easy to check:



commutes. □

Proposition 25. *If $f : H \times A \to A, f(e,h) = e = f(g,e) \forall g,h \in H$, and H acts on A, then a product defined by*

$$(n_1, h_1)(n_2, h_2) = (\varphi(n_1)(h_1 h_2) f(h_1, h_2),\ h_1 h_2)$$

is associative if and only if

$$f(h_1, h_2) + f(h_1 h_2, h_3) = h_1 \cdot f(h_2, h_3) + f(h_1, h_2 h_3) \tag{1}$$

Proof. **Calculation** □

Corollary 26. *In the situation of Proposition 4, if f satisfies* (**??**)*, then the product defines a group structure on $A \times H$.*

Proof. Check: $((\varphi(h^{-1})(n)^{-1} f(h^{-1}, h)^{-1}, h^{-1})(n, h) = (e, e)$. □

Definition 27. If A, H are groups and A is abelian, we say a function $f : H \times H \to A$ satisfying (**??**) is called a **2-cocycle** of H with coefficients in A. (**??**) is called the **cocycle condition.**

If $f(e,h) = e = f(g,e) \forall g,h \in H$, we say f is **normalized**.

Let $s' : H \to G$ be a different set-theoretic normalized section of our extension $1 \to A \to G \to H \to 1$. If $h \in H, s'(h) = c(h)s(h)$ for some $c(h) \in A, c(e) = e$. (this is because $\pi(s'(h)s(h)^{-1} = \pi(s'(h))\pi(s(h))^{-1} = hh^{-1} = e$ so if $c(h) := s'(h)s(h)^{-1}, c(h) \in \ker \pi = A$).

We calculate f' associated to s':

$$\begin{aligned} f'(h_1, h_2) = s'(h_1)s'(h_2)s'(h_1 h_2)^{-1} = \\ c(h_1)s(h_1)c(h_2)s(h_2)\left(c(h_1, h_2)s(h_1, h_2)\right)^{-1} = \\ c(h_1)s(h_1)c(h_2)s(h_1)^{-1}s(h_2)s(h_1 h_2)^{-1}c(h_1 h_2)^{-1} = \\ c(h_1) \cdot \left(s(h_1)c(h_2)s(h_1)^{-1}\right) \cdot f(h_1, h)_2) \cdot c(h_1 h_2)^{-1} \in A \end{aligned}$$

(recall the action $\varphi(h_1)(c(h_2))$ from Lemma 2)

$$= c(h_1)s(h_1)c(h_2)s(h_1)^{-1}c(h_1 h_2)^{-1} f(h_1 h_2),$$

because A is abelian.

Note 28. $c : H \to A$ defines a 2-cocycle $\delta_c : H \times H \to A$ as follows:

$$\delta_c(h_1, h_2) = c(h_1)s(h_1)c(h_2)s(h_1)^{-1}c(h_1 h_2)^{-1}$$

This works even if $c(e) \neq e$.

Definition 29. A 2-cocycle $f : H \times H \to A$ is called a **2-coboundary** (of H with coefficients in A) is there exists a function $c : H \to A$ so that $\delta_c = f$.

Denote by $Z^2(H, A)$ the group of 2-cocycles of H with coefficients in A and by $B^2(H, A)$ the group of 2-coboundaries of H with coefficients in A.

These are abelian groups, and we define the **second cohomology group of H with coefficients in A:**

$$H^2(H, A) = Z^2(H, A)/B^2(H, A).$$

Note 30. This depends on $\varphi : H \to \operatorname{Aut} A$.

Lemma 31. *Every class in $H^2(H, A)$ has a normalized representative.*

Proof. Let $[f] \in H^2(H, A)$ of $f : H \times H \to A$ be a 2-cocyle. Define $c(h) = h \cdot f(e, e)$. Check: $\tilde{f} := f \circ \delta_c^{-1}$ is normalized in $[f]$. □

Theorem 32 (Schreier)**.** *Let H be a group, A an abelian group with fixed action $\varphi : H \to \operatorname{Aut}(A)$. Then there is a one-to-one correspondence*

$$H^2_\varphi(H, A) \leftrightarrow \{\textit{extensions } 1 \to A \to G \to H \to 1 \textit{ which induce } \varphi\}/\sim$$

$$1 \leftrightarrow \textit{split extensions}$$

Note 33. If we can show $H^2(H, A) = 1$ whenever $|A|$ and $|H|$ are coprime, we'll know that $|A|$, $|H|$ coprime $\implies$ every extension splits.

Example 34. Let $C_6 = \mathbb{Z}/6\mathbb{Z} = \langle g \rangle$ be considered as an extension of $C_2 = \mathbb{Z}/2\mathbb{Z} = \{e, h\} (= H)$ by $C_3 = \langle g^2 \rangle = \mathbb{Z}/3\mathbb{Z} (= A)$.

C_6 is abelian $\implies$ φ is trivial.

Set theoretic section $s : C_2 \to C_6, s(e) = e, s(h) = g$.

$f(h, h) = s(h)s(h)s(h^2)^{-1} = g^2 \cdot e = g^2 \in C_3$ (determines f, since f is normalized).

Let $c : C_2 \to C_3, c(h) = g^4$. Check: $\delta_c(h, h) = c(h)^2 = (g^4)^2 = g^2 = f(h, h) \implies [f] = 1$ in $H^2(C_2, C_3)$.

1.7.4 Group Cohomology

As before, let H, A be groups, A abelian, fix $\varphi : H \to \operatorname{Aut} A$.

Let $F(H^n, A)$ be the abelian group of functions $f : H^n \to A$ under multiplication ($H^0 := \{1\}$).

If $n \in \mathbb{N} \cup \{0\}$, $d^n := d^n_\varphi : F(H^n, A) \to F(H^{n+1}, A)$,

$$f \mapsto \varphi(x_0)(f(x_1, \ldots, x_n)) \cdot \Big(\prod_{i=1}^{n} f(x_0, \ldots, x_{i-1}x_i, \ldots, x_n)^{(-1)^i}\Big) \cdot f(x_0, \ldots, x_{n-1})^{(-1)^{n+1}} ((d^0_\varphi)(f))(x) := \varphi(x)(f(e$$

Lemma 35. *1. d^n is a group homomorphism for all n*

2. $d^{n+1} \circ d^n = 1$ for all n, i.e. $\mathrm{Im}(d^n) \subseteq \mathrm{Ker}(d^{n+1})$.

Proof. **Exercise.** □

Definition 36. Let H, A, φ be as before. Define

$$Z^n(H,A) = \ker d^n(\text{n-cocycles}), \quad B^n(H,A) = \mathrm{im}\, d^{n-1}(\text{n-coboundaries})$$

$$H^n(H,A) = Z^n(H,A)/B^n(H,A) \quad (\text{n-th cohomology group}).$$

For $f \in Z^n(H,A)$, let $[f]$ denote the class in $H^n(H,A)$.

Note 37. 1. $F(H^0, A) \cong A$ via $e \mapsto f(e)$

$d^0_\varphi(f)(h) = \varphi(h)(f(e)) \cdot f(e)^{-1}$, hence
$Z^0(H,A) = H^0(H,A) = A^H = \{\text{elements fixed under } H\}$

2. $f : H \to A$ is in $Z^1(H,A) \iff f(h_1h_2) = \varphi(h_1)(f(h_1)) \cdot f(h_1)$ for all $h_1, h_2 \in H$.
Such f's are called **crossed homomorphisms**

$f : H \to A$ is in $B^1(H,A) \iff \exists a \in A$ such that $f(h) = (\varphi(h)(a))a^{-1}$.
Such f's are called **principal crossed homomorphisms**

If the action is trivial, $H^1(H,A) = Z^1(H,A) = \mathrm{Hom}(H,A)$

3. For $n = 2$, both definitions of $H^2(H,A)$ coincide

Example 38. $\mathbb{Z}/2\mathbb{Z}$ acts on $\mathbb{C}^\times \setminus \{0\}$ via complex conjugation. Calculate $Z^1(\mathbb{Z}/2\mathbb{Z}, \mathbb{C}^\times)$.

$d^1 : F^1(\mathbb{Z}/2\mathbb{Z}, \mathbb{C}^\times) \to F^2(\mathbb{Z}/2\mathbb{Z}, \mathbb{C}^\times)$
$f \mapsto \varphi(x_0)(f(x_1)) \cdot f(x_0) \cdot f(x_0x_1)^{-1}$

$f : \mathbb{Z}/2\mathbb{Z} = \langle x \rangle \to \mathbb{C}^\times$
$f(1) = 1, f(x) = x \in \mathbb{C}^\times$

$(x, x) \mapsto \bar{z} \cdot z = 1 = z \cdot \bar{z} = |z|^2$

$Z^1 = \{f : \mathbb{Z}/2\mathbb{Z} \to \mathbb{C}^\times \mid f(x) = z \text{ where } |z| = 1\} \cong S^1$.

Lemma 39. *$|A| < \infty \implies f^{|A|} = 1$ for all $f \in F^n(H,A)$ (and all n). Hence every cohomology class has order dividing $|A|$.*

Proof. **Exercise.** □

Proposition 40. *If $|H| < \infty$, $f^{|H|} = 1$ for all $f \in Z^n(H,A)$, for all n.*

Proof. $f \in Z^n(H, A), x_0, \ldots, z_n \in H$.

$F \in \ker d^n \implies$
$f(x_0, \ldots, x_{n-1})^{(-1)^n} = \varphi(x_0)(f(x_1, \ldots, x_n)) \cdot \prod_{i=1}^{n} f(x_0, \ldots, x_{i-1}x_i, \ldots, x_n)^{(-1)^i}$.

Fix $x_0, \ldots, x_{n-1} \in H$, vary $x = x_n \in H$. Multiply: $f(x_0, \ldots, x_{n-1})^{(-1)\cdot|H|\cdot n} =$
$\varphi(x_0)(\prod_{x\in H} f(x_1, \ldots, x_{n-1}, x))(\prod_{i=1}^{n} \prod_{x\in H} f(x_0, \ldots, x_{i-1}x_i, \ldots, x_n)^{(-1)^i}$.

Define $g : H^{n-1} \to A$ via $g(x_1, \ldots, x_{n-1}) = (x_1, \ldots, x_{n-1}, x)$.

$f^{|H|} = d^{n-1}(g^{(-1)^n}) \implies [f] = 1$ by definition. □

Corollary 41. *Suppose $|H|$ and $|A|$ are coprime. Then $H^n(H, A) = 1$ for all $n \in \mathbb{N}$ (and all actions φ).*

Proof. $1 = |H| \cdot a + |A| \cdot b$ for some $a, b \in \mathbb{Z}$.

$[f] = [f]^1 = [f]^{|H|\cdot a+|A|\cdot b} = [f]^{|H|\cdot a}[f]^{|A|\cdot b} = 1 \cdot 1 = 1$ by Lemma 2 and Prop 3.

(Proves Schur Zassenhaus) □

Question: How unique are complements when they exist?

Proposition 42. *Let $A \trianglelefteq G$ and assume A has a complement H in G. Then there exists a bijection*

$$H^1(H, A) \leftrightarrow \{\textit{conjugacy classes of complements of } A \textit{ in } G\}.$$

Proof. If K is a complement, write $h \in H$ as $h = f(h) \cdot k$ for some $k \in K$. (where $f(h) \in A$; this defines $f(h)$).

$$h_1h_2 = h_1f(h_2)k_2, \text{ where } h_2 = f(h_2)k_2,$$

$$h_1 = f(h_1)k_1 = h_1f(h_2)h_1^{-1}h_1k_2 = h_1f(h_2)h_1^{-1}(f(h_1)k_1)k_2 = f(h_1h_2) \cdot k_{12} \in AK$$

(because $h_1h_2 = f(h_1h_2)k_{12}$ since $h_1h_2 \in H$).

Hence $k_{12} = k_1k_2$ by uniqueness and $f(h_1h_2) = h_1f(h_2)h_1^{-1}f(h_1) = (\varphi(h_1)(f(h_2)) \cdot h_1 \implies$
$f \in Z^1(H, A)$.

On the other hand, if $f \in Z^1(H, A)$ is normalized, define $K := \{f(h)^{-1}h \mid h \in H\}$

$(f(h_1)^{-1}h_1)(f(h_2)^{-1}h_2) = \varphi(f(h_1)^{-1}h_1)(f(h_2)^{-1}h_1h_2 = f(h_1h_2)^{-1}h_1h_2 \in K$ (since $f \in Z^1(H, A)$).
$f(h)^{-1}h = (f(h^{-1}))^{-1}h^{-1} \in K$. (So we can conclude that $K \leq G$).

If $f(h)^{-1}h \in A$, then $h \in H \cap A = \{e\}$ (since H was a complement), thus $K \cap A = \{e\}$.

Now $g \in G \implies g = n \cdot h$ $(n \in A, h \in H) = nf(h)f(h)^{-1}h \in AK$. Conclude K is a complement in G.

One checks: conjugate elements correspond to equivalent cocycles (actually, A-conjugate complements = G-conjugate complements). □

Corollary 43. *Suppose* $\gcd(|A|, [G : A]) = 1$. *Then any 2 complements of* A *in* G *are conjugate in* G.

Proof. $H^1(H, A) = 1$ for H some complement. □

Theorem 44 (Schur–Zassenhaus, part II)**.** *Let* G *be a finite group,* $N \trianglelefteq G$ *with* $|N|$ *and* $|G/N|$ *coprime. Assume either* N *or* G/N *is solvable. Then all complements of* N *in* G *are conjugate in* G.

Note 45. Note that we don't really need the hypothesis that either N or G/N are solvable because this is always satisfied by Feit-Thompson (odd order groups are solvable).

Proof. We know that if N is abelian, this holds. Reduce to this case by induction on $|G|$. Let H_1, H_2 be complements to N, $\{e\} \neq M \leq N$ minimal normal subgroup of G. Then H_1M/M and H_2M/M are complements of N/M in G/M and $\gcd(|H_1M/M|, |N/M|) = 1$ and $\gcd(|H_2M/M|, |N/M|) = 1$.

Note that $|G/M| < |G|$. By induction, H_1M/M and H_2M/M are conjugate in G/M. Hence $H_1M \subsetneq H_1N = G \implies H_1$ and gH_2g^{-1} are conjugate in G.

Wlog assume N is a minimal normal subgroup of G.

Case 1: N is solvable $\implies$ N abelian (solvable minimal normal subgroup $\implies$ abelian).

Case 2: G/N is solvable. Take composition series $G/N \trianglerighteq N_1/N \trianglerighteq \cdots \trianglerighteq N_r/N = \{e\}$ where quotients $(N_i/N)/(N_{i+1}/N)$ are cyclic of prime order (by being simple abelian).

Let $A := N_1 \implies A \trianglelefteq G$ and $N \trianglelefteq A$. Then $|G/A| = \frac{|G/N|}{|A/N|} = p$ prime.

$(H_1 \cap A) \cap N - \{e\}$

$(H_1 \cap A) \cdot N = H_1N \cap AN = H_1N \cap A = G \cap A = A \implies H_1 \cap A$ is complement to N in A. We can do the analagous construction with H_2 so $H_2 \cap A$ is complement to N in A.

By induction, $H_1 \cap A$ and $H_2 \cap A$ are conjugate in A ($|A| < |G|$). Wlog assume $H_1 \cap A = H_2 \cap A =: D$.

$A \trianglelefteq G \implies D \trianglelefteq H_1, D \trianglelefteq H_2$.

$H_1/D \cong H_1A/A = G/A$ and $H_2/D \cong H_2A/A = G/A$. And $|G/A| = p$ is prime, meaning $p \mid |H_1|$ and $p \mid |H_2|$.

Hence H_1, H_2 have nontrivial p-Sylow subgroups P_1, P_2.

$|P_1D| = \frac{|P_1||D|}{|P_1 \cap D|} \geq \frac{p^r \cdot p^{r-1} \cdot m}{p^{r-1}} = p^r m = |H_1|$ (if $|H_1| = p^r m$ and $p \nmid m$).

But $P_1D \leq H_1$ so $P_1D = H_1$. Similarly, $P_2D = H_2$. Since $p \mid |H_1|, p \nmid |N|$. Similarly for H_2. Thus P_1, P_2 are p-Sylows of G.

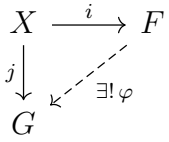
Then P_1 and P_2 are p-Sylows of $N_G(D) \implies \exists f, g \in N_G(D)$ such that $P_2 = gP_1g^{-1}$. So $H_2 = DP_2 = D(gP_1g^{-1}) = gH_1g^{-1}$.

□

1.8 Free Groups and Presentations

1.8.1 Definition and Existence

Definition 1. Let X be a set. A **free group** F on X is a group F equipped with a map $i : X \to F$ with the following universal mapping property (UMP):
For any group G and any map $j : X \to G$, there exists a unique group homomorphism $\varphi : F \to G$ such that $\varphi \circ i = j$.



Theorem 2. *For every set X, there exists a free group on X, unique up to isomorphism. It is generated as a group by $i(X)$.*

Proof. Uniqueness: **Exercise. Use UMP.**

Existence: Let $\mathcal{W} = \{x_1^{e_1} \dots x_n^{e_n} \mid n \geq 0, e_i \in \{\pm 1\}, x_i \in X\}$ be the set of all **words** in X.

Define an equivalence relation on $\mathcal{W}$ by declaring that two words u, v are equivalent if and only if u can be obtained from v by inserting and deleting strings of the form xx^{-1} or $x^{-1}x$ for $x \in X$. Check: this defines an equivalence relation.

Let $[u]$ denote the equivalence class of a word $u \in \mathcal{W}$. Define $F_X := W/\sim$ with product $[u] \cdot [v] = [uv]$ (uv is concatenation of u and v). Then F_X is a group with identity the empty word [].

Let $i : X \to F_X, x \mapsto [x]$. Let G be any group with $j : X \to G$. Define $g_x := j(x)$ for all $x \in X$.

Define $f : \mathcal{W} \to G$ by $x_1^{e_1} \dots x_n^{e_n} \mapsto g_{x_1}^{e_1} \dots g_{x_n}^{e_n}$. If $u \sim v$, then $f(u) = f(v)$ because inverses will cancel (G is a group). So f induces a map $\varphi : F_X \to G$. φ is a homomorphism:
$\varphi([u])\varphi([w]) = f(u)f(w) = f(uw) = \varphi([uw])$.

The diagram commutes: Let $x \in X$. Then $\varphi i(x) = \varphi([x]) = f(x) = g_x = j(x)$. Let

$\gamma : F_X \to G$ be another homomorphism such that $\gamma i = j$. Then $\gamma i = \varphi i \implies \gamma$ and φ agree on the image of i. F_X is generated by $i(X)$ by definition, hence $\varphi = \gamma$. □

Note 3. 1. $i : X \to F$ is injective (exercise)

2. Every equivalence class in F_X contains a unique **reduced word**, i.e. a word that does not contain subwords of the form xx^{-1} or $x^{-1}x$ for $x \in X$.

   Let $R :=$ set of reduced words. The uniqueness assertion can be seen as follows: $x \in X, e \in \{\pm 1\} \implies$:

   $$|x^e| : R \to R, x_1^{e_1} \dots x_n^{e_n} \mapsto \begin{cases} x_2^{e_2} \dots x_n^{e_n} & x^e = x_1^{-e_1} \\ x^e x_2^{e_2} \dots x_n^{e_n} & \text{otherwise} \end{cases}$$

   Then $|x^e|$ is a permutation of R with inverse $|x^{-e}|$.

   By UMP there exists a unique homomorphism $\varphi : F \to S(R)$ such that $\varphi i(x) = |x|$. Let v, w be reduced and $[v] = [w]$. WTS: $v = w$.

   $[v] = [w] \implies \varphi([v]) = \varphi([w])$.

   $v = x_1^{e_1} \dots x_n^{e_n} \implies [v] = [x_1^{e_1}] \dots [x_n^{e_n}] \implies \varphi([v]) = |x_1^{e_1}| \dots |x_n^{e_n}|$. Then apply to the empty word []:

   $\varphi([v])([]) = (|x_1^{e_1}| \dots |x_n^{e_n}|)([]) = |x_1^{e_1}| \dots |x_{n-1}^{e_{n-1}}| x_n^{e_n} = \dots = x_1^{e_1} \dots x_n^{e_n} = v$, and $\varphi([w])([]) = w$. Conclude $v = w$. (This is called the Van der Waerden trick)

Example 4. Let $X = \{x\}$. If F_X were finite, the UMP would give us

$$\begin{array}{ccc} x & & X \xrightarrow{\ i\ } F_X \\ \big\downarrow & & \big\downarrow j \quad \swarrow \varphi \\ z & & \mathbb{Z} \end{array}$$

which is a contradiction since elements in $\mathbb{Z}$ have infinite additive order. So F_X is infinite and, by definition, cyclic. Hence $F_X = \{x^k \mid x \in \mathbb{Z}\} \cong \mathbb{Z}$

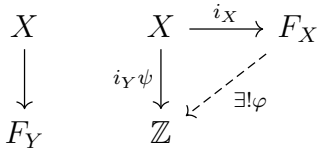
We note that the only examples for abelian F_X are $X = \{x\}$ or $X = \varnothing$.

1.8.2 The rank of a free group

Theorem 5. *Let X, Y be sets and F_X, F_Y the corresponding free groups. Then*

$$F_X \cong F_Y \iff |X| = |Y|.$$

Proof. ($\Longleftarrow$) : $|X| = |Y| \implies \exists$ bijection $\psi : X \to Y \implies$



In fact φ is an isomorphism.

($\implies$) : (Sketch) Suppose $F_X \cong F_Y$. Then $\exists 2^{|X|}$ maps $X \to \mathbb{Z}/2\mathbb{Z} \implies \exists 2^{|X|}$ homomorphisms $F_X \to \mathbb{Z}/2\mathbb{Z}$ (by UMP). Similarly for Y and F_Y.

But $F_Y \cong F_X$ so $\mathrm{Hom}(F_X, \mathbb{Z}/2\mathbb{Z}) \cong \mathrm{Hom}(F_Y, \mathbb{Z}/2\mathbb{Z}) \implies 2^{|X|} = 2^{|Y|} \implies$ if X or Y is finite, $|X| = |Y|$. □

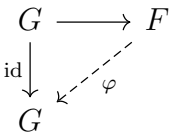
Definition 6. A group F is **free** if $F \cong F_X$ for some set X. The **rank** of F is $\mathrm{rk}(F) := |X|$. (Theorem 1 implies that rank is well-defined)

Example 7. $G = \left\langle \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \right\rangle \leq SL_2(\mathbb{C})$ is a free group of rank 2.

1.8.3 Presentations of groups

Theorem 8. *Every group is a quotient of a free group.*

Proof. Consider a group G as a set. Let F_G be the free group on the set G. Then we have a map $\mathrm{id} : G \to G$ mapping G (set) to G (group). Apply UMP:



Then $G \cong F/\ker\varphi$. □

Definition 9. Let X be a set and I a family of words on X. Let F be the free group on X, and N the normal subgroup generated by I. Then the group $G = F/N$ is said to have **presentation**

$$G = \langle X \mid I \rangle.$$

Here X are **generators** and I are **relations**.

Example 10. 1. $G = \langle x, y \mid x^2 = y^3, xyx^{-1} = y^2 \rangle$ i.e. $G = \{e, x, y, xy, xy^2, y^2\}$. $G \cong S_3$.

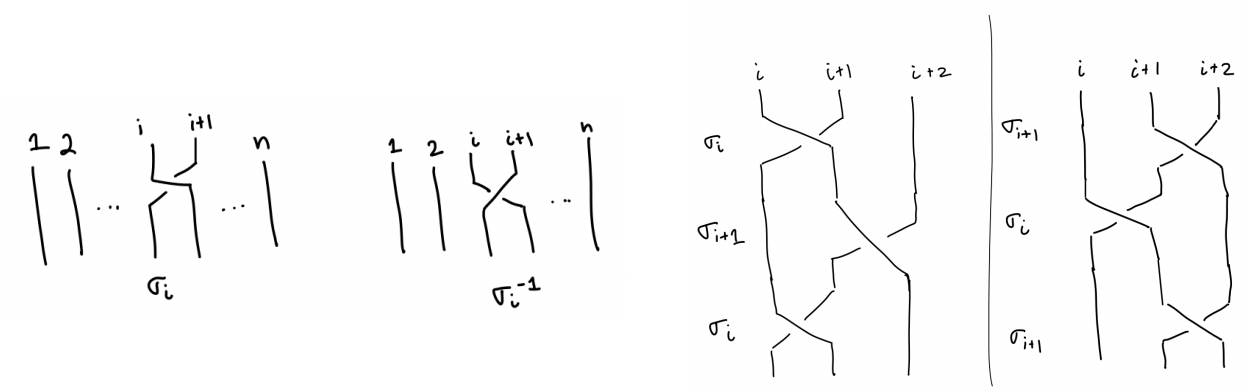
Define $j : \{x, y\} \to S_3$ so that $j(x) = (12)$ and $j(y) = (123)$. Then there exists a unique $\varphi : F_{\{x,y\}} \to S_3$, surjective because the image contains the generators of S_3. In fact, $x^2, y^3.xyxy \in \ker\varphi$. So

$$G \cong F_{\{x,y\}}/\langle x^2, y^3, xyxy \rangle \twoheadrightarrow F_{\{x,y\}}/\ker\varphi \twoheadrightarrow S_3$$

but $G \leq 6 \implies G \cong S_3$.

2. Braid group on n strings:

$$B_n = \langle \sigma_1, \ldots, \sigma_{n-1} \mid \sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1} \text{ for all } 1 \leq i \leq n-2, \sigma_i\sigma_j = \sigma_j\sigma_i \text{ if } |i-j| > 1 \rangle.$$



1.8.4 Properties of Free Groups

Definition 11. A group P is **projective** if for any surjective homomorphism $\psi : G \twoheadrightarrow H$ and any homomorphism $\pi : P \to H$, there exists a homomorphism $\varphi : P \to G$ such that $\psi \circ \varphi = \pi$:

$$\begin{array}{ccc} & & P \\ & \overset{\varphi}{\swarrow} & \downarrow \pi \\ G & \xrightarrow{\psi} & H \end{array}$$

Note 12. Free groups are projective **(exercise; use UMP)**

Definition 13. Let $S \leq G$. A surjective homomorphism $p : G \twoheadrightarrow S$ is a **retraction** if $p^2 = p$ (i.e. $p|_S = \mathrm{id}_S$). In this case, S is a **retract** of G.

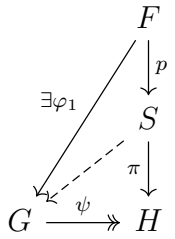
Proposition 14. *A group is projective if and only if it is a retract of a free group.*

Proof. ($\Rightarrow$) Let P be projective. Set $H = P, \pi = \mathrm{id}$. Let G be free with $\varphi : G \to P$. We get:

$$\begin{array}{ccc} & & P \\ & \overset{\exists!\gamma}{\swarrow} & \downarrow \mathrm{id} \\ G & \twoheadrightarrow & G \end{array}$$

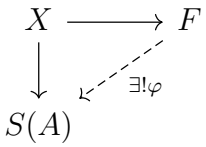
with $\gamma\varphi = \mathrm{id}$. Let $R = \varphi(P) \leq G, P = \varphi\gamma : G \to P$. Then $p^2 = \varphi\gamma\varphi\gamma = \varphi\,\mathrm{id}\,\gamma \implies R$ is a retract of G, isomorphic to P.

($\Leftarrow$) Let F be free and $S \leq F$ a retract of F via p. Then $\exists\varphi_1 : F \to G$ so that $\psi\varphi_1 = \pi p$. Define $\varphi := \varphi_1|_S$. Then $\psi\varphi = \psi\varphi_1|_S = \pi p|_S = \pi$.



□

Proposition 15. *Let X be a set, F a free group with $i : X \to F$. Then F is free on $F \iff F$ has the following UMP:*
For any set A and any map $X \to S(A), \exists$ unique homomorphism $\varphi : F \to S(A)$ such that



commutes. (That is, we only have to check UMP on symmetric groups)

Next aim is to prove: **Schreier–Nielsen:** Subgroups of free groups are free.

Definition 16. Let F be free on X and $H \leq F$. A **Schreier transversal** for H in F is a set T of reduced coset representatives of H in F which is closed under taking suffixes. That is, if $x^{\pm 1}u \in T$ is reduced for $x \in X$ then $u \in T$.

Note 17. Schreier transversals should always contain the empty word.

Example 18. For $F_2 = \langle a, b\rangle$ and $H = \langle a^2, b^2, aba^{-1}b^{-1}\rangle$, $F_2/H \cong V_4$. Candidates for Schreier transversal:

- $\{1, a, b, ab^{-1}\}$ doesn't work
- $\{1, a, b, ab\}$ works
- $\{1, a, b, ba^{-1}\}$ works

Zorn's Lemma: Let S be a partially ordered set, $S \neq \varnothing$. Assume that every totally ordered subset (chain) of S has an upper bound in S. Then S has a maximal element.

- Totally ordered: all elements are comparable
- Upper bound: $T \subseteq S$ has an upper bound if there exists $s \in S$ such that $t \leq s$ for all $s \in S$ (note that in general, we don't require $s \in T$)

Example 19. Every vector space has a basis (= maximal linearly independent generating set)

$S =$ set of linearly independent subsets of V over K.

$S \neq \varnothing$ since $0 \in S$.

Take any chain (where $\leq$ is inclusion) T_α. The upper bound is the union $T := \cup_\alpha T_\alpha$. The union is linearly independent: Say $\lambda_1 t_1 + \cdots + \lambda_n t_n = 0$ where $\lambda_i \in K$ and $t_i \in T$. Then there exists T_{α_0} containing all the $t_i \implies$ they're linearly independent $\implies \lambda_1 = \cdots = \lambda_n = 0$. So T is linearly independent.

Zorn's $\implies$ there exists a maximal element B in S maximal linearly independent subset of V. Claim: this generates V (if not there is some $v \in V \setminus \langle B \rangle$ such that $v \cup B$ is linearly independent).

Proposition 20. *Let F be a free group, $H \leq F$. Then there exists a Schreier transversal for H in F.*

Proof. F free $\implies$ F is free on some set X.

Let $S = \{$suffix-closed subsets of reduced words in $X \cup X^{-1}$ which intersect each coset of H in ≤ 1 element and contain $e\}$.

$S \neq \varnothing$ because $\{e\} \in S$. The union in a chain is the upper bound. So by Zorn, there is a maximal element $T \in S$. We claim T is a Schreier transversal. Need to show: T contains a representative for every coset of H in F.

Suppose not. Then there is a reduced word w of minimal length so that $wH \cap T = \varnothing$. $w \neq e$ by definition of S. So $w = xu$ (xu reduced form) where $x \in X \cup X^{-1}$. wH has a representative in T by minimality of w so there is some $t \in T$ so that $uH = tH$. Then $wH = xuH = xtH$.

If xt is reduced, $T \sqcup \{xt\} \in S \to\leftarrow$ (contradiction because T is maximal).

If xt is not reduced, the first letter of t must be x^{-1} : $xt = xx^{-1}t' \implies t' \in T$ (since elements in S are suffix-closed).
$wH = xtH = xx^{-1}t'H = t'H \to\leftarrow$ contradicts assumption that w does not have coset representative in T. □

Theorem 21 (Schreier–Nielsen)**.** *Subgroups of free groups are free.*

Proof. (Schreier-Nielson) Let F be free on $X, H \leq F$. WTS: H is free.

Let T be a Schreier transversal for H in F. Define

$$B := \{\overline{(xt)}^{-1} xt \mid x \in X, t \in T, \overline{(xt)}^{-1} xt \neq 1\} \subseteq H$$

Here for $g \in F, \overline{g} \in T$ is the representative of gH in T.

Claim: H is free on B. Use Proposition 2: we only have to check UMP for maps to symmetric groups. Let A be a set, $\sigma : B \to S(A)$. Define $\sigma(e) := \mathrm{id}_A$. Define an action of F on $(F/H) \times A$ by

$$x(wH, a) := (xwH, \sigma(\overline{xw}^{-1}xw)(a))$$

This is an action because F is free on X.

Note: $\{H\} \times A \subseteq (F/H) \times A$ is stable under the action of $H \leq F$.

Induces action of H on A : so we get a homomorphism $\varphi : H \to S(A)$. Need to show:

$$\begin{array}{ccc} B & \longrightarrow & H \\ {\scriptstyle\sigma}\downarrow & \swarrow_{\varphi} & \\ S(A) & & \end{array}$$

commutes and φ is unique (**exercise**).

We'll show $b(H, a) = (H, \sigma(b)(a))\forall b \in B$. Suppose $t \in T, a \in A$. Show: $t(H, a) = (tH, a)$ by induction on length of t.

$t = e$ ✓

Assyme $t = xu, x \in X$ ($t = x^{-1}u$ is analagous). Then $u \in T$ by Schreier property, by having shorter length.

$t(H, a) = xu(H, a) =$ (by induction) $x(uH, a) =$ (by definition) $(xuH, \sigma((\overline{xu})^{-1}xu)(a)) =$ (noting that $(\overline{xu})^{-1} = (xu)^{-1}$ and $xu \in T$) $(xuH, \sigma(e)(a)) = (xuH, a) = (tH, a)$.

Now take $b \in B, b = (\overline{xt})^{-1}xt, x \in X.t \in T, (\overline{xt})^{-1}xt \neq 1$. Want to show: $b(H, a) = (H, \sigma(b)(a))$.

We observe: $b(H, a) = (\overline{xt})^{-1}xt(H, a)$ and $(H, \sigma(b)(a)) = (H, \sigma((\overline{xt})^{-1}xt)(a))$.

But: $\overline{xt}b(H, a) = \overline{xt}(\overline{xt})^{-1}xt(H, a) = xt(H, a) = x(tH, a) = (xtH, \sigma((\overline{xt})^{-1}xt)(a)) = (xtH, \sigma(b)(a))$.

And $\overline{xt}(H, \sigma(b)(a)) = \overline{xt}(H, \sigma((\overline{xt})^{-1}xt)(a)) = ((\overline{xt})H, \sigma((\overline{xt})^{-1}xt)(a)) = (xH, \sigma(b)(a))$.

So we've proven $b(H, a) = (H, \sigma(b)(a))$. □

Note 22. From the proof: $|T| = [F : H]$. It is not hard to see that if $|X| = n$ and $|T| = [F : H] = m$, then $|B| = 1 + m(n - 1)$.

Corollary 23. *Projective groups are free.*

2 Rings

2.1 Rings and ideals

2.1.1 Rings and homomorphisms

Definition 1. A **ring** is a set R with two operations $+ : R \times R \to R$ and $\cdot : R \times R \to R$ such that:

1. $(R, +)$ is an abelian group with neutral element 0
2. $\cdot$ is associative
3. $\cdot$ is distributive over $+$:

$$x \cdot (y + z) = x \cdot y + x \cdot z, \quad (x + y) \cdot z = x \cdot z + y \cdot z.$$

R is **commutative** if $x \cdot y = y \cdot x$ for all $x, y \in R$.

Unless stated otherwise, assume R has an identity $1 \in R$ such that $1x = x1 = x$ for all $x \in R$.

We do not require $1 = 0$. The only ring with $1 = 0$ is the zero ring $\{0\}$.

Definition 2. A **ring homomorphism** $\varphi : R \to S$ is a map such that:

1. $\varphi : (R, +) \to (S, +)$ is a homomorphism of abelian groups: $\varphi(x + y) = \varphi(x) + \varphi(y)$
2. $\varphi(xy) = \varphi(x)\varphi(y)$
3. $\varphi(1_R) = 1_S$.

A bijective ring homomorphism is an **isomorphism**. (Notation: $R \cong S$)

Definition 3. A subset $S \subseteq R$ is a **subring** if it is a ring with restrictions of $\cdot, +$. (Notation: $S \leq R$)

Note 4. Compositions of ring homomorphisms are ring homomorphisms.

Definition 5. Let R be a ring. A subset $I \subseteq R$ that is an additive subgroup is called:

- a **left ideal** if $rI \subseteq I$ for all $r \in R$;
- a **right ideal** if $Ir \subseteq I$ for all $r \in R$;
- an **ideal** (two-sided) if it is both left and right ideal.

Notation: $I \trianglelefteq R$.

Proposition 6. *Let R be a ring and $I \trianglelefteq R$ an ideal. Then the quotient group R/I is a ring with operations*

$$(x + I)(y + I) = xy + I.$$

Moreover, the map $R \to R/I$, $x \mapsto x + I$, is a ring homomorphism.

Definition 7. $\pi : R \to R/I$ is called the **canonical homomorphism to the quotient.**

Proposition 8. *If $\varphi : R \to S$ is a ring homomorphism, then $\ker \varphi = \{r \in R : \varphi(r) = 1\}$ is an ideal of R, $\varphi(R) \leq S$, and φ induces an isomorphism*

$$R/\ker \varphi \cong \varphi(R).$$

Example 9. Find the problem:

Claim: $\mathbb{R}$ is countable.

Let $\mathcal{I}$ be the set of countable subsets of $\mathbb{R}$, partially ordered by inclusion.

$\mathcal{I}$ is nonempty (e.g. $\mathbb{Z} \in \mathcal{I}$).

Let $\{A_\alpha\}_{\alpha \in A}$ be a chain in $\mathcal{I}$.Let $A = \bigcup_\alpha A_\alpha$. Then A is countable. It is an upper bound, so by Zorn's Lemma, there exists a maximal element M.

Suppose $M \subsetneq \mathbb{R}$. Let $x \in \mathbb{R} \setminus M$.

Then $M' := M \cup \{x\} \implies M'$ is countable, but $M \subsetneq M'$ contradicting maximality.

Hence $M = \mathbb{R}$.

This is a misuse of Zorn's Lemma: the hypothesis requires all totally ordered subsets to have upper bounds, not just countable chains.

Note 10. If $I \trianglelefteq R$, the relation $x \sim y \iff x - y \in I$ is an equivalence relation on R that respects the ring operations. Such relations are called **congruence relations** (Notation: $x \equiv y$).

Proposition 11. *Let $I \trianglelefteq R$ and let $\pi : R \to R/I$ be the canonical homomorphism. Then there is a one-to-one correspondence between ideals of R containing I and ideals of R/I.*

Example 12.

1. $\mathbb{Z}$ is a commutative ring. For each $n \in \mathbb{N}$, $\mathbb{Z}/n\mathbb{Z}$ is a ring. It is the quotient of $\mathbb{Z}$ by the ideal $n\mathbb{Z}$.
2. $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are fields; every field is a ring
3. $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ (Gaussian integers) is a ring
4. $R[x]$, $R[[x]]$ (polynomials and power series) are rings
5. $\mathbb{M}_n(R)$ (matrix ring) is a ring

Definition 13. An ideal $I \trianglelefteq R$ is **maximal** if it is not properly contained in any other proper ideal $J \trianglelefteq R$.

Theorem 14 (Krull, 1929)**.** *Every nonzero ring has a maximal ideal.*

Proof. Let S be the set of proper ideals of R, partially ordered by inclusion. S is nonempty ($\{0\} \in S$).

Let $\{I_\alpha\}$ be a totally ordered subset (chain). Let $I = \bigcup_\alpha I_\alpha$. Then I is an ideal:

If $x \in I, r \in R$ then there is some α so that $x \in I_\alpha$ so $rx, xr \in I_\alpha$ hence $rx, xr \in I$.

If $x, y \in I$, then $x, y \in I_\alpha, I_\beta$ for some α, β respectively. Since we're in a totally-ordered subset, either $I_\alpha \subseteq I_\beta$ or $I_\beta \subseteq I_\alpha$. Without loss of generality, assume $I_\alpha \subseteq I_\beta$. Then $x + y \in I_\beta \implies x + y \in I$.

$1 \notin I$ (if it were, $1 \in I_\alpha$ for some $\alpha \implies I_\alpha = R \rightarrow\leftarrow$), so $I \subsetneq R \implies I$ is a proper ideal.

By Zorn's Lemma, there exists a maximal element i.e. a maximal ideal. □

Note 15. The theorem fails for rings without identity. It is also false for groups: $(\mathbb{Q}, +)$ has no maximal proper subgroup.

Corollary 16. *Every proper ideal in a nonzero ring is contained in a maximal ideal.*

Proof. Let $I \subsetneq R$, $R/I \neq 0$. Apply Theorem 4 and Proposition 3. □

2.1.2 Zero Divisors and Units

Definition 17. Let R be a ring.

- $0 \neq a \in R$ is a **left zero divisor** if $\exists b \neq 0$ such that $ab = 0$.
- $0 \neq b \in R$ is a **right zero divisor** if $\exists a \neq 0$ such that $ab = 0$.
- $x \in R$ is **nilpotent** if $x^n = 0$ for some $n \in \mathbb{N}$.

Example 18. In $\mathrm{Mat}_2(\mathbb{R})$,

$$A = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

are zero divisors (and nilpotent).

Definition 19. Let R be a ring. If $a, b \in R$ satisfy $ab = 1$, then a is **left invertible** and b is **right invertible**.

An element that is both left and right invertible is a **unit**.

Notation: $R^\times$ is the set of units in R. For $x \in R^\times$, the unique inverse of x is denoted x^{-1}.

Definition 20. A ring R is a **division ring** (or **skew field**) if every nonzero element is a unit (i.e. $R^\times = R \setminus \{0\}$.)

A commutative division ring is a **field**.

Note 21. Division rings have no zero divisors.

Example 22.
1. $\mathbb{Z}^\times = \{\pm 1\}$
2. $(\mathbb{Z}/n\mathbb{Z})^\times = \{a + n\mathbb{Z} \mid \gcd(a, n) = 1\}$

3. $\mathbb{H} = \{a + bi + cj + dk \mid a, b, c, d \in \mathbb{R}, i, j, k \in Q_8\}$ (real Hamiltonian quaternions) with multiplication the $\mathbb{R}$-linear extension of multiplication in Q_8 is a noncommutative division ring. Once can check: $(a + bi + cj + dk)^{-1} = \frac{a-bi-cj-dk}{a^2+b^2+c^2+d^2}$.

Definition 23. A nonzero commutative ring R is an **integral domain** if R has no zero divisors.

Proposition 24. *A finite integral domain is a field.*

Proof. Let R be a finite integral domain. Let $0 \neq a \in R$. The map $\phi : R \to R$, $\phi(x) = a \cdot x$ is injective (by there being no zero divisors), hence surjective (since $|R| < \infty$). So there exists $b \in R$ such that $ab = 1$. Thus a is a unit and R is a field. □

2.1.3 Generation of ideals, prime ideals

Definition 25. Let R be a nonzero ring. Let $X \subseteq R$. The **ideal generated by X** is

$$(X) := \bigcap_{I \trianglelefteq R, X \subseteq I} I$$

(Check: The intersection of any family of ideals is an ideal).

If $X = \{a\}$ write (a) for the ideal generated by X. Such ideals are called **principal**.

An integral domain in which every ideal is principal is called a **principal ideal domain (PID)**.

If $I = (X)$ for some finite X, then I is **finitely generated.**

Example 26. 1. In $\mathbb{Z}$, $(12, 16) = (4)$.

2. In $\mathrm{Mat}_2(\mathbb{R})$, $0 \neq A \in \mathrm{Mat}_2(\mathbb{R}) \implies (A) = \mathrm{Mat}_2(\mathbb{R})$ (**exercise**)

Definition 27. Let R be a ring, let $A, B \trianglelefteq R$. We define the **product of ideals**

$$A \cdot B := (\{a \cdot b \mid a \in A, b \in B\})$$

$P \trianglelefteq R$ is a **prime ideal** if $P \neq R$ and P has the following property: for any $A, B \trianglelefteq R, A \cdot B \subseteq P \implies A \subseteq P$ or $B \subseteq P$.

Proposition 28. *1. If $P \subsetneq R$ is an ideal such that for all $a, b \in R$, $ab \in P$ implies $a \in P$ or $b \in P$, then P is prime.*

2. If R is commutative and P is prime, then for all $a, b \in R$, $ab \in P$ implies $a \in P$ or $b \in P$.

Example 29. In $\mathrm{Mat}_2(\mathbb{R}), 0$ is the only proper ideal so is prime. But $\mathrm{Mat}_2(\mathbb{R})$ has zero divisors. (When R is not commutative it is **not** true that $[ab \in P \implies a \text{ or } b \in P] \iff [P \subsetneq R \text{ is prime}]$).

Proposition 30. *Let R be a nonzero ring, let I be an ideal in R.*

1. *If I is maximal and R is commutative, then R/I is a field.*
2. *If R is commutative, every maximal ideal is prime.*
3. *If $I \cap R^\times \neq \varnothing$, then $I = R$.*
4. *If R/I is a division ring, I is maximal.*
5. *If R is commutative, I is maximal $\iff$ R/I is a field.*

2.1.4 Examples of Rings

1. **Polynomial rings**:
   If R is commutative, we let $R[x_1, \ldots, x_n]$ be the polynomial ring in n commuting variables

   e.g. $R[x]$:

   $p = \sum_{i=0}^{n} a_i x^i, a_n \neq 0 \in R[x]$,
   $\deg p = n$,
   leading coefficient $\mathrm{LC}(p) = a_n$,
   leading term $\mathrm{LT}(p) = a_n x^n$,
   p is monic if $\mathrm{LC}(p) = 1$

   Note:

   - $R \leq R[X]$
   - If R is an integral domain, $\deg(p \cdot q) = \deg(p) + \deg(q)$
   - If R is an integral domain, $R[X]^\times = R^\times$.
   - If R is an integral domain, so is $R[x]$.

2. **Matrix rings**:
   R ring, $n > 0$: define $\mathrm{Mat}_n(R) = n$ by n matrices with entries in R.

   $\mathrm{Mat}_n(R)$ is not commutative for $n \geq 2$ and has zero divisors.

   R embeds into $\mathrm{Mat}_n(R)$ as scalar matrices.

   If R is commutative, $\mathrm{GL}_n(R) := \mathrm{Mat}_n(R)^\times = \{A \in \mathrm{Mat}_n(R) \mid \det A \in R^\times\}$

3. **Quadratic integer rings**:
   Let $D \in \mathbb{Z}$ be square-free.

   $\mathbb{Z}[\sqrt{D}] = \{a + b\sqrt{D} \mid a, b \in \mathbb{Z}\}$ is a subring of $\mathbb{Q}(\sqrt{D}) = \{a + b\sqrt{D} \mid a, b \in \mathbb{Q}\}$.

   If $D \equiv 1 \pmod 4, \mathbb{Z}[\frac{1+\sqrt{D}}{2}]$ is a subring.

   Define the norm $N : \mathbb{Q}(\sqrt{D}) \to \mathbb{Q}, a + b\sqrt{D} \mapsto a^2 - Db^2$. Check: $N(xy) = N(x)N(y)$.

   If $a \in \mathbb{Z}[\sqrt{D}], N(a) \in \mathbb{Z}$.

   And note: $1 + x$ is invertible in $R[[x]]$.

4. **Group rings**:
   If R is a ring and G is a group (under multiplication), the **group ring** $R[G]$ is the set

$$\{(r_g)_{g\in G} \mid r_g \in R, r_g = 0 \text{ for all but finitely many } g\} \leq \prod_{g\in G} R$$

   with componentwise addition and multiplication extended linearly from the group multiplication:

$$(r_g, g)(r_h, h) = r_g r_h \cdot (g \cdot h)$$

   We can think of this as R-linear combinations of elements in G.

   Example 31. $\mathbb{C}[\mathbb{Z}/2\mathbb{Z}] \subseteq \mathbb{C} \times \mathbb{C}$
   $1 \mapsto (1, 0)$ $-1 \mapsto (0, 1)$
   $(0, 1)(0, 4) = (4, 0)$

   $R[G]$ is called the **group ring of G over R**.

   $R[G]$ is commutative $\iff$ both R and G are commutative.

   The inclusion $R \hookrightarrow R[G]$ is given by $r \mapsto r \cdot e_G$.

   Group rings are important in representation theory: e.g. $\mathbb{Q}[S_3] \cong \mathbb{Q} \oplus \mathbb{Q} \oplus M_2(\mathbb{Q})$

5. **Endomorphism rings**: A abelian group. $\mathrm{End}(A)$ is a ring (with 1)

2.1.5 Prime and Maximal Ideals

From now on, all rings are commutative.

Proposition 32. *If $0 \neq I \trianglelefteq R$, then R/I is an integral domain $\iff$ I is prime.*

Proof. **Exercise.** □

Proposition 33. *Let $R \neq 0$. The following are equivalent:*

1. *R is a field.*
2. *The only ideals in R are 0 and (1).*
3. *Every nonzero ring homomorphism $R \to S$ is injective.*

Proof. (1) $\Rightarrow$ (2): If R is a field, $R^\times = R \setminus \{0\}$ so every nonzero element of R generates (1), hence 0 and R are the only ideals.

(2) $\Rightarrow$ (3): Let $\varphi : R \to S$ be a nonzero homomorphism. Then $\ker \varphi$ is a proper ideal of R. Since only 0 and R are ideals, $\ker \varphi = 0$, so φ is injective.

(3) $\Rightarrow$ (1): Let $x \in R \setminus R^\times$. Then $(x) \neq R$ and $S := R/(x) \neq 0$. $\pi : R \to S$ is a ring homomorphism. By assumption π is injective, so $(x) \subseteq \ker \pi = (0) \implies (x) = (0) \implies x = 0$. □

Proposition 34. *Every nonzero ring R has at least one prime ideal.*

Proof. R has a maximal ideal (by Theorem 1.4), and every maximal ideal is prime (by Proposition 3.2). □

Proposition 35. *If $\varphi : R \to S$ is a ring homomorphism and $P \trianglelefteq S$ is prime, then $\varphi^{-1}(P) \trianglelefteq R$ is prime.*

Proof. $R/\varphi^{-1}(P) \to S/P, r \mapsto \varphi(r) \pmod P$ is injective. $P \trianglelefteq S$ is prime so S/P is a domain. Then $R/\varphi^{-1}(P)$ is a domain (by being a subring of S/P) so $\varphi^{-1}(P)$ is prime. □

Example 36. (0) is maximal in $\mathbb{Q}$ but not maximal in $\mathbb{Z}$. There is no direct analogue for maximal ideals as for maximal subgroups in groups.

Proposition 37. *Every element $a \in R \setminus R^\times$ is contained in a maximal ideal.*

Definition 38. A ring with exactly one maximal ideal is called a **local ring**. If I is the unique maximal ideal of a local ring R, then R/I is called the **residue field** of R.

Example 39. $\mathbb{C}[[t]]$ is a local ring.

Proposition 40. *Let R be a nonzero ring and $I \lneq R$.*

1. *If $R/I \subseteq R^\times$ then R is a local ring with maximal ideal I.*
2. *If I is maximal and $1 + I \subseteq R^\times$ then R is local.*

Proof. (1) If $J \lneq R$ is any ideal, then $b \in J \implies b \notin R^\times \implies b \in I \implies J \subseteq I$.

(2) Let $x \in R \setminus I$. Since I is maximal, $(x, I) = R$. So there exist $a \in R$, $r \in I$ with $ax + r = 1$. Then $ax = 1 - r \in 1 + I \subseteq R^\times$. Hence $x \in R^\times$ so by (1), R is local. □

Proposition 41. *If R is a PID, every nonzero prime ideal is maximal.*

Proof. Let $P = (x)$ be a nonzero prime ideal.
If $(x) \subseteq (y) \subseteq R$, then $x = yz$ for some $z \in R$, so $yz \in (x)$.
If $(y) \subsetneq (x)$ then $y \notin (x) \implies z \in (x)$ since (x) is prime.
Then $z = xr$ for some $r \in R$, hence $x = yz = yrx \implies yr = 1$ since R is a domain.
So $(y) = R$. So (x) is maximal. □

Theorem 42 (Cohn)**.** *Suppose R is a ring in which every prime ideal is finitely generated. Then every ideal of R is finitely generated.*

Proof. Let S be the set of ideals of R that are not finitely generated. Assume (for contradiction) that $S \neq \varnothing$.

Partially order S by inclusion.
Let $\{I_\alpha\}_{\alpha \in A}$ be a totally ordered subset of S. Let $I = \bigcup_{\alpha \in A} I_\alpha$. Then I is an upper bound for $\{I_\alpha\}_{\alpha \in A}$ and I cannot be finitely generated since no I_α is, hence $I \in S$. By Zorn's lemma, there exists a maximal element P of S.

By hypothesis P is not prime. So there exist $x, y \in R$ such that $xy \in P$ but $x \notin P$, $y \notin P$. Then $P \subsetneq (P, (x))$ so $(P, (x))$ is finitely generated. Write $(P, (x)) = (x_1, \ldots, x_k) =$

$(c_1x + p_1, \ldots, c_kx_k + p_k)$, where $p_i \in P$.

Take $p \in P$. Then $p = c \cdot x + \sum_{i=1}^{k} a_i p_i$ where $a_i, c \in R$. Then $c \in J = \{r \in R \mid rx \in P\}$. Since $xy \in P, y \in J$. But $y \notin P$ so $P \subsetneq J$. So J is finitely generated, say $J = (j_1, \ldots, j_s)$.

But then $P \in xJ + (p_1, \ldots, p_k) \implies P$ is finitely generated by $\{x_{j_1}, \ldots x_{j_s}, p_1, \ldots, p_k\}$. But $P \notin S \rightarrow\leftarrow$. Hence every ideal is finitely generated. □

2.1.6 Nilradical and Jacobson Radical

Again all rings are commutative.

Proposition 43. *Let* $\mathrm{Nil}(R) = \{x \in R \mid x^n = 0 \text{ for some } n\}$*, the set of all nilpotent elements. Then* $\mathrm{Nil}(R)$ *is an ideal, and* $\mathrm{Nil}(R/\mathrm{Nil}R) = 0$.

Proof. Let $I := \mathrm{Nil}(R)$.
If $x, y \in I$, then $x^n = y^m = 0$ for some n, m. Then $(x+y)^{n+m-1} = 0$ so $x + y \in I$.
If $x \in I$ then $x^n = 0$ some n so if $r \in R$, $(rx)^n = r^n x^n = 0$, meaning $rx \in I$.
Hence I is an ideal.

Now let $\bar{x} \in R/I$ with preimage x so that $\bar{x}^n = 0$ for some n. Then $\overline{x^n} = \bar{x}^n = 0 \implies x^n \in I \implies x \in I$. □

Example 44. $\mathbb{Z}/12\mathbb{Z}$. $12 = 2 \cdot 2 \cdot 3$
$\mathrm{Nil}\, R = \{\bar{6} + 12\mathbb{Z}, 0\} = 6\mathbb{Z}/12\mathbb{Z}$
Prime ideals of $R : 3\mathbb{Z}/12\mathbb{Z}, 2\mathbb{Z}/12\mathbb{Z}$

Definition 45. $\mathrm{Nil}(R)$ is called the **nilradical** of R.

Proposition 46. *The nilradical of* R *is the intersection of all prime ideals of* R.

Proof. Let

$$I := \bigcap_{P \trianglelefteq R \text{ prime}} P$$

If $x \in \mathrm{Nil}(R)$, say $x^n = 0$, then for all primes P, since $0 \in P$, $x \in P$. Hence $\mathrm{Nil}\, R \subseteq I$.

Conversely, say $x \notin \mathrm{Nil}\, R$. We show there exists a prime ideal $P \trianglelefteq R$ so that $x \notin P$.

Let $S := \{J \trianglelefteq R \mid \forall n > 0 : x^n \notin J\}$. $x \notin \mathrm{Nil}\, R \implies 0 \in S \implies S \neq \varnothing$. Partially order S by inclusion. **Check:** Zorn's Lemma applies.

Then we can take a maximal element $P \in S$. We will show P is prime:

Let $a, b \notin P$. Then $P \subsetneq (P, a)$ and $P \subsetneq (P, b) \implies (P, a), (P, b) \notin S$ by maximality. $x^m \in (P, a)$ some m and $x^n \in (P, b)$ some $n \implies x^{m+n} \in (P, ab) \implies$ by definition of $S, (P, ab) \notin S \implies (P, ab) \neq P \implies ab \notin P \implies P$ is prime. □

Definition 47. The **Jacobson radical** of R, denoted $\mathrm{Jac}(R)$, is the intersection of all maximal ideals of R.

Proposition 48. $x \in \mathrm{Jac}(R) \iff 1 - xy \in R^\times$ *for all* $y \in R$.

Proof. ($\Rightarrow$) : If $x \in \operatorname{Jac} R$ and $1 - xy \notin R^\times$ for some $y \in R$, $(1-xy) \subsetneq R \implies 1 - xy \in I$ some I maximal $\implies x \in I$ since $x \in \operatorname{Jac} R \implies xy \in I \implies 1 \in I \to\leftarrow$

($\Leftarrow$) : If $1 - xy \in R^\times$ for all $y \in R$ but $x \notin I$ for some maximal ideal I of R, then $(x, I) = R \implies u + xy = 1$ for some $y \in R, u \in I \implies 1 - xy = u \in I$ is not a unit $\to\leftarrow$. □

2.1.7 Operations on ideals

Definition 49. Let $I, J \trianglelefteq R$.

The **ideal sum** $I + J = \{a + b \mid a \in I, b \in J\} = (I, J) = (I \cup J)$ is an ideal.

More generally, for $(I_k)_k$ a family of ideals,

$$\sum I_k = (\cup I_k) = \left\{ \sum x_k \mid x_k \in I_k, x_k \neq 0 \text{ for only finitely many } k \right\}$$

Note 50. 1. $(I_k)_k$ ideals $\implies \cap I_k$ is an ideal.

2. From $I, J \trianglelefteq R$ we can define the **ideal product** $I \cdot J := (a \cdot b \mid a \in I, b \in J)$
3. Product of finitely many ideals is an ideal
4. Powers of ideals are ideals

Example 51. 1. $R = \mathbb{Z}$:

$$(a) + (b) = (\gcd(a, b)), \quad (a) \cap (b) = (\operatorname{lcm}(a, b)), \quad (a) \cdot (b) = (ab)$$

In particular, $(a) \cdot (b) = (a) \cap (b) \iff a, b$ are coprime.

2. $R = K[x_1, \ldots, x_n], K$ field

$I = (x_1, \ldots, x_n)$ **augmentation ideal**

$I^n = \{f \in R \mid f \text{ only has terms of degree} \geq n\}$, so $I^n \neq \{f^n \mid f \in I\} \neq (f^n \mid f \in I)$

Note 52. If $R_1, \ldots, R_n$ are rings then $\prod_{i=1}^n R_i =: R$ (defined as a group) is a ring with componentwise multiplication. There are projections $p_i : R \to R_i, i = 1, \ldots, n$ (homomorphisms)

Definition 53. $I, J \trianglelefteq R$ are **coprime** if $I + J = R$.

Proposition 54 (Chinese Remainder Theorem)**.** *Let $I_1, \ldots, I_n \trianglelefteq R$ be ideals and let $\varphi : R \to \prod_{i=1}^n R/I_i, x \mapsto (x + I_1, \ldots, x + I_n)$. Then*

1. *I_i, I_j are coprime for all $i \neq j \implies \prod_{i=1}^n I_i = \cap_{i=1}^n I_i$*
2. *φ is surjective $\iff$ I_i, I_j are coprime for all $i \neq j$.*
3. *φ is injective $\iff \bigcap I_i = (0)$*

Proof. **Exercise.** □

Proposition 55 (Prime Avoidance)**.** 1. *If $P_1, \ldots, P_n$ are prime then if $I \subseteq \bigcup_{i=1}^n P_i \implies I \subseteq P_i$ for some i.*

2. *$I_1, \ldots, I_n \trianglelefteq R$ and P prime so that $P \supseteq \bigcap_{i=1}^n I_i \implies P_i \supseteq I$ for some i. And if $P = \bigcap_{i=1}^n I_i$ then $P = I_i$ for some i.*

Proof. (1): We show: $I \not\subseteq P_i$ for all $i \implies I \not\subseteq \cap_{i=1}^n P_i$ by induction on n:
$n = 1$ ✓
Inductive step: Fix i. $I \not\subseteq \cap_{i \neq j} P_j$ by induction $\implies \exists x_i$ such that $x_i \notin P_j \forall j \neq i$.
If $x_i \notin P_i$ (some i) then we're done. So we can assume $x_i \in P_i$ for all i.
Define $Y := \sum_{i=1}^n x_1 \ldots x_{i-1} x_{i+1} \ldots x_n \in I$
Fix $j \implies x_j \in P_j \implies$ all terms except one are in $P_j \implies y \notin P_j \implies$ done.

(2): Suppose $P \not\supseteq I_i$ for all $i = 1, \ldots, n$. Then $\exists x_i \in I_i, x_i \notin P \implies$
$x_1 \ldots x_n \in \prod_{i=1}^n x_i \subseteq \cap_{i=1}^n I_i$ but $x_1 \ldots x_n \notin P$ since P is prime. So $P \not\supseteq \cap_{i=1}^n I_i$.
If $P = \cap_{i=1}^n I_i, I_i \subseteq P$ for some $i \implies P = I_i$, since $P \subseteq I_i$. □

Definition 56. Let $I, J \trianglelefteq R$. We define the **ideal quotient**

$$(I : J) := \{x \in R \mid xJ \subseteq I\}$$

(Easy to check: this is an ideal)
$(0 : I) = \{x \in R \mid xI = 0\} = \mathrm{Ann}_R(I)$ **annihilator of I in R**
$(I : x) := (I, (x))$

Proposition 57. *1. $I \subseteq (I : J)$*

2. *$(I : J)J \subseteq I$*

3. *$((I : J_1) : J_2) = (I : J_1 J_2)$*

4. *$(\cap I_i : J) = \cap (I_i : J)$*

5. *$(I : \sum_i J_i) = \cap (I : J_i)$*

Proof. **Exercise.** □

Definition 58. For an ideal $I \trianglelefteq R$, the **radical** of I is

$$\sqrt{I} = \{x \in R \mid x^n \in I \text{ for some } n\}.$$

I is a **radical ideal** if $\sqrt{I} = I$.

Example 59. $\mathrm{Nil}\, R = \sqrt{(0)}$ is a radical ideal

Proposition 60. *$\sqrt{I}$ is an ideal. It is the intersection of all prime ideals containing I.*

Proof. Let $\varphi : R \to R/I$. Then $\sqrt{I} = \varphi^{-1}(\mathrm{Nil}(R/I))$ is an ideal.

$\mathrm{Nil}(R/I)$ is the intersection of all prime ideals in R/I. These correspond to prime ideals in R containing I, hence the second assertion. □

Proposition 61. *1. $\sqrt{I} \supseteq I$*

2. $\sqrt{\sqrt{I}} = \sqrt{I}$

3. $\sqrt{IJ} = \sqrt{I \cap J} = \sqrt{I} \cap \sqrt{J}$

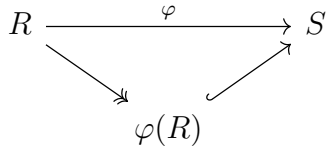
4. $\sqrt{I} = 1 \iff I = 1$

5. $\sqrt{I + J} = \sqrt{\sqrt{I} + \sqrt{J}}$

6. *P prime* $\implies \sqrt{P^n} = P$ *for all* $n \geq 1$ *(prime ideals are radical ideals)*

Definition 62. If $\varphi : R \to S$ is a ring homomorphism and $I \trianglelefteq R$, the **extension** of I to S is $I^e := (\varphi(I))$, and if $J \trianglelefteq S$, its **contraction** to R is $J^c := \varphi^{-1}(J)$.

Question:



In this scenario what happens to prime ideals?

Example 63. $\mathbb{Z} \subseteq \mathbb{Z}[i]$:

$$(2)^e = (1+i)^2 \text{ (ramified)}, \quad (5)^e = (2+i)(2-i) \text{ (split)}, \quad (3)^e \text{ prime (inert)}.$$

2.2 Localization and rings of fractions

2.2.1 Rings of fractions

Again all rings are commutative with identity.
Recall how to construct $\mathbb{Q}$ from $\mathbb{Z}$:

$$(a, s) \in \mathbb{Z} \times \mathbb{Z} \setminus \{0\}$$

with equivalence relation $(a, s) \sim (b, t) \iff at - bs = 0$

Definition 1. Let R be a commutative ring, $S \subseteq R$ a **multiplicatively closed subset** (i.e. $1 \in S$ and $S \cdot S \subseteq S$). We define an equivalence relation on $R \times S$ by

$$(a, s) \sim (b, t) \iff \exists u \in S \text{ such that } u(at - bs) = 0.$$

Proposition 2. *The above relation is an equivalence relation.*

Proof. Symmetric, reflexive: easy to check
Transitivity: Say $(a, s) \sim (b, t)$ and $(b, t) \sim (c, r)$. Then there exist $u, v \in S$ such that $(at - bs)u = 0$ and $(br - ct)v = 0$. So $(at - bs)urv = 0$ and $(br - ct)vsu = 0 \implies a + urv - ctsuv = 0 \implies tuv(ar - cs) = 0 \implies (a, s) \sim (c, r)$. □

Note 3. Denote the equivalence class of (r, s) by $\frac{r}{s}$ and let $S^{-1}R :=$ set of equivalence classes.

Proposition 4. *$S^{-1}R$ forms a commutative ring with 1. And $\varphi : R \to S^{-1}R, x \mapsto \frac{x}{1}$ is a ring homomorphism.*

Example 5. $R = \mathbb{Z}/12\mathbb{Z} = \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$
Let $S_1 = \{3^i \mid i \in \mathbb{N}\}$ and $S_2 = \{6^i \mid i \in \mathbb{N}\}$.
So $S_1 = \{1, 3, 9\}$ and $S_1^{-1}R \cong \mathbb{Z}/4\mathbb{Z}$
$S_2 = \{1, 6, 0\}$ and $S_2^{-1}R = \{0\}$

Note 6. φ in Proposition 2 need not be injective (e.g. if $0 \in S$ then $S^{-1}R = 0$)

Definition 7. $S^{-1}R$ is called the **ring of fractions of R with respect to S.**
If $S =$ set of all non zero-divisors of $R, S^{-1}R$ is called the **total ring of fractions**.
If R is a domain, $S^{-1}R$ is its **fraction field** ($S = R \setminus \{0\}$).

Proposition 8. *Let R, T be rings, let $S \subseteq R$ be multiplicatively closed. Let $\varphi : R \to T$ be a ring homomorphism with $\varphi(S) \subseteq T^\times$ for all $s \in S$. Then $\exists! \psi : S^{-1}R \to T$ such that the following diagram commutes:*

$$\begin{array}{ccc} R & \xrightarrow{i} & S^{-1}R \\ {\scriptstyle\varphi}\downarrow & \swarrow_{\psi} & \\ T & & \end{array}$$

Proof. Uniqueness: Let $\varphi : R \to T$ be a homomorphism so that $\varphi(S) \in T^\times$ for all $s \in S$. Then:
$\psi(\frac{a}{1}) = \psi(i(s)) = \varphi(a)$ for all $a \in R$, so $\psi(\frac{1}{s}) = \psi((\frac{s}{1})^{-1}) = \psi(\frac{s}{1})^{-1} = \varphi(s)$ for all $s \in S$. So $\psi(\frac{a}{s}) = \varphi(a)\varphi(s)^{-1}$ is uniquely determined by φ.

Existence: Define $\psi(\frac{a}{s}) := \varphi(a)\varphi(s)^{-1}$. This is well-defined: if $\frac{a}{s} = \frac{b}{t}$ so there exists some $u \in S$ so that $(at - bs)u = 0$. Then $\varphi(at - bs)\varphi(u) = 0$. $\varphi(u) \in T^\times$ since $u \in S$ so $\varphi(at - bs) = 0 \iff \varphi(a)\varphi(s)^{-1} = \varphi(b)\varphi(t)^{-1}$. □

Note 9. R ring, S multiplicative subset, $i : R \to S^{-1}R$. Then

1. $s \in S \implies i(s) \in (S^{-1}R)^\times$.
2. $a \in R, i(a) = 0 \implies as = 0$ some $s \in S$.
3. Every element of $S^{-1}R$ is of the form $i(a)i(s)^{-1}$ for some $a \in A, s \in S$.

Corollary 10. *Let $\varphi : R \to T$ be a ring homomorphism. If:*

1. *$s \in S \implies \varphi(s) \in T^\times$*
2. *$\varphi(a) = 0 \implies as = 0$ for some $s \in S$*
3. *$T = \{\varphi(a)\varphi(s)^{-1} \mid a \in R, s \in S\}$*

Then there exists a unique isomorphism $\psi : S^{-1}R \to T$ such that the following diagram commutes:

$$\begin{array}{ccc} R & \xrightarrow{i} & S^{-1}R \\ {\scriptstyle\varphi}\downarrow & \swarrow_{\psi} & \\ T & & \end{array}$$

Proof. Property 1 $\implies \exists!$ homomorphism $\psi : S^{-1}R \to T$ making the diagram commute, by Proposition 3, and $\psi(\frac{a}{s}) = \varphi(a)\varphi(s)^{-1}$. So property 3 $\implies \psi$ is surjective.

If $\psi(\frac{a}{s}) = 0, \varphi(a)\varphi(s)^{-1} = 0 \implies \varphi(a) = 0 \implies at = 0$ for some $t \in S$ (by 2) $\implies \frac{a}{s} = 0$ in $S^{-1}R \implies \psi$ is injective. □

2.2.2 Localization at a prime ideal

Definition 11. If $P \trianglelefteq R$ is a prime ideal, $R_P := S^{-1}R$ where $S = R \setminus P$ is called the **localization of R at P**.

Proposition 12. *If $P \trianglelefteq R$ is prime, R_P is a local ring with maximal ideal $I := \{\frac{a}{s} \mid a \in P, s \notin P\}$.*

Proof. I is an ideal (easy to check). Say $\frac{b}{t} \notin I$. Then $b \notin P$ so $b \in S = R \setminus P \implies b \in R^\times$ so $\frac{b}{t}$ is a unit in R_P. So I is the only maximal ideal and R_P is local. □

Example 13. For $R = \mathbb{Z}$, if $P = (p)$ where p is a prime, the localization $\mathbb{Z}_{(p)} = \{\frac{m}{n} \mid \gcd(n, p) = 1\}$, and $\mathbb{Z}_{(0)} = \mathrm{Frac}(\mathbb{Z}) = \mathbb{Q}$.

Example 14. $f \in R, S := \{f^k \mid k \in \mathbb{N}\}$
$R_f = S^{-1}R$ is called the **localization of R away from f**

Example 15. $R = \mathbb{Z}, m \in \mathbb{Z} \setminus \{0\}$
$\mathbb{Z}_m = \{\frac{a}{b} \mid b = m^k \text{ some } k \geq 0\}$

2.2.3 Extension and contraction in rings of fractions

R ring, S multiplicative subset, $i : R \to S^{-1}R$:
$C := \{i^{-1}(I) \mid I \trianglelefteq S^{-1}R\}$ **contracted ideals**
$E := \{I \cdot (S^{-1}R) \mid I \trianglelefteq R\}$ **extended ideals**
For $I \trianglelefteq R, S^{-1}I := I \cdot (S^{-1}R) = \{\sum \frac{a_i}{s_i} \mid a_i \in I, s_i \in S\}$

Proposition 16. *1. Every ideal in $S^{-1}R$ is an extended ideal*

2. $I \trianglelefteq R \implies I^{ec} = \bigcup_{s \in S}(I : s)$, hence $I^e = (1) \iff I \cap S \neq \varnothing$

3. $I \in C \iff$ no elements of S map to zero divisors in R/I

4. There is a one-to-one correspondence between prime ideals P of R with $P \cap S = \varnothing$ and prime ideals Q of $S^{-1}R$

5. S^{-1} commutes with finite sums, finite intersections, and radicals

Proof. 1. $J \trianglelefteq S^{-1}R$. We know $J^{ec} \subseteq J$. If $\frac{x}{s} \in J, \frac{x}{1} \in J \implies x \in J^c \implies \frac{x}{s} \in J^{ce}$ so $J = J^{ce}$.

2. $x \in I^{ec} = (S^{-1}I)^c \iff \frac{x}{1} = \frac{a}{s}$ some $a \in I, s \in S \iff (xs - a)t = 0$ some $t \in S, a \in I, s \in S \iff xst \in I$ some $s, t \in S \iff x \in \bigcup_{s \in S}(I : s)$

3. $I \in C \iff I = I^{ec} \iff I \supseteq I^{ec} \iff$ (by (2)) $\iff I \supseteq \cup_{s \in S}(I : s) \iff (sx \in I$ some $s \in S \implies x \in I) \iff$ no $s \in S$ maps to a zero-divisor in R/I

4. $P \trianglelefteq S^{-1}R$ prime $\implies P^c \trianglelefteq R$ prime. We have:

$$S^{-1}R/S^{-1}P \cong \bar{S}^{-1}(R/P), \bar{S} := S \bmod P$$

and either $\bar{S}^{-1}(R/P) = 0 \implies S^{-1}P = (1) \iff S \cap P \neq \varnothing$ or $\bar{S}^{-1}(R/P)$ integral domain $\implies S^{-1}P$ prime ideal

5. **Exercise**

□

Corollary 17. $\mathrm{Nil}(S^{-1}R) = S^{-1}(\mathrm{Nil}\, R)$

Corollary 18. *Let $P \trianglelefteq R$ be a prime ideal. Then there is a one-to-one correspondence between prime ideals of R_P and prime ideals in R contained in P.*

Note 19. All prime ideals contained in P can be studied in R_P, and all prime ideals containing P can be studied in R/P.

So if P, Q are prime in R and $Q \leq P$, then $(R_P)/Q^e \cong (R/Q)_{\overline{P}}$ has prime ideals coming from those between Q and P.

Proposition 20. *Let $\varphi : R \to T, P \trianglelefteq R$ be prime. Then:*
$P = Q^c$ some prime $Q \trianglelefteq T$ prime $\iff P^{ec} = P$

Proof. ($\Rightarrow$) : If $P = Q^c, P^{ec} = Q^{cec} = Q^c = P$.
($\Leftarrow$): If $P^{ec} = P$, $S := \varphi(R \setminus P) \subseteq T$ is multiplicatively closed.

Suppose $r \in R \setminus P$. $\varphi(r) \in P^e \cap S \implies r \in P^{ec} = P \to\leftarrow$. So $S \cap P^e = \varnothing$.

Then $S^{-1}P^e \lneq S^{-1}T$ is contained in proper maximal ideal $I \lneq S^{-1}T$. $Q := I^c \trianglelefteq T$ is prime. Hence $P^e \subseteq Q \subseteq T \setminus S \implies P \subseteq Q^c \subseteq P \implies P = Q^c$. □

2.3 Factorization in commutative rings

2.3.1 Unique factorization domains

Let R be a commutative ring.

Definition 1. Let $a, b \in R$. We say **a divides b** if $b = ax$ for some $x \in R$. Notation: $a \mid b$.

a, b are **associates** if $a \mid b$ and $b \mid a$. Notation: $a \simeq b$.

Proposition 2. *Suppose $a, b \in R, u \in R$.*

1. $a \mid b \iff (b) \subseteq (a)$
2. $a \simeq b \iff (a) = (b)$
3. $u \in R^\times \iff u \mid r$ *for all* $r \in R \iff (u) = R \iff u \simeq 1$
4. $a \simeq b$ *defines an equivalence relation on* R
5. $a = b \cdot r$ *for* $r \in R^\times \implies a \simeq b$
6. R *integral domain,* $a \simeq b \implies a = b \cdot r$ *some* $r \in R^\times$

Proof. **Exercise** □

Example 3. $R = \mathbb{Z}/6\mathbb{Z}$.
$\bar{2} \cdot \bar{2} = \bar{4}$ so $\bar{2} \mid \bar{4}$.
$\bar{4} \cdot \bar{2} = \bar{2}$ so $\bar{4} \mid \bar{2}$.
So $\bar{2} \simeq \bar{4}$. But $R^\times = \{\bar{1}, \bar{5}\}$ so there is no way to write $\bar{2} \cdot r = \bar{4}$ for any $r \in R^\times$.
Hence if R is not an integral domain, $a \simeq b \not\Rightarrow a = bu_1$ and $b = au_2$ for units u_1, u_2.

Definition 4. We say $a \in R$ is **irreducible** if

1. $a \notin R^\times, a \neq 0$
2. $a = bc \implies b \in R^\times$ or $c \in R^\times$

We say $p \in R$ is **prime** if

1. $p \notin R^\times, p \neq 0$
2. $p \mid ab \implies p \mid a$ or $p \mid b$

Example 5.
1. $R = \mathbb{Z}$, p prime number $\implies p$ and $-p$ are irreducible and prime
2. $R = \mathbb{Z}/6\mathbb{Z}$, $\bar{2}$ is prime but $\bar{2} = \bar{4} \cdot \bar{2}$ so is not irreducible

Theorem 6. $p, c \in R \setminus \{0\}, R$ *integral domain.*

1. p *prime* $\iff (p) \neq (0)$ *is a prime ideal*
2. c *irreducible* $\iff (c)$ *maximal proper principal ideal*
3. *prime elements are irreducible*
4. *If* R *is PID, prime* $\iff$ *irreducible*
5. *Every associate of an irreducible (resp. prime) element is irreducible (resp. prime)*
6. *The only divisors of an irreducible element are its associates and elements of* $R^\times$

Proof. (1) ✓

(2) (c) irreducible $\implies$ $c \notin R^\times$ so $(c) \lneq R$. If $(c) \subseteq (d), c \in (d)$ so $c = dx$ some $x \in R \implies$ either $d \in R^\times$ or $x \in R^\times \implies$ either $(d) = R$ or $(d) = (c) \implies (c)$ is maximal proper principal.

Conversely, (c) maximal proper principal $\implies$ $c \notin R^\times$. If $c = ab, (c) \subseteq (a)$ so $(a) = R$ or $(a) = (c) \implies$ wither $a \in R^\times$ or $a = cs$ some $s \in R$. If $a \notin R^\times, c = ab = csb \implies c(1 - sb) = 0 \implies 1 = sb \implies b \in R^\times$. So c is irreducible.

(3) Say p is prime and $p = ab$. Wlog $p \mid a$. $px = a$ some $x \in R \implies p = ab = pxb \implies xb = 1 \implies b \in R^\times$.

(4) Let p be irreudcible, and let R be a PID. By (2), (p) is a maximal proper principal ideal. R PID $\implies$ (p) is a maximal ideal $\implies$ (p) is a prime ideal $\implies$ by (1), p is prime.

(5) Say a is irreducible and $a \simeq b$. Then $a = bu, u \in R^\times$. Suppose $b = cd$. $a = bu = cdu \implies$ either $c \in R^\times$ or $du \in R^\times \implies$ either $c \in R^\times$ or $d \in R^\times \implies b$ is irreducible.

Prime case of (5) and (6) = **exercise.** □

Note 7. So in a domain, prime elements are irreducible, and in a PID, these concepts are equivalent.

Example 8. In $R = \mathbb{Z}[i\sqrt{5}]$,

$$2 \cdot 3 = (1 + i\sqrt{5})(1 - i\sqrt{5}) \implies 2 \mid (1 + i\sqrt{5})(1 - i\sqrt{5})$$

$N : \mathbb{Z}[i\sqrt{5}] \to \mathbb{Z}, a + i\sqrt{5}b \mapsto a^2 + 5b^2$. N is a homomorphism so if $x \mid y, N(x) \mid N(y)$,
$N(2) = 4, N(1 \pm i\sqrt{5}) = 6$. 4 does not divide 6 so 2 does not divide $1 + i\sqrt{5}$.

2 is a prime in $\mathbb{Z}$ but not in $\mathbb{Z}[i\sqrt{5}]$. One can show that 2 is irreducible in $\mathbb{Z}[i\sqrt{5}]$ (since the norm of a proper divisor of 2 is of the form $a^2 + 5b^2$ and a proper divisor of 4 so equals $2 \neq a^2 + 5b^2$).

So in $\mathbb{Z}[i\sqrt{5}]$ irreducible $\not\Rightarrow$ prime, hence $\mathbb{Z}[i\sqrt{5}]$ is not a PID.

Definition 9. Let R be an integral domain. Then R is a **unique factorization domain (UFD)** if:

1. Every nonzero nonunit can be written as $a = c_1 \dots c_n$ with c_i irreducible
2. If $a = c_1 \dots c_n = d_1 \dots d_m$ are two such decompositions, then $m = n$ and $c_i \simeq d_{\sigma(i)}$ for some $\sigma \in S_n, \forall i = 1, \dots, n$.

Example 10. $\mathbb{Z}$ is a UFD.

Note 11. In a UFD prime = irreducible.

Lemma 12. *Let R be a PID, and $(a_1) \subseteq (a_2) \subseteq \dots$ a chain of ideals. Then $\exists N$ such that $(a_i) = (a_N)$ for all $i \geq N$.*

Proof.

$$I = \bigcup_{i \geq 1} (a_i) \trianglelefteq R$$

and R is a PID $\Rightarrow I = (a)$ some $a \Rightarrow a \in (a_N)$ some $N \Rightarrow (a_i) = (a_N) = (a)$ for all $i \geq N$. □

Theorem 13. *Every PID is a UFD.*

Proof. Let R be a PID.

Existence of factorization: Let $a \in R \setminus \{0\}, a \notin R^\times$. Without loss of generality assume a is reducible. So $a = b_1c_1$ $(b_1 \simeq a, c_1 \not\simeq a) \Rightarrow (a) \subsetneq (b_1)$.

If b_1 is reducible, $b_1 = b_2c_2$:

$$(a) \subsetneq (b_1) \subsetneq (b_2)$$

This can be iterated, and by Lemma 3, this process must stop. So there exists some $b_2 =: d_1$ irreducible. $d_1 \mid a$. Let $d_1a_1 = a$. If a_1 is not irreducible, $a_1 = d_2a_2$ where d_2 is also irreducible (same argument as previous step). Iteration gives

$$(a) \subsetneq (a_1) \subsetneq (a_2) \subsetneq \dots$$

By Lemma 3, a_r is irreducible for some r. So $a = a_1 \dots a_r$.

Uniqueness of factorization: Say $a = c_1 \dots c_n = d_1 \dots d_m$ where c_i, d_i are irreducible. We'll proceed by induction on n:

$n = 1$ is clear.

For the inductive step, $c_1 \mid a \Rightarrow c_1 \mid d_1 \dots d_m \Rightarrow$ by Theorem 2.4, $\exists i$ such that $c_1 \mid d_i$ (irreducible $\Rightarrow$ prime) i.e. $c_1d_i' = d_i$. Note $d_i' \in R^\times$ since d_i is irreducible. So $c_2 \dots c_n = d_1 \dots d_{i-1}d_i'd_{i+1} \dots d_m$. (we cancelled c_1 on each side, which is allowed via R being an integral domain)

Now the claim follows by induction. □

2.3.2 Euclidean domains and greatest common divisor

Definition 14. Let R be an integral domain. We say R is a **Euclidean domain** if there is a function $\varphi : R \setminus \{0\} \to \mathbb{N}$ so that

1. $a, b \in R \setminus \{0\} \Rightarrow \varphi(a) \leq \varphi(ab)$
2. $a, b \in R, b \neq 0 : \exists q, r \in R$ such that $a = qb + r$ with either $r = 0$ or $\varphi(r) < \varphi(b)$

Example 15.
1. $\mathbb{Z}, \varphi(x) = |x|$
2. K field, $\varphi(x) = 1$ for all $x \in K^\times = K \setminus \{0\}$
3. K field, $R = K[x], \varphi(f) = \deg f$

Theorem 16. *Every Euclidean domain is a PID. (hence a UFD)*

Proof. Let $0 \neq I \trianglelefteq R$. Take $a \in I$ such that $\varphi(s) = \min\{\varphi(x) \mid x \in I \setminus \{0\}\}$. Easy to check: $I = (a)$ (**exercise**). □

Definition 17. Let R be an integral domain, $a_1, \dots, a_n \in R$. We say d is a **greatest common divisor** of $a_1, \dots, a_n$ if

1. $d \mid a_i \forall i = 1, \dots, n$
2. $t \mid a_i \forall i \Rightarrow t \mid d$

Notation: $d \in \gcd(a_1, \ldots, a_n)$ = set of all greatest common divisors

Example 18. $R = \mathbb{Z}$
$\gcd(6, 16) = \{2, -2\}$

Note 19. 1. $d, d' \in \gcd(a_1, \ldots, a_n) \Rightarrow d \simeq d'$

2. In many cases (e.g. $R = \mathbb{Z}, R = K[x]$) we can normalize (e.g. $d \in \mathbb{N}$, monic)
Then d is uniquely defined and we write $d = \gcd(a_1, \ldots, a_n)$.

Theorem 20. *Let R be an integral domain, $a_1, \ldots, a_n \in R \setminus \{0\}$*

1. *R UFD $\Rightarrow \gcd(a_1, \ldots, a_n) \neq \varnothing$*
2. *R PID, $d \in \gcd(a_1, \ldots, a_n) \Rightarrow \exists r_i \in R$ such that $d = \sum_{i=1}^{n} r_i a_i$*

Proof. **Exercise** □

Summary:

- Euclidean ⇒ PID ⇒ UFD
- All converse implications are false: $\mathbb{Z}[x]$ is a UFD but not a PID, $\mathbb{Z}[\frac{1+\sqrt{-19}}{2}]$ is a PID but not Euclidean

2.3.3 Gauss's Lemma and unique factorization in polynomial rings

Let R be a UFD. Recall: R domain $\Rightarrow R[x]$ is an integral domain.

Definition 21. Let $f = a_0 + \cdots + a_n x^n \in R[x], d \in \gcd(a_0, \ldots, a_n)$. Then (d) is called the **content** of f. Notation: $c(f)$. We say f is **primitive** if $(c) = (1)$.

Note 22. In general, irreducible in $\mathbb{Q}[x] \not\Rightarrow$ irreducible in $\mathbb{Z}[x]$ because e.g. $2x$ is irreducible in $\mathbb{Z}[x]$ but not $\mathbb{Q}[x]$ (constants are units in $\mathbb{Q}[x]$)

Proposition 23. *$f, g, \in R[x]$ primitive $\Rightarrow f \cdot g$ is primitive.*

Proof. Suppose not. Then there exists $d \in R$ not a unit such that d divides all coefficients of $f \cdot g$. Since R is a UFD, there is some irreducible (hence prime) $p \in R$ so that $p \mid d$.

$R/(p)$ is an integral domain $\Rightarrow R/(p)[x]$ is an integral domain.

Let $\varphi : R[x] \to R/(p)[x]$ denote coefficientwise reduction mod p. So $\varphi : a_0 + \ldots a_n x^n \mapsto (a_0 + (p)) + \ldots (a_n + (p))x^n$.

$\varphi(fg) = 0 \Rightarrow \varphi(f)\varphi(g) = 0$. Since $R/(p)[x]$ is a domain, p dibides every coefficient of $g \to\leftarrow$ (we said f, g were primitive) □

Theorem 24 (Gauss's Lemma)**.** *Let R be a UFD with* $\mathrm{Frac}(R) =: F$ *field of fractions. Let $f \in R[x], \deg f \geq 1$. Then f is irreducible in $R[x] \iff f$ is primitive and irreducible in $F[x]$.*

Proof. ($\Leftarrow$) : Suppose f is primitive and irreducible in $F[x]$. Then $f = g \cdot h$ in $R[x] \implies g \in F[x]^\times$ or $h \in F[x]^\times$. Without loss of generality say $g \in F[x]^\times = F^\times = F \setminus \{0\} \implies g \in F \cap R[x] \subseteq R$. Since f is primitive, $g \in R^\times \implies g \in R[x]^\times \implies f$ is irreducible in $R[x]$.

($\Rightarrow$) : Suppose f is irreducible in $R[x]$. Recall: we showed that f_1, f_2 primitive $\implies f_1 \cdot f_2$ is primitive.

Suppose $f(x) = \sum_{i=0}^n a_i x^i$. Let $(d) = c(f)$ where $c(f)$ is the content of f.

Write $d\alpha_i = a_i$. So $f = d\sum_{i=0}^n \alpha_i x^i$. The degree of $\sum_{i=0}^n \alpha_i x^i$ is ≥ 1 so $\sum_{i=0}^n \alpha_i x^i$ is not a unit. Then since f is irreducible, $d \in R^\times$. So f is primitive.

Now suppose $f = f_1 f_2$ in $F[x]$.

Write $f_1 = \frac{1}{r_1}\tilde{f}_1, \tilde{f}_1 \in R[x], r_1 \in R$.

$f_2 = \frac{1}{r_2}\tilde{f}_2, \tilde{f}_2 \in R[x], r_2 \in R$.

$\tilde{f}_1 = \alpha_1 \tilde{\tilde{f}}_1$ and $\tilde{f}_2 = \alpha_2 \tilde{\tilde{f}}_2$ where $\tilde{\tilde{f}}_1, \tilde{\tilde{f}}_2 \in R[x]$ are primitive and $\alpha_1, \alpha_2 \in R$.

So $r_1 r_2 f = \tilde{f}_1 \tilde{f}_2 = \alpha_1 \alpha_2 \tilde{\tilde{f}}_1 \tilde{\tilde{f}}_2$ (f primitive and $\tilde{\tilde{f}}_1 \tilde{\tilde{f}}_2$ primitive).

Without loss of generality, $r_1 r_2 = \alpha_1 \alpha_2 \in R$ i.e. $f = \tilde{\tilde{f}}_1 \tilde{\tilde{f}}_2$ in $R[x] \implies \tilde{\tilde{f}}_i \in R \setminus \{0\}$ some i, since f is irreducible $\implies f_i \in F[x]^\times$ some $i \implies f$ is irreducible in $F[x]$. □

Corollary 25. *Let R be a UFD. If $f \in R[x]$ is irreducible, then f is prime.*

Proof. Suppose $f \mid gh$ in $R[x]$ for some $g, h \in R[x]$. Then $gh = cf$, some $c \in R[x]$.

Gauss's lemma implies f is irreducible in $F[x]$ where $F = \mathrm{Frac}(R)$ (and f is primitive). Since $F[x]$ is a UFD, f is prime in $F[x]$. Thus $f \mid g$ or $f \mid h$ in $F[x]$.

Without loss of generality say $fk = g$ for some $k \in F[x]$.

$k = (\frac{\alpha}{\beta})k'$ for $\alpha, \beta \in R, \beta \neq 0, k' \in R[x]$ primitive. Then $\beta g = \alpha f k'$ (where RHS is primitive) $\implies \beta \mid \alpha$ in R since irreducible factprs of β cannot divide primitive fk'.

Thius $g = \gamma f k'$ i.e. $f \mid g$ in $R[x] \implies f$ is prime. □

Theorem 26. *R is a UFD $\implies$ $R[x]$ is a UFD.*

Proof. We will use the following characterization of UFDs (will prove in homework): An integral domain is a UFD $\iff$ every nonzero prime ideal contains a nonzero principal prime ideal.

Let $P \subset R[x]$ be a nonzero prime ideal. Strategy: use $R[x] \to F[x] = S^{-1}R[x]$ where $S = R \setminus \{0\}$.

Case 1: $P \cap R \neq \{0\}$.

Then $P \cap R$ is a prime ideal in P. Since R is a UFD, there is some (p) nonzero principal prime ideal contained in $P \cap R$. By homework, $(p)R[x]$ is a prime nonzero principal ideal contained in $P \implies P$ contains a nonzero principal prime.

Case 2: $P \cap R = (0)$.

Let $F = \mathrm{Frac}\, R$. Then $PF[x] \trianglelefteq F[x]$ and is prime since $F[x] = S^{-1}R[x]$ as above and $P \cap S = \varnothing$. Since $F[x]$ is a PID, we write $PF[x] = (f)F[x]$.

Without loss of generality assume $f \in R[x]$ and is primitive. Then since $(f)F[x]$ is prime, f is a prime element of $F[x]$ hence is irreducible in $F[x]$ (and still primitive). By Gauss, f is irreducible in $R[x]$.

Also: $f \in PF[x] \cap R[x] = P$ (by 2.3.1). Corollary 3 $\implies$ f is prime (in $R[x]$) $\implies$ $(f) \subseteq P$ is nonzero principal prime ideal.

So every prime $P \trianglelefteq R[x]$ nonzero contains a nonzero principal prime ideal $\implies$ $R[x]$ is a UFD. $\square$

Theorem 27 (Eisenstein Criterion)**.** *Let R be a UFD and let $F = \operatorname{Frac} R$. Let $f(x) = a_0 + a_1x + \cdots + a_nx^n \in R[x]$ and suppose that $\deg f \geq 1$. If there is some irreducible $p \in R$ such that:*

- *$p \mid a_i$ for $i = 0, \ldots, n-1$,*
- *$p \nmid a_n$,*
- *$p^2 \nmid a_0$.*

Then f is irreducible in $F[x]$. (So if f is also primitive, then f is irreducible in $R[x]$ also).

Example 28. 1. $f = 2x^5+6x^3+9x^2-15 \in \mathbb{Z}[x]$ is Eisenstein at $p = 3$ $\implies$ is irreducible in $\mathbb{Q}[x]$

2. $f = x^2 + 1 \in \mathbb{Z}[x]$ is Eisenstein at $p = 2$ via the change of variables $x = y - 1 : f = (y-1)^2 + 1$
3. $f = y^3 + x^2y^2 + x^3y + x \in \mathbb{Z}[x, y]$ is Eisenstein via $p = x$ in $\mathbb{Z}[x, y] = \mathbb{Z}[x][y]$. (since $\mathbb{Z}[x]$ is a UFD, $\mathbb{Z}[x, y]$ is a UFD and hence x is irreducible.

Proof. Without loss of generality assume f is primitive.

If $d \mid a_i$ for $i = 0, \ldots, n$, then $p \nmid d$ $\implies$ the new f after divifing by d still satisfies the hypotheses.

By Gauss, we only need to show f is irreducible in $R[x]$. Now suppose $f = gh \in R[x]$.

Say $g = b+rx^r+\ldots b_1x+b_0, \deg(g) = r \geq 1$. Say $h = c_sx^s+\cdots+c_1x+c_0, \deg(h) = s \geq 1$. And say $r, s \leq n-1$.

$p \mid a_0$ $\implies$ $p \mid b_0c_0$ $\implies$ $p \mid b_0$ or $p \mid c_0$. Without loss of generality assume $p \mid b_0$. Then $p \nmid c_0$ (otherwise $p^2 \mid a_0$).

f primitive $\implies$ $\exists k$ such that $p \nmid b_k$. Let k be minimal so that $p \mid b_i$ for $i < k$ and $p \nmid b_k$.

Then $a_k = b_0c_k + \cdots + b_kc_0$.

p divides $b_0c_k + \cdots + b_{k-1}c_1$ but not b_kc_0 $(p \nmid b_k, p \nmid c_0)$ $\implies$ $p \nmid a_k$. $\rightarrow\leftarrow$

So f must be irreducible. $\square$

2.4 Noetherian Rings

In this section all rings are commutative and unital.

2.4.1 Chain conditions and Hilbert Basis Theorem

Definition 1. A ring R is **Noetherian** if every ideal is finitely generated.

Example 2. R PID $\implies$ R Noetherian.

Proposition 3. *R is Noetherian $\iff$ every increasing sequences $I_1 \subseteq I_2 \subseteq \ldots$ of ideals is eventually constant.*

Proof. ($\Rightarrow$) : If R is Noetherian and $I_1 \subseteq I_2 \subseteq \dots$ is an increasing sequence of ideals, then $I := \bigcup I_j$ is an ideal in R. So I is finitely generated, i.e. $I = (x_1, \dots, x_n)$ for some n and some $x_i \in R$. So there is some j such that $x_i \in I_j$ for all $i = 1, \dots n$. Thus $I_j = I_{j+1} = \dots$

($\Leftarrow$) : If R is not Noetherian, then there is some ideal $I \trianglelefteq R$ that is not finitely generated. Take $0 \neq x_1 \in I$. Then we can take $x_1 \in I \setminus (x_1), \dots$ and we get

$$(x_1) \subsetneq (x_1, x_2) \subsetneq (x_1, x_2, x_3) \subsetneq \dots$$

which gives an infinite increasing sequence. □

Theorem 4 (Hilbert Basis Theorem)**.** *If R is Noetherian, then $R[x]$ is Noetherian.*

Proof. Let $0 \neq J \trianglelefteq R[x]$.

Let $m := \min\{\deg f \mid f \in J \setminus \{0\}\}$.

For $n \geq m$ define $I_n := \{a \in R \mid a = \mathrm{LC}(f) \text{ for some } f \in J \text{ where } \deg f = n\} \cup \{0\}$ the ideal of leading coefficients of degree-n polynomial in J.

Easy to check: $I_n \trianglelefteq R$ for all $n \geq m$ and $I_n \subseteq I_{n+1} \subseteq \dots$

R Noetherian $\implies$ this sequence eventually becomes constant, so $I_k = I_{k+1} = \dots$ for some k.

Also, R Noetherian $\implies$ $I_m \subseteq I_{m+1} \subseteq \cdots \subseteq I_k$ are all finitely generated.

For $m \leq n \leq k$: Let $A_n \subseteq J$ be a finite set of polynomials of degree n such that $(\mathrm{LC}(f) \mid f \in A_n\}) = I_n$.

Define

$$A := \bigcup_{n=m}^{k} A_n$$

Note that A is finite.

Claim: $(A) = J$.

$(A) \subseteq J$ is clear. For the reverse inclusion, let $p \in J$. We'll prove $p \in (A)$ by induction on $\deg p$.

If $\deg p = m, \exists q_i \in A_m$ and $a_i \in R$ such that $\mathrm{LC}(p) = \sum a_i \,\mathrm{LC}(q_i) (\in I_m) \implies \deg(p - \sum a_i q_i) < m$. By definition of m this forces $p - \sum a_i q_i = 0$ i.e. $p = \sum a_i q_i \in (A)$.

Now for the inductive step, assume the statement holds fo $\deg \leq n$.

If $n+1 \leq k, \deg p = n+1 \implies p - \sum a_i q_i$ has degree $\leq n$ for some suitable $a_i \in R_i, q_i \in A_{n+1} \implies p - \sum a_i q_i \in (A)$ by induction, hence $p \in (A)$.

If $n+1 > k$, $I_{n+1} = I_n$ so $\mathrm{LC}(p) = \mathrm{LC}(x \cdot f)$ some $f \in J, \deg f = n \implies p - x \cdot f \in (A)$ by induction and $f \in (A)$ by induction $\implies p \in (A)$. □

In particular, $K[x_1, \dots, x_n]$ is Noetherian when K is a field.

Theorem 5. *Let R be Noetherian.*

1. *If $\varphi : R \to T$ is a ring homomorphism, then $\varphi(R)$ is Noetherian. In particular, every quotient ring of a Noetherian ring is Noetherian.*

2. *If $S \subseteq R$ is multiplicatively closed, then $S^{-1}R$ is Noetherian. In particular, $R_{\mathfrak{p}}$ is Noetherian for every prime ideal $\mathfrak{p}$ in R.*

Corollary 6. *If K is a field, every finitely generated K-algebra (every ring of the form $K[x_1, \ldots, x_n]/I$) is Noetherian. (these are called affine rings)*

Note 7. R Noetherian $\iff$ every generating set of an ideal contains a finite generating set $\iff$ any non-empty set of ideals has a maximal element.

2.4.2 Primary decomposition

Definition 8. Let R be a ring and $Q \trianglelefteq R$ a proper ideal. Q is **primary** if $xy \in Q \implies x \in Q$ or $y^n \in Q$ for some n.

Note 9. Q is primary $\iff$ $R/Q \neq 0$ and every zero divisor of R/Q is nilpotent.

Proposition 10. *If R is a ring, and $Q \trianglelefteq R$ is primary, then $\sqrt{Q}$ is the smallest prime ideal containing Q.*

Proof. If $xy \in \sqrt{Q}$, then $(xy)^n \in Q$ some n. So $x^n \in Q$ or $(y^n)^m \in Q$ some m. Thus $x \in \sqrt{Q}$ or $y \in \sqrt{Q}$. So $\sqrt{Q}$ is prime. But

$$\sqrt{Q} = \bigcap_{\text{primes } P \supseteq Q} P$$

hence the claim. □

Definition 11. If $Q \trianglelefteq R$ is primary and $P := \sqrt{Q}$, we say Q is P**-primary**.

Example 12.

1. $R = \mathbb{Z}$. Then $(0), (p^n)$ for p prime are the primary ideals.
2. Let $R = K[x, y]$ where K is a field and $Q = (x, y^2)$. Then $R/Q \cong K[x, y]/(x, y^2) \cong K[y]/(y^2)$.

   $(x, y) = \sqrt{Q} =: P$ but $P^2 \subsetneq Q \subsetneq P$.

   Q is primary but not a power of a prime ideal.
3. Let $R = K[x, y, z]/(xy - z^2)$. Let $\bar{x}, \bar{y}, \bar{z}$ be images of x, y, z. Then $P = (\bar{x}, \bar{z})$ is prime in R. (to see this: mod out and get $K[y]$ which is an integral domain). But P^2 is not primary: $\bar{x}\bar{y} \in P^2$ but $\bar{x} \notin P^2$ and no power of $\bar{y}$ lies in P^2.

Proposition 13. *If $\sqrt{I} \trianglelefteq R$ is maximal, then I is primary. In particular, all powers of a maximal ideal are primary.*

Proof.

$$\sqrt{I} = \bigcap_{I \subset P \trianglelefteq R, P \text{ priime}} P$$

But also if $\sqrt{I} = M$ is maximal, then M is the only prime containing $I \implies M + I$ is the only maximal ideal in $R/I \implies R/I$ is a local ring.

Then every element of R/I is either a unit or a nilpotent. So every zero divisor of R/I is nilpotent, hence I is primary. □

Definition 14. Let R be Noetherian and $I \trianglelefteq R$.

1. $\text{Ass}(I) := \{P \trianglelefteq R \mid P \text{ prime}, P = (I : b) \text{ some } b \in R\}$ is the set of **associated primes** of I
2. If $P, Q \in \text{Ass}(I)$ and $P \subsetneq Q$, then P is called an **embedded prime** ideal of I
3. 3. A **primary decomposition** of I is a decomposition
$$I = Q_1 \cap \cdots \cap Q_s$$
with each Q_i primary. It is called **irredundant** if no Q_i can be omitted and $\sqrt{Q_i} \neq \sqrt{Q_j}$ for $i \neq j$.

Example 15. $R = \mathbb{Z}/12\mathbb{Z}$ has prime ideals $(3), (2)$. $(0) = (3) \cap (4)$ is a primary decomposition.

Example 16. If $R = K[x, y]$, find the following primary decompositions:

- $I = (x^2, y)$ is actually primary: $(x, y) = \{f \in K[x, y] \mid xf \in (x^2, y)\}$ $\sqrt{(x^2, y)} = (x, y)$
- $J = (x^2, xy)$ $\{f \in K[x, y] \mid xf \in (x^2, xy)\} = (x, y)$ $\{f \in K[x, y] \mid yf \in (x^2, xy)\} = (x)$
  $J = (x^2, y) \cap (x) = (x, y)^2 \cap (x)$ and note: $\sqrt{(x^2, y)} = (x, y)$, $\sqrt{(x)} = (x)$.
  We see that in general, primary decomposition is not unique: $(x) \subsetneq (x, y)$ is an embedded prime.

Lemma 17. *Let R be Noetherian and let Q be P-primary.*

1. *If Q' is P-primary, then $Q \cap Q'$ is P-primary.*
2. *If $b \in R$ and $b \notin Q$ then $(Q : b)$ is P-primary and: $b \in P \iff Q \subsetneq (Q : b)$*
3. *If $P' \supseteq Q$ is prime then $QR_{P'} \cap R = Q$ (i.e. $Q^{ec} = Q$)*
4. *$\exists b \in R$ such that $P = (Q : b)$. In particular, $P \in \text{Ass}(Q)$.*

Proof. (Sketch)

1. **Exercise**
2. $(\Leftarrow)$: If $b \notin P$ we claim $(Q : b) = Q$:
   If $a \in (Q : b)$, $ab \in Q \implies$ either $a \in Q$ or $b \in P$, hence $a \in Q$. Thus $(Q : b) \subsetneq Q \implies b \in P$.

   $(\Rightarrow)$: $b \in P \implies b^n \in Q$ some n.
   Without loss of generality say $b^{n-1} \notin Q$ (we can do this since $b \notin Q$). So $b^{n-1} \in (Q : b) \setminus Q$ hence $Q \subsetneq (Q : b)$. So if $b \in R, b \notin Q, b \in P \iff Q \subsetneq (Q : b)$.
   (Q:b) is primary: $xy \in (Q : b)$ and $x \notin (Q : b) \implies xyb \in Q$ but $bx \notin Q \implies$ since Q is primary, $y^n \in Q$ some $n \implies y^n b \in Q \implies y^n \in (Q : b)$.
   P = (Q:b): $\sqrt{Q} = P$. $Q \subseteq (Q : b)$ so $P = \sqrt{Q} \subseteq \sqrt{(Q : b)}$.
   To prove $\sqrt{(Q : b)} \subseteq P$, let $x \in \sqrt{(Q : b)}$. Then $bx^n \in Q$ for some n. $b \notin Q \implies x^n \in \sqrt{Q} = P \implies x \in P$ since P is prime. So $P = (Q : b)$.

3. $x \in QR_{P'} \cap R \implies sx \in Q$ some $s \in R \setminus P'$. Assume $x \notin Q$ (for contradiction). Then Q primary $\implies s \in \sqrt{Q}.Q \subseteq P' \implies \sqrt{Q} \subseteq \sqrt{P'} = P \implies s \in P' \to\leftarrow$. So $x \in Q$, hence $QR_{P'} \cap R \subseteq Q$. This proves ($\subseteq$), and ($\supseteq$) is clear.

4. Without loss of generality $Q \subsetneq P$ (otherwise clear).

   $g_1 \in P \setminus Q \implies (Q : g_1) \supsetneq Q$ and is P-primary (by part 2). If $(Q : g_1) \subsetneq P$, take $g_2 \in P \setminus (Q : g_1)$. $((Q : g_1) : g_2) = (Q : g_1 g_2)$. So we get

   $$Q \subsetneq (Q : g_1) \subsetneq (Q : g_1 g_2) \subsetneq \dots$$

   R Noetherian $\implies \exists g_1, \dots, g_n \in R$ such that $P = (Q : g_1 \dots g_n)$.

□

Theorem 18 (Primary Decompositon I)**.** *If R is Noetherian, every proper ideal of R has an irredundant primary decomposition.*

Proof. By Lemma 3.1, it is sufficient to show there exists a primary decomposition.

Assume not; let $\mathcal{M}$ be the set of ideals without primary decomposition. So suppose $\mathcal{M} \neq \varnothing$. By Zorn's lemma, let I be maximal in $\mathcal{M}$.
Notice: I is not primary, so $\exists\, a, b$ with $ab \in I$, $a \notin \sqrt{I}$, and $b \notin I$.
Consider the chain:

$$(I : b) \subsetneq (I : b^2) \subsetneq \dots$$

Since R is Noetherian, $(I : b^n) = (I : b^{n+1})$ for some n.
Claim:

$$I = (I : b^n) \cap (I, b^n).$$

If we can show this, since $I \subsetneq (I : b^n), I \subsetneq (I, b^n)$, we know $(I : b^n) \notin \mathcal{M}, (I, b^n) \notin \mathcal{M}$ so they have a primary decomposition $\to\leftarrow$.

It is clear that $(I : b^n)$ and (I, b^n) contain I.

For $\supseteq$, say $d \in (I : b^n) \cap (I, b^n)$. Then $db^n \in I$ and $d = r + cb^n, r \in I, c \in R$. So $db^n = b^n r + cb^{2n} \implies cb^{2n} \in I \implies c \in (I : b^{2n}) \implies c \in (I : b^n) \implies cb^n \in I \implies d = r + cb^n \in I$. □

Theorem 19 (Primary Decomposition II)**.** *Let R be Noetherian and $I \subsetneq R$. $I = Q_1 \cap \dots \cap Q_r$ irredundant $\implies$ $\operatorname{Ass}(I) = \{\sqrt{Q_1}, \dots, \sqrt{Q_r}\}, r = \#\operatorname{Ass}(I)$.*

Moreover, if $P \in \operatorname{Ass}(I)$ with $\operatorname{Ass}(I, P) = \{\sqrt{Q_{i_1}}, \dots, \sqrt{Q_{i_s}}\}$ then $Q_{i_1} \cap \dots \cap Q_{i_s}$ is independent of the decomposition.

Proof. Say $I = Q_1 \cap \dots \cap Q_r, P \in \operatorname{Ass}(I)$. Then $P = (I : b)$ for some $b \in R$.
Write $P = (I : b) = (Q_1 : b) \cap \dots \cap (Q_r : b)$.

By prime avoidance, $(Q_j : b) \subseteq P$ for some j. But $P = (I : b) \subseteq (Q_j : b) \subseteq P \implies P = (Q_j : b)$. By Lemma 3.2, P is $\sqrt{Q_j}$-primary. So $P = \sqrt{Q_j}$ since P is prime and contains Q_j. Hence $\operatorname{Ass}(I) \subseteq \{\sqrt{Q_1} \dots, \sqrt{Q_r}\}$.

Conversely, let $J := Q_1 \cap \dots Q_{i-1} \cap Q_{i+1} \cap \dots \cap Q_n$ some fixed i.

Take $d \in J \setminus Q_i$. Easy to check: $(I : d) = (Q_i : d) \implies \sqrt{Q_i} = \sqrt{(Q_i : d)} = ((Q_i : d) : g)$ some $g = (I : dg) \implies \sqrt{Q_i} \in \operatorname{Ass}(I)$. So $\{\sqrt{Q_1}, \dots, \sqrt{Q_r}\} \subseteq \operatorname{Ass}(I)$.

Now just left to check uniqueness: If $\mathrm{Ass}(I,P) = \{\sqrt{Q_{i_1}}, \ldots, \sqrt{Q_{i_s}}\}$ i.e. $P \supseteq \sqrt{Q_{i_k}} \supseteq Q_{i_k}$ for all i_k, then $Q_{i_j} R_P \cap R - Q_j$ by Lemma 3 part 3.

On the other hand, $k \in \{i_1, \ldots, i_s\} \implies Q_k \not\subseteq P \implies Q_k R_P = R_p$. Then

$$IR_P \cap R = \cap(Q_j R_P \cap R) = Q_{i_1} \cap \cdots \cap Q_{i_s},$$

independent of decomposition. □

Example 20. $I = (f) \trianglelefteq K[x_1, \ldots, x_n]$. $f = f_1^{n_1} \ldots f_s^{n_s}$ irreducible factorization. Then $I = (f_1^{n_1}) \cap \cdots \cap (f_s^{n_s})$ is the primary decomposition.

2.4.3 Krull dimension and height

Definition 21. Let R be a ring. The supremum of lengths n of chains

$$P_0 \subsetneq P_1 \subsetneq \cdots \subsetneq P_n$$

of prime ideals is denoted $\dim R$ and called the **Krull dimension** of R.

Example 22.

1. If K is a field, $\dim K = 0$.
2. If R a PID but not a field, $\dim R = 1$ (every nonzero prime ideal is maximal)
3. If $0 \neq R$: $\dim R \geq 0$.
4. $\dim K[x_1, \ldots, x_n] \geq n$ because e.g. $(0) \subsetneq (x_1) \subsetneq (x_1, x_2) \subsetneq \cdots$

Note 23. Let R be a ring, P prime.

1. Chains of prime ideals that start with P correspond bijectively to chains starting at 0 in R/P.
2. Chains of prime ideals that end with P correspond bijectively to chains in R_P ending at PR_P.

Definition 24. Let $P \trianglelefteq R$. The **height** of P is

$$\mathrm{ht}(P) = \dim R_P.$$

For $I \trianglelefteq R$,

$$\mathrm{ht}(I) = \inf\{\mathrm{ht}(P) \mid P \text{ prime}, \ I \subseteq P\}.$$

Facts (in Noetherian rings):

- If $I = (a_1, \ldots, a_r)$, $\mathrm{ht}(P) \leq r$ for every minimal $P \in \mathrm{Ass}(I)$. In particular, $\mathrm{ht}(I) < \infty$.
- $\dim R[x_1, \ldots, x_n] = \dim R + n$.

2.5 Normalization and integral extensions

2.5.1 Integral extensions and algebraic extensions

Definition 1. $A \subseteq B$ rings. $b \in B$ is **integral over** A if there exists a monic polynomial $f \in A[x]$ such that $f(b) = 0$.

If A is a field then an element which is integral over A is called **algebraic** over A.

The monic polynomial f of minimal degree such that $f(b) = 0$ is the **minimal polynomial** of b.

$A^C := \{b \in B \mid b \text{ is integral over } A\}$ is the **integral closure** of A in B.

B is **integral over** A if $A^C = B$ i.e. every element of B is integral over A.

If A is a field and B is integral over A, we say B is **algebraic over** A

A is **integrally closed in** B if $A^C = A$.

Some facts about fields:

Definition 2. Let $K \subseteq L$ be fields. We say $t_1, \ldots, t_n \in L$ are **algebraically independent over** K if $f(t_1, \ldots, t_n) = 0$ for $f \in K[T_1, \ldots, T_n] \implies f = 0$.

The subfield of L generated by $t_1, \ldots, t_n \in L$ over K is

$$\bigcap_{L \geq L' \geq K, t_1, \ldots, t_n \in L'} L'$$

L/K is **finitely generated** if L is generated by $t_1, \ldots, t_n$ for some n and suitable t_i.
L/K is **finite** if L/K is algebraic over K and finitely generated.

Note 3. If $t_1, \ldots, t_n$ are algebraically independent, the ring generated by $t_1, \ldots, t_n$ over K is (isomorphic to) a polynomial ring in n variables. Write: $K[x_1, \ldots, x_n] \subseteq L$.

Lemma 4. *$L \geq K \geq M$ fields. If L/K is finite and K/M is finite, L/M is finite.*

Proposition 5. *Suppose $A \subseteq B$ is an integral extension of integral domains. Then A is a field $\iff$ B is a field.*

Proof. ($\Rightarrow$) : Let b be nonzero in B.

$b^n + \cdots + a_1 b + a_0 = 0$, some $a_0, \ldots, a_{n-1} \in A$. Without loss of generality assume $a_0 \neq 0$.

Then $b^{-1} = -a_0^{-1}(b^{n-1} + a_{n-1}b^{n-2} + \cdots + a_1) \in B$ (since $a_0^{-1} \in A$ and $b^{n-1} + a_{n-1}b^{n-2} + \cdots + a_1 \in B$).

($\Leftarrow$) : Let $a \in A \setminus \{0\}$. Then $a^{-1} \in B$ since $a \in B$ and B is a field. So a^{-1} is integral over A. Thus $a^{-n} + \cdots + a_1 a^{-1} + a_0 = 0$ some $a_0, \ldots, a_{n-1} \in A$ some n.
So $a^{-1} = -(a_0 a^{n-1} + \cdots + a_{n-1}) \in A$. □

Proposition 6. *Let L/K be a finitely generated field extension. Then there exists a unique $n \in \mathbb{N}, t_1, \ldots, t_n \in L$ such that $K[t_1, \ldots, t_n]$ is a polynomial ring and $L/K(t_1, \ldots, t_n)$ is finite.*

Proof. Let $\{t_1, \ldots, t_n\} \subseteq L$ be a maximal subset so that $t_1, \ldots, t_n$ are algebraically independent. Without loss of generality, assume n is minimal among sizes of such subsets.

Then $L/K(t_1, \ldots, t_n)$ is algebraic $\implies$ $L/K(t_1, \ldots, t_n)$ is finite.

Say $\{u_1, \ldots, u_m\} \subseteq L$ is another maximal subset so that $u_1, \ldots, u_m$ are algebraically independent. Then $m \geq n$. Want to show: $m = n$. We will show this by induction on n.

$n = 0 : L/K$ algebraic $\implies$ $m = 0$.

$n > 0 : \exists 0 \neq f(T_1, \ldots, T_n, u_1) \in K[T_1, \ldots, T_n, u_1]$ such that $f(t_1, \ldots, t_n, u_1) = 0$. ($u_1$ is algebraic over $K(t_1, \ldots, t_n)$; clear denominators)

Note that u_1 occurs in f and some T_i occurs (wlog we assume T_1).

Then t_1 is algebraic over $K(t_2, \ldots, t_n, u_1)$ $\implies$ $L \geq K(t_1, \ldots, t_n, u_1) \geq K(t_1, \ldots, t_n, u_1)$ are both finite extensions $\implies$ L is finite over $K(t_2, \ldots, t_n, u_1)$ $\implies$ by minimality of $n, \{t_2, \ldots, t_n, u_1\}$ are algebraically independent.

By induction, $m = n$. □

Definition 7. A set of elements as in Prop 2 is called a **transcendence basis of L/K** and n is the **transcendence degree of L/K.** Notation: $\mathrm{trdeg}(L/K)$ or $\mathrm{trdeg}_K(L)$.

2.5.2 Noether Normalization

Lemma. Let $A \subseteq B \subseteq C$ be rings.

1. A^C is a subring of B
2. If B is finite over A and C is finite over B then C is finite over A (here finite = finitely generated and integral.)

Theorem 8 (Noether Normalization)**.** *Let K be a field. Let A be an affine K-algebra, say $A = K[x_1, \ldots, x_m]/I$, and assume A is an integral domain. Let $F = \mathrm{Frac}(A)$.*
Then there exist $t_1, \ldots, t_n \in A$ such that:

1. *$t_1, \ldots, t_n$ form a transcendence basis of F/K,*
2. *A is integral over the polynomial ring $K[t_1, \ldots, t_n]$.*

Example 9. This is not easy to show: $A = K[x, y]/(xy - 1)$.

$F = K(x)$ has transcendence degree 1.

But $y = x^{-1}$ is not integral over $K[x]$ since y satisfies $xy - 1 = 0$ which is *not* monic in y.

Proof. (Of Noether Normalization)

Choose $t_1, \ldots, t_n \in A$ so that A is integral over the subring B generated by $t_1, \ldots, t_n$ and so that n is minimal with this property.

Check: $t_1, \ldots, t_n$ are algebraically independent.

Assume $\exists 0 \neq f \in K[T_1, \ldots, T_n]$ so that $f(t_1, \ldots, t_n) = 0$. Let d be such that f is of degree $< d$ in every variable.

Let $U_1 := T_1 - T_n^{d^{n-1}}, \ldots, U_{n-1} := T_{n-1} - T_n^d$ (in general $U_i = T_i - T_n^{d^{n-1}}$)
$u_i := U_i(t_1, \ldots, t_n) \in B \implies f(u_1 - T_n^{d^{n-1}}, \ldots, u_{n-1} + T_n^d, T_n) \in K[u_1, \ldots, u_{n-1}, T_n]$.
Easy to see that this gives a monic polynomial in $B'[T_n]$ where B' is the subring of A generated by $u_1, \ldots, u_{n-1}$ over K.

This polynomial has t_n as a zero $\implies$ t_n is integral over $B' \implies A$ is finite over the subring of B generated by B' and t_n which in turn is finite over $B' \implies A \geq B'$ is finite by the lemma $\implies A \geq B'$ is integral. $\rightarrow\leftarrow$ (B' is generated by $n-1$ elements so this contradicts minimality of n). □

Example 10. $A = K[x, y]/(xy - 1)$.
Find $z \in A$ such that A is integral over $K[z]$.
$z = x + y$ works:
If $f(t) = t^2 + (x - y)t - 1, f(y) = 0$. And if $g(t) = t^2 - (x - y)t - 1, g(x) = 0$.
Thus A is integral over $K[x + y]$.

2.5.3 Hilbert's Nullstellensatz

Lemma 11 (Zariski's Lemma)**.** *Let L/K be a field extension. If L is an affine ring over K then L/K is finite.*

Proof. Choose $t_1, \ldots, t_n$ so that L is integral over $K[t_1, \ldots, t_n]$ and $t_1, \ldots, t_n$ are algebraically independent (by Noether normalization). By Prop 1, $K[t_1, \ldots, t_n]$ is a field so $n = 0$. □

Theorem 12 (Weak Nullstellensatz)**.** *Let K be a field. Let $I \subsetneq K[x_1, \ldots, x_n]$. Then there exists a finite extension L over K and $a \in L^n$ such that $f(a) = 0 \forall f \in I$.*

Proof. Let $I \subseteq M \trianglelefteq K[x_1, \ldots, x_n]$ where M is a maximal ideal. Then $L := K[x_1, \ldots, x_n]/M$ is a field extension of K and is finite by Zariski.

Let $\varphi : K[x_1, \ldots, x_n] \to L$ be the canonical quotient map. Then $a := (\varphi(x_1), \ldots, \varphi(x_n)) \in L^n$ and $f(a) = f(\varphi(x_1), \ldots, \varphi(x_n)) = 0$. □

Note 13. If $K = \mathbb{C}$ there are no finite extensions because $\mathbb{C}$ is integrally closed. So f_i's don't have a common zero for all $i = 1, \ldots, r \iff (f_1, \ldots, f_r) = 1$.

Example 14. $f_1 = x$, $f_2 = 1 - x^2$.
$(f_1, f_2) = 1$
$1 = 1 \cdot (1 - x^2) + x \cdot x$

Theorem 15 (Hilbert's Nullstellensatz)**.** *Let $I \trianglelefteq K[x_1, \ldots, x_n]$. Let $f \in K[x_1, \ldots, x_n]$ be such that for all finite extensions L/K and for all $a \in L^n$ having $g(a) = 0$ for all $g \in I$, we have $f(a) = 0$. Then $f \in \sqrt{I}$.*

Proof. Without loss of generality, $f \neq 0$.
Rabinowitsch trick: $I' = (I, 1 - fY) \trianglelefteq K[x_1, \ldots, x_n, Y]$.
Suppose (for contradiction) that I' is proper. Theorem 5 $\implies \exists L/K$ finite extension and $a \in L^{n+1}$ such that $g(a) = 0 \forall g \in I'$.
Then by assumption: $f(a) = 0$ ($I \subseteq I'$) and $(1 - fY)(a) = 0 \implies 1 = 0$ (since $f(a)Y = 0$), $\rightarrow\leftarrow$.

Thus I' is not proper. In particular: $1 \in I'$. Let $I = (h_1, \ldots, h_r)$. Then write $1 = g_0(1 - Yf) + g_1h_1 + \cdots + g_rh_r, g_i \in K[x_1, \ldots, x_n, Y]$.

Now set $Y := f^{-1} \in K(x_1, \ldots, x_n)$. Plugging $Y = f^{-1}$,

$$1 = g_0(1 - f^{-1}f) + g_1(x_1, \ldots, x_n, f^{-1})h_1 + \ldots g_r(x_1, \ldots, x_n, f^{-1})h_r.$$

There exists m so that (clearing denominators) $f^m = \tilde{g_1}(x_1, \ldots, x_n)h_1 + \cdots + \tilde{g_r}(x_1, \ldots, x_n)h_r$ where $\tilde{g_i} \in K[x_1, \ldots, x_n] \implies f^m \in (h_1, \ldots, h_r) = I \implies f \in \sqrt{I}$. □

2.6 Basic Algebraic Geometry

2.6.1 Affine algebraic sets

Definition 1. Let K be a field. $\mathbb{A}^n(K) := K^n$ **affine n–space.**
$\mathbb{A}^1(K)$ affine line, $\mathbb{A}^2(K)$ affine plane, etc.
For $f \in K[x_1, \ldots, x_n]$, define the **zero set/ locus of f in** $\mathbb{A}^n(K)$

$$V(f) := \{a \in \mathbb{A}^n(K) \mid f(a) = 0\}.$$

If $f \notin K$, $V(f)$ is a **hypersurface**.
$S \subseteq K[x_1, \ldots, x_n] \implies$

$$V(S) := \bigcap_{f \in S} V(f)$$

is an **affine algebraic set.**

Example 2. $K = \mathbb{R}$:

1. $f = x^2 + y^2 - 1$ in $\mathbb{R}^2$: $V(f)$ is the unit circle.
2. $f = x_{11}x_{22} - x_{12}x_{21} - 1 = \det X - 1$: $V(f) = \mathrm{SL}_2(\mathbb{R})$.
3. $f = \det(X)$: singular matrices.
4. $f = (x_{11}x_{22} - x_{12}x_{21}) \cdot x - 1$: $V(f) = \mathrm{GL}_2(\mathbb{R}) \subseteq \mathbb{A}^5(K)$

$K = \mathbb{C}$:

1. $$f = \prod_{i<j | i,j \in \{1,\ldots,n\}} (r_i - r_j)^2$$

   $V(f)$ is the set of monic polynomials of degree n with repeated roots
2. $\mathrm{ch} : \mathbb{A}^{n^2}(K) \to \mathbb{A}^n(K), A \mapsto (a_1, \ldots, a_n)$ where a_i are the coefficients in the characteristic polynomial. Combining with (1): Set of matrices with repeated eigenvalues is affine algebraic.

Proposition 3. *Let K be a field.*

1. $S \subseteq K[x_1, \ldots, x_n], V(S) = V((S))$

2. $A, B \trianglelefteq K[x_1, \ldots, x_n], A \subseteq B \implies V(A) \supseteq V(B)$

3. $A_i \trianglelefteq K[x_1, \ldots, x_n \implies$

$$\bigcap_{k \in I} V(A_i) = V(\bigcup_{i \in I} A_i) = V(\sum_{i \in I} A_i)$$

4. $A, B \trianglelefteq K[x_1, \ldots, x_n] \implies V(A) \cup V(B) = V(A \cap B)$

5. $V(0) = \mathbb{A}^n(K), V((1)) = \varnothing, V(x_1 - \alpha_1, \ldots, x_n - \alpha_n) = \{(\alpha_1, \ldots, \alpha_n)\}$

Proof. **Exercise** □

Definition 4 (2)**.** Let K be a field. For $\mathcal{X} \subseteq \mathbb{A}^n(K)$, define

$$I(\mathcal{X}) = \{f \in K[x_1, \ldots, x_n] \mid f(a) = 0 \ \forall a \in \mathcal{X}\} \trianglelefteq K[x_1, \ldots, x_n]$$

the **vanishing ideal of** $\mathcal{X}$

Proposition 5. 1. $\mathcal{X} \subseteq \mathcal{Y} \implies I(\mathcal{Y}) \subseteq I(\mathcal{X})$.

2. $I(\varnothing) = K[x_1, \ldots, x_n]$. *If* $\#K = \infty, I(\mathbb{A}^n(K)) = 0$.

3. $S \subseteq K[x_1, \ldots, x_n] \implies I(V(S)) \supseteq S$ *and* $V(I(V(S))) = V(S)$

4. $\mathcal{X} \subseteq \mathbb{A}^n(K) \implies V(I(\mathcal{X})) \supseteq \mathcal{X}$ *and* $I(V(I(\mathcal{X}))) = I(\mathcal{X})$.

Corollary 6. *Every affine algebraic subset is the intersection of finitely many hypersurfaces.*

Proposition 7. $\mathcal{X} \subseteq \mathbb{A}^n(K) \implies I(\mathcal{X})$ *is a radical ideal.*

Proof. If $f^m \in I(\mathcal{X})$ for some m, then for any $\alpha \in \mathcal{X}$, $f(\alpha)^m = 0$, hence $f(\alpha) = 0$ so $f \in I(\mathcal{X})$. □

Recall: Nullstellensatz ($K = \bar{K}$): $A \trianglelefteq K[x_1, \ldots, x_n] \implies I(V(A)) = \sqrt{A}$.
From now on assume K algebraically closed unless stated otherwise.

Corollary 8. $A \trianglelefteq K[x_1, \ldots, x_n] \implies I(V(A)) = \sqrt{A}$. *So we have an inclusion-reversing bijection*

$$\{\textit{affine algebraic subsets of } \mathbb{A}^n(K)\} \leftrightarrow \{\textit{radical ideals in } K[x_1, \ldots, x_n]\}$$

2.6.2 The Zariski topoology

Assume $K = \bar{K}$.

Definition 9. The complements of affine algebraic subsets of $\mathbb{A}^n(K)$ satisfy the axioms for open subsets in a topology, which we all the **Zariski topology.**

That is, $\mathcal{X} \subset \mathbb{A}^n(K)$ is **Zariski-closed (Z-closed)** if $\mathcal{X}$ is affine algebraic.

Proposition 10. *Let $\mathcal{X} \subseteq \mathbb{A}^n(K)$. Let*

$$\bar{\mathcal{X}} := \bigcap_{\mathcal{X} \subseteq \mathcal{Y}, \mathcal{Y} \text{ } Z\text{-closed}} \mathcal{Y}$$

*be the **Z-closure** of $\mathcal{X}$. Then $\bar{\mathcal{X}} = V(I(\mathcal{X}))$.*

Definition 11. Let $\mathcal{X} \subseteq \mathcal{Y} \subseteq \mathbb{A}^n(K)$.
We say $\mathcal{X}$ is **dense in** $\mathcal{Y}$ if $\bar{\mathcal{X}} = \mathcal{Y}$.
We say $\mathcal{X}$ is **quasicompact** if every covering of $\mathcal{X}$ by open subsets has a finite subcovering (i.e. if $\mathcal{X} = \cup_{\alpha \in A} U_\alpha, U_\alpha$ open, then there is $B \subseteq A$ finite such that $\mathcal{X} = \cup_{b \in B} U_b$)

Theorem 12. *$\mathcal{X} \subseteq \mathbb{A}^n(K)$ closed $\implies$ $\mathcal{X}$ is quasicompact.*

Definition 13. A space $\mathcal{Y}$ is **reducible** there are some $\mathcal{Y}_1, \mathcal{Y}_2 \subseteq \mathcal{Y}$ closed in $\mathcal{Y}$ such that $\mathcal{Y} \neq \mathcal{Y}_1, \mathcal{Y} \neq \mathcal{Y}_2$ and $\mathcal{Y} = \mathcal{Y}_1 \cup \mathcal{Y}_2$. Otherwise we call $\mathcal{Y}$ **irreducible.**

$\mathcal{X}$ is **Noetherian** if every descending chain of closed subsets stabilizes.
$\dim \mathcal{X} := \sup\{r \in \mathbb{N} \cup \{\infty\} \mid \exists \mathcal{Y}_0 \subsetneq \mathcal{Y}_1 \subsetneq \cdots \subsetneq \mathcal{Y}_r$ where $\mathcal{Y}_i$ are irreducible closed $\}$.

Theorem 14. *$\mathcal{X}$ Noetherian topological space, $\mathcal{Y} \subseteq \mathcal{X}$ closed subset $\implies$ $\mathcal{Y} = \cup_{i=1}^r \mathcal{Y}_i$ where each $\mathcal{Y}_i$ is irreducible and closed and $\mathcal{Y}_i \not\subseteq \mathcal{Y}_j$ for $i \neq j$.*

Note 15. This is unique up to order. It is called the decomposition into irreducible components.

Proof. Let $A := \{\mathcal{Y} \subseteq \mathcal{X}$ closed $\mid \mathcal{Y}$ violates the statement$\}$.
$A \neq \varnothing \implies A$ contains a minimal element $\mathcal{Z}$. $\mathcal{Z}$ is not irreducible. So $\mathcal{Z} = \mathcal{Z}' \cup \mathcal{Z}'' \implies \mathcal{Z}', \mathcal{Z}'' \subsetneq \mathcal{Z}$. So $\mathcal{Z}', \mathcal{Z}'' \in A \implies \mathcal{Z}' = \bigcup \mathcal{Z}'_j$ and $\mathcal{Z}'' = \bigcup \mathcal{Z}''_k$. So $\mathcal{Z}' = (\bigcup \mathcal{Z}'_j) \cup (\bigcup \mathcal{Z}''_j) \rightarrow\leftarrow$. □

Proposition 16. *Let $\mathcal{X} \subseteq \mathbb{A}^n(K)$ be affine algebraic. Then*
$\mathcal{X}$ is irreducible $\iff$ $I(\mathcal{X})$ is prime.

Proof. We will show that $\mathcal{X}$ is reducible $\iff$ $I(\mathcal{X})$ is not prime.
($\Rightarrow$) : Suppose $\mathcal{X} = \mathcal{X}_1 \cup \mathcal{X}_2$, where $\mathcal{X}_1, \mathcal{X}_2$ are proper and closed.
Take $f \in I(\mathcal{X}_1) \setminus I(\mathcal{X})$ and $g \in I(\mathcal{X}_2) \setminus I(\mathcal{X})$.
Then $fg \in I(\mathcal{X})$ so $I(\mathcal{X})$ is not prime.
($\Leftarrow$) : If $fg \in I(\mathcal{X})$ but $f \notin I(\mathcal{X})$ and $g \notin I(\mathcal{X})$, define

$$\mathcal{X}_1 = V(I(\mathcal{X}) + (f)), \quad \mathcal{X}_2 = V(I(\mathcal{X}) + (g))$$

These are proper closed sets. $x \in \mathcal{X} \implies fg(x) = 0 \implies f(x) = 0$ or $g(x) = 0 \implies x \in \mathcal{X}_1$ or $x \in V_2 \implies \mathcal{X} = \mathcal{X}_1 \cup \mathcal{X}_2$. □

Example 17. Find irreducible components of $V(x^2, xy)$:
$(x^2, xy) = (x) \cap (x, y)^2$ and $V(x^2, xy) = V(x)$ is irreducible.

Example 18. 1. If $f \in K[x_1, \ldots, x_n]$ and $f = c \cdot \prod_{i=1}^{r} f_i^{e_i}$ with f_i distinct prime and c a unit, then

$$V(f) = \bigcup V(f_i)$$

is the decomposition into irreducible components.

2. Say $I \trianglelefteq K[x_1, \ldots, x_n]$ and $I = Q_1 \cap \cdots \cap Q_r$ is a primary decomposition, then

$$V(I) = \bigcup V(Q_i)$$

If I is radical, then each Q_i minimal prime and this is the decomposition into irreducible components.

Example 19. Irreducible components of $V(xy, x - yz)$:
$x = 0 \implies yz = 0 \implies$ either $y = 0$ or $z = 0$
$x \neq 0 \implies y = 0 \rightarrow\leftarrow$
$V(xy, x - yz) = V(x, y) \cup V(y, z)$

Example 20. $\mathcal{X} = V(xz - y^2, x^3 - yz) \subseteq \mathbb{A}^3(\mathbb{C})$
$x = 0 \implies y = 0 \implies$ the z-axis is contained in $\mathcal{X}$
$x \neq 0 \implies z = \frac{y^2}{x}, \quad x^3 = yz = \frac{y^3}{x} \implies x = (\frac{y}{x})^3$.
Define $t = \frac{y}{x}$ So $x = t^3, y = t^4, z = t^5$.
$C = \{(t^3, t^4, t^5) \mid t \in \mathbb{C}\}$ is a curve (the image of $\mathbb{A}^1(\mathbb{C}) \to \mathbb{A}^3(\mathbb{C}), t \mapsto (t^3, t^4, t^5)$).
If $\varphi : \mathbb{A}^m(\mathbb{C}) \to \mathbb{A}^n(\mathbb{C})$ is a polynomial map, $\mathcal{X} \subseteq \mathbb{A}^m(\mathbb{C})$ irreducible affine $\implies$ $\varphi(\mathcal{X})$ is irreducible affine. So C is irreducible.
$\mathcal{X} = C \cup \{(0, 0, z) \mid z \in \mathbb{C}\}$ is the decomposition.

Prime avoidance:
$X_j \subseteq A^n$ irreducible $\iff$ $P_j \trianglelefteq K[x_1, \ldots, x_n]$ prime.
Let $f_1, \ldots, f_m \in K[x_1, \ldots, x_n]$ not all vanish on any X_j. $(f_1, \ldots, f_m) \not\subseteq P_j \implies$ there exists $g \in (f_1, \ldots, f_m)$ which does not vanish on any $X_j, (f_1, \ldots, f_m) \not\subseteq \cup P_j$.

2.6.3 Coordinate rings

Assume $K = \bar{K}$.

Definition 21. Let $\mathcal{X} \subseteq \mathbb{A}^n(K)$ be an affine algebraic set.
The **coordinate ring** of $\mathcal{X}$ is

$$k[\mathcal{X}] := K[x_1, \ldots, x_n]/I(\mathcal{X})$$

A ring R of this form is **reduced** if $\operatorname{Nil} R = 0$.

Proposition 22.

1. *$\mathcal{X} \subseteq \mathbb{A}^n(K)$ affine algebraic subset $\implies$ $k[\mathcal{X}]$ is a reduced affine ring.*
2. *R reduced affine ring $\implies$ $\exists n, \mathcal{X} \subseteq \mathbb{A}^n(K)$ such that $R = k[\mathcal{X}]$.*

Proof. 1) We know $I(\mathcal{X})$ is a radical ideal. So $0 \leq k[\mathcal{X}]$ is radical hence $\operatorname{Nil} k[\mathcal{X}] = 0$.
2) $R = K[x_1, \ldots, x_n]/I$ is reduced $\implies I = \sqrt{I}$ since R is reduced. Take $\mathcal{X} := V(I)$. □

Example 23. 1. $k[\mathbb{A}^n(K)] = K[x_1, \ldots, x_n]$.

2. $k[(\alpha_1, \ldots, \alpha_n)] = K[x_1, \ldots, x_n]/(x_1 - \alpha_1, \ldots, x_n - \alpha_n) \cong K$

3. $\mathcal{X} = V(xy - 1) \subseteq \mathbb{A}^2(K) \implies k[\mathcal{X}] = K[z, z^{-1}]$

Proposition 24. *$\mathcal{X}$ affine* $\implies \dim \mathcal{X} = \dim(k[\mathcal{X}])$

Proof.

$$\mathcal{Y} \subseteq \mathcal{X} \text{ irreducible and algebraic}$$

$$\updownarrow$$

$$I(\mathcal{Y}) \supseteq I(\mathcal{X}) \text{ prime ideal}$$

$$\updownarrow$$

$$\text{prime ideals in } k[\mathcal{X}]$$

□

2.6.4 Example: Topological proof of Cayley-Hamilton

Again assume $K = \bar{K}$.

Theorem 25 (Cayley-Hamilton)**.** $A \in \operatorname{Mat}_n(K)$.
$p_A(t) := \det(tI - A)$ *characteristic polynomial* $\implies p_A(A) = 0$.

Lemma 26. *Let* $D_n := \{A \in \operatorname{Mat}_n(K) \mid$ *all eigenvalues of* A *are distinct*$\}$. *Then* $D_n \subseteq \operatorname{Mat}_n(K)$ *is Zariski dense.*

Proof. (Sketch) $D_n \neq \varnothing$. Not hard to show: D_n is open (matrices with repeated eigenvalues form an affine algebraic set). But any Zariski open (nonempty) subset of $\mathbb{A}^n(K)$ is dense. □

Proof. (Of Theorem 1) Let $\varphi : \mathbb{A}^{n^2} \to \mathbb{A}^{n^2}, A \mapsto p_A(A)$. Then φ is Z-continuous.
$D_n \subseteq \mathbb{A}^{n^2}$ dense $\implies$ sufficient to show $\varphi = 0$ on D_n.

But $p_{T^{-1}AT}(T^{-1}AT) = T^{-1}p_A(A)T$ (check), so it is sufficient to show result for diagonal matrices with distinct eigenvalues, which is clear.

□

2.7 Introduction to category theory

2.7.1 Categories

Definition 1. A category consists of the following:

1. A class of **objects** $\mathrm{Obj}\,\mathcal{C}$.
2. For every pair of objects $X, Y \in \mathrm{Obj}\,\mathcal{C}$ a class $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ (or $\mathrm{Mor}_{\mathcal{C}}(X, Y)$) of **morphisms** from X to Y; write $f : X \to Y$.
3. Composition maps $\mathrm{Hom}_{\mathcal{C}}(Y, Z) \times \mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{C}}(X, Z)$ with $(f, g) \mapsto f \circ g$ for $X, Y, Z \in \mathrm{Obj}\,\mathcal{C}$

subject to the following axioms:

1. $(f \circ g) \circ h = f \circ (g \circ h)$.
2. For each $X \in \mathrm{Obj}\,\mathcal{C}$ there is an identity $1_X \in \mathrm{Hom}_{\mathcal{C}}(X, X)$ such that for all f, g where defined, $1_X \circ f = f$ and $g \circ 1_X = g$.

Note 2. 1. 1_x is called the **identity morphism on X.**

2. We often write $X \in \mathcal{C}$ instead of $X \in \mathrm{Obj}\,\mathcal{C}$.
3. We assume $\mathcal{C}$ is locally small, i.e. each $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ is a set.

Example 3. 1. **Set** (sets with arbitrary maps)

2. **Grp**, **Ring** (groups, rings with homomorphisms)
3. **Top** (topological spaces with continuous maps)
4. If $|\mathrm{Obj}\,\mathcal{C}| = 1$ then the category is just a monoid.

Definition 4. Let $\mathcal{C}$ be a category and $X, Y \in \mathrm{Obj}\,\mathcal{C}$. $f \in \mathrm{Hom}_{\mathcal{C}}(X, Y)$ is a **monomorphism** if for any object Z and any $g, h \in \mathrm{Hom}_{\mathcal{C}}(Z, X)$ we have $f \circ g = f \circ h \iff g = h$. (i.e. we can cancel f as the second arrow in composition.)

A morphism $f \in \mathrm{Hom}_{\mathcal{C}}(X, Y)$ is an **epimorphism** if for any object Z and any $g, h \in \mathrm{Hom}_{\mathcal{C}}(Y, Z)$ we have $g \circ f = h \circ f \iff g = h$. (i.e. can cancel f as the first arrow in composition.)

A morphism $f \in \mathrm{Hom}_{\mathcal{C}}(X, Y)$ is an **isomorphism** if there exists $g \in \mathrm{Hom}_{\mathcal{C}}(Y, X)$ such that $g \circ f = 1_X$ and $f \circ g = 1_Y$.

Example 5. 1. In **Grp**: if f is mono, f is injective:

If $f : X \to Y$ is a group homomorphism and mono but not injective then $\ker f \neq 1$. So let $g \in \mathrm{Hom}(\ker f, X) : g(x) = x$ for all $x \in \ker f$ and let $h \in \mathrm{Hom}(\ker f, X) : h(x) = 1$ for all $x \in \ker f$. Then $g \neq h$ since $1 \neq \ker f$. $f \circ g = f \circ h$. But then f mono $\implies g = h \to\leftarrow$.

2. In **Ring**, epi need not be surjective: e.g. the inclusion $\mathbb{Z} \to \mathbb{Q}$ (any homomorphism $\mathbb{Q} \to X$ is determined by what it does to $\mathbb{Z}$)

3. In general: iso $\Rightarrow$ epi and mono.

4. Epi and mono $\not\Rightarrow$ iso.

Note 6. In many categories, isomorphism = bijective morphism. But in many, bijective morphism is not enough.

2.7.2 Functors

Definition 7. A **functor** $F : \mathcal{C} \to \mathcal{D}$ between categories is:

1. A map on objects: $\mathrm{Obj}\,\mathcal{C} \to \mathrm{Obj}\,\mathcal{D}$.

2. For each $X, Y \in \mathrm{Obj}\,\mathcal{C}$ a map $F = F_{X,Y} : \mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{D}}(F(X), F(Y))$ which preserves composition and identity maps.

This is a *covariant* functor. There are also contravariant functors (which reverse arrows).

Example 8. 1. The forgetful functor $\mathbf{Grp} \to \mathbf{Set}, \mathbf{Ring} \to \mathbf{AbGroup}$, etc.

2. The abelianization functor $G \mapsto G^{\mathrm{ab}}$ sending $G \mapsto G^{ab} := G/[G, G]$

3. For a fixed object $X \in \mathcal{C}$, the Hom-functor $\mathrm{Hom}_{\mathcal{C}}(X, -) : \mathcal{C} \to \mathbf{Set}$ sending $Y \mapsto \mathrm{Hom}_{\mathcal{C}}(X, Y)$

Definition 9. Given a category $\mathcal{C}$, the **opposite category** $\mathcal{C}^{op}$ has the same objects and $\mathrm{Hom}_{\mathcal{C}^{op}}(X, Y) = \mathrm{Hom}_{\mathcal{C}}(Y, X)$. A **contravariant functor** $\mathcal{C} \to \mathcal{D}$ is a covariant functor $\mathcal{C}^{op} \to \mathcal{D}$.

Example 10. 1. $\mathrm{Hom}_{\mathcal{C}}(-, Y) : \mathcal{C} \to \mathbf{Set}$

2. $X \mapsto \mathrm{Maps}(X, \mathbb{Z})$ is a functor $\mathbf{Set} \to \mathbf{Ring}^{op}$

3. $F : \mathbf{Vect_K} \to \mathbf{Vect_K}^{op}, V \mapsto V^*, f \mapsto f^*$ (dual) is a contravariant functor on $\mathbf{Vect}_k$.

2.7.3 Natural transformations and equivalence of categories

Definition 11. Let $F, G : \mathcal{C} \to \mathcal{D}$ be functors. A **natural transformation or morphism** $\alpha : F \to G$ is a collection of morphisms

$$\{\alpha_X : F(X) \to G(X)\}_{X \in \mathrm{Obj}\mathcal{C}}$$

which is **functorial** in X, meaning for every $f : X \to Y$ in $\mathcal{C}$ the diagram

$$\begin{array}{ccc} F(X) & \xrightarrow{F(f)} & F(Y) \\ {\scriptstyle \alpha_X}\downarrow & & \downarrow{\scriptstyle \alpha_Y} \\ G(X) & \xrightarrow[G(f)]{} & G(Y) \end{array}$$

commutes, i.e. $\alpha_Y \circ F(f) = G(f) \circ \alpha_X$.

α is an **isomorphism** of functors if each α_X is an isomorphism in $\mathcal{D}$.

Example 12. Let $\mathcal{C} = \mathrm{Vect}_{\mathrm{K}}$. Then $\mathrm{id} \cong **$ as functors $\mathcal{C} \to \mathcal{C}$.

Example 13. Define two categories $\mathcal{C}_1$ and $\mathcal{C}_2$ so that:
$\mathrm{Obj}\,\mathcal{C}_1 = \{X\}, \quad \mathrm{Hom}_{\mathcal{C}_1}(X,X) = \{1_X\}$
$\mathrm{Obj}\,\mathcal{C}_2 = \{X, Y\}$, and
$\mathrm{Hom}_{\mathcal{C}_2}(X,X) = \{1_X\}, \quad \mathrm{Hom}_{\mathcal{C}_2}(Y,Y) = \{1_Y\}, \quad \mathrm{Hom}_{\mathcal{C}_2}(X,Y) = \{a\}, \quad \mathrm{Hom}_{\mathcal{C}_2}(Y,X) = \{b\}$
with $a \circ b = 1_Y, \quad b \circ a = 1_X$

If $\mathcal{D}$ is any category then $F(X) \cong G(X)$ via $\varphi \implies F(Y) \cong G(Y)$ via $G(a) \circ \varphi \circ F(b)$.

Definition 14. Two categories $\mathcal{C}$ and $\mathcal{D}$ are **equivalent** if there exist functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ and natural isomorphisms $GF \cong \mathrm{Id}_{\mathcal{C}}$, $FG \cong \mathrm{Id}_{\mathcal{D}}$. In this case G is a **quasi-inverse** to F (and vice versa).

Example 15. $\mathcal{C}_1$ and $\mathcal{C}_2$ are equivalent categories.
Define $F : \mathcal{C}_1 \to \mathcal{C}_2, X \mapsto X, 1_X \mapsto 1_X$.
Define $G : \mathcal{C}_2 \to \mathcal{C}_1, X \mapsto X, Y \mapsto X, f \mapsto 1_X \forall$ morphisms f in $\mathcal{C}_2$
One can check: G is a quasi-inverse.

Example 16.
1. $\mathbf{Vect}_k$ is equivalent to the category $\mathcal{C}$ with $\mathrm{Obj}\,\mathcal{C} = \mathbb{N}, \mathrm{Hom}(m,n) := K^{n \times m}$ matrix multiplication
2. Category of affine algebraic sets with "polynomial maps" is equivalent to AlgRed^{op}, the opposite of the category of reduced affine algebras.

2.7.4 Representable functors

Definition 17. A (covariant) functor $F : \mathcal{C} \to \mathbf{Set}$ is **representable** if $F \cong \mathrm{Hom}(X, -)$ for some $X \in \mathrm{Obj}\,\mathcal{C}$; then we say F is **represented by X.**

Similar definition for contravariant functors.

Lemma 18 (Yoneda Lemma)**.** *If a functor is represented by an object, that object is unique up to unique isomorphism.*

Proof. (Sketch) WTS: If $X, Y \in \mathcal{C}$ so that $\mathrm{Hom}_{\mathcal{C}}(X, -) \cong \mathrm{Hom}(Y, -)$ via φ then there exists a unique isomorphism $\alpha_\varphi : X \to Y$ which induces φ.

From φ, we evaluate at X to get $\mathrm{Hom}(X,Y) \cong \mathrm{Hom}(Y,Y)$ via some φ_Y and $\mathrm{Hom}(X,X) \cong \mathrm{Hom}(Y,X)$ via some φ_X.

Let $\alpha_\varphi := \varphi_Y^{-1}(1_Y)$ and $\beta\varphi := \varphi_X^{-1}(1_X)$. One can check: α_φ is an isomorphism with inverse β_φ.

Exercise: uniqueness. □

Example 19. 1. Power set: $F : \mathbf{Set} \to \mathbf{Set}, X \mapsto \mathcal{P}(X)$

This is a contravariant functor:

$F(-) \cong \mathrm{Hom}(-, \{0,1\})$ via (for every set A)
$\varphi_X : \mathrm{Hom}(X, A) \to \mathcal{P}(X), \quad f \mapsto \{x \in X \mid f(x) = 1\}$

2. For a fixed set A, $X \mapsto \mathrm{Hom}(X, A)$ is a contravariant functor.

3. Forgetful functor $F : \mathbf{Grp} \to \mathbf{Set}$

This is a covariant functor:

$\mathrm{Hom}(\mathbb{Z}, X) = F(X)$ via $f \mapsto f(1)$

2.7.5 Adjoint functors

Definition 20. Let $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$.
F, G are a pair of **adjoint functors** if for all $X \in \mathcal{C}, Y \in \mathcal{D}$ we are given isomorphisms

$$\zeta_{X,Y} : \mathrm{Hom}_{\mathcal{D}}(F(X), Y) \to \mathrm{Hom}_{C}(X, G(Y))$$

functorial in X and Y, i.e. we have an isomorphism of functors $\mathrm{Hom}_{\mathcal{D}}(F(-), -) \to \mathrm{Hom}_{C}(-, F(-))$ as functors $\mathcal{C} \times \mathcal{D} \to \mathbf{Set}$.

We say F is **left adjoint to** G and G is **right adjoint to** F. Note: By Yoneda, adjoints are unique if they exist.

Example 21. 1. $G : \mathbf{AbGrp} \to \mathbf{Grp}$ (forgetful functor) has abelianization as left adjoint.

$F : \mathbf{Grp} \to \mathbf{AbGrp}, H \mapsto H^{ab} = H/[H, H]$

So for any group H and any abelian group A,

$$\mathrm{Hom}(H^{ab}, A) \cong \mathrm{Hom}(H, A)$$

2. The forgetful functor $\mathbf{Field} \to \mathbf{Dom}$ has a left adjoint the field of fractions functor

3. The forgetful functor $G : \mathbf{Ring}\bullet \to \mathbf{Ring}$ has as its left adjoint
$F : \mathbf{Ring} \to \mathbf{Ring}\bullet, R \mapsto (R[x], x)$

So $F(R) := (R[x], x])$ and then $\mathrm{Hom}(R, A) \cong \mathrm{Hom}((R[x], x), (A, a))$.

3 Linear Algebra

3.1 Introduction

3.1.1 Vector spaces

Let K be a field. A **vector space** is $(V, +, \cdot)$ where $(V, +)$ is an abelian group and $\cdot : K \times V \to V$ scalar multiplication, satisfying a list of axioms.

Example 1. 1. 0 zero vector space

2. $(K, +, \cdot)$ is a K-vector space

3. $(\mathbb{C}, +, \cdot)$ is an $\mathbb{R}$-vector space

4. M set: $K^M := \{f : M \to K\}$ is a K-vector space (write K^n for $|M| = n$)

$W \subseteq V$ is a **subspace** if W is a vector space with respect to the induced operations ($\iff$ it is closed under addition and scalar multiplication). Notation: $W \leq V$.

3.1.2 Constructions

1. Intersection of subspaces is a subspace

2. $S \subseteq V$ subset of K-vector space:

$$\langle S \rangle = \bigcap_{W \leq V | S \subseteq W} W$$

   is the **subspace generated by S**

   If $\langle S \rangle = V$ we say **S generates V**

3. A **linear combination** of elements of S is $v \in V$ of the form

$$v = \sum_{i=1}^{n} a_i s_i$$

   for some n, $a_i \in K, s_i \in S$

4. If $W_i \subseteq V$ for $i \in I$:

$$\langle \cup W_i \rangle =: \sum W_i$$

   vector space sum

5. If $S \neq \varnothing$ then $\langle S \rangle$ is the set of all linear combinations of elements of S

   $\text{Span}(\varnothing) = 0$

6. $w \in \sum_{i \in I} W_i \implies w = \sum_{i \in I} w_i, w_i \in W_i$, and this sum is finite

7. The set of subspaces of V with $\leq, \cap, \sum$ forms a modular lattice

3.1.3 Direct sum and complements

1. If V is a K-vector space and $W_i \leq V$ for $i \in I$, then V is the **direct sum** of the W_i if

   (a) $V = \sum W_i$

   (b)

$$i \in I \implies W_i \cap \sum_{j \in I \setminus \{i\}} W_j = \{0\}$$

Notation:

$$V = \bigoplus_{i \in I} W_i$$

Exercise: Find UMP for $\bigoplus$

2. $V = \oplus W_i \iff$ every element in V has a *unique* representation $v = \sum_{i \in I} w_i$' $(w_i \in W_i)$
   (Note: ' = all but finitely many terms in the sum are zero)

3. If $W \leq V$ then $X \leq V$ is a **complement** to $W \iff X \oplus W = V$ (Not always unique but always exists)

3.1.4 Linear independence and basis

Let $S = \{s_1, \ldots, s_n\} \subseteq V$ be finite. S is **linearly independent** if for any $a_1, \ldots, a_n \in K$ so that $\sum a_i n_i = 0$, we must have $a_1 = \cdots = a_n = 0$.
$S \subseteq V$ arbitrary is **linearly independent** if every finite subset is linearly independent. Otherwise $S \subseteq V$ is **linearly dependent**

Note 2.
1. We say "$s_1, \ldots, s_n$" are linearly (in)dependent

2. $\varnothing$ is linearly independent

3. S is linearly independent $\iff$ every vector in $\langle S \rangle$ is a unique linear combination of elements in S

4. A **basis** of V is a linearly independent generating set

   $S \subseteq V$ is a basis $\iff$ S is a minimal generating set $\iff$ S is a maximal linearly independent set $\iff$ every element in V is a unique linear combination of elements of S.

   Every vector space has a basis (Zorn's).

   - In fact, every linearly independent subset can be extended to a basis
   - Every generating subset contains a basis $\implies$ any linearly independent subset can be extended to a basis using elements from a given generating subset

5. If $V = \langle S \rangle$ for S finite and $T \subseteq V$ linearly independent $\implies |T| \leq |S|$. So every basis has the same number of elements (finite or infinite). This is called **dimension**. Notation: $\dim V$.

6. $W \leq V \implies \dim W \leq \dim V$ and $\dim W < \dim V$ if $W \neq V$. $\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2)$

3.2 Homomorphisms

3.2.1 Linear maps

1. V, W K-vector spaces. $\varphi : V \to W$ is K-linear or a K-homomorphism if $\varphi(av_1 + v_2) = a\varphi(v_1) + \varphi(v_2)$ for all $a \in K, v_1, v_2 \in V$
2. $\varphi(V) \leq W$ is called the image and $\ker \varphi = \varphi^{-1}(0) \leq V$ is called the kernel

   φ is injective $\iff \ker \varphi = 0$

3.2.2 Isomorphism Theorem

1. Let V be a K-vector space and $S = \{s_1, \ldots, s_n\} \subseteq V$ a basis for V. Let W be a K-vector space and $T = \{t_1, \ldots, t_n\}$ a subset of W. Then there exists a unique homomorphism $\varphi : V \to W, s_i \mapsto t_i$

   φ is surjective $\iff \langle T \rangle = W$

   φ is injective $\iff T$ is linearly independent

   Hence, φ is bijective $\iff T$ is a basis
2. $V \cong W \iff \dim V = \dim W$

3.2.3 Congruence relations and quotients

1. Let V be a vector space. An equivalence relation $\equiv$ is a **congruence relation** if $v \equiv \tilde{v}, w \equiv \tilde{w}$ means $v + w \equiv \tilde{v} + \tilde{w}$ and $av \equiv a\tilde{v}$ for all $a \in K$
2. A congruence relation $\equiv$ on V gives a subspace $W := \{v \in V \mid v \equiv 0\} \leq V$
3. Similarly, $W \leq V$ defines $X \equiv Y \iff x - y \in W$
4. V with $\equiv$ gives $\bar{V} := V/W$

   $W = \{v \in V \mid v \equiv 0\}$ **quotient space**

   $\pi : V \to V/W$ canonical homomorphism
5. Homomorphism theorem:

   If $\varphi \in \mathrm{Hom}(V, W) \implies V/\ker \varphi \cong \varphi(V)$

   This isomorphism is induced by φ

$$\begin{array}{ccc} V & \xrightarrow{\varphi} & W \\ \downarrow & & \uparrow \\ V/\ker \varphi & \xrightarrow{\bar{\varphi}} & \varphi(V) \end{array}$$

   $\bar{\varphi}(\bar{v}) = \varphi(v)$ is well-defined; $\bar{v} := v + \ker \varphi$
6. $\dim V/W = \dim V - \dim W$ if $\dim V < \infty$

7. $\dim \varphi(V) = \dim V - \dim \ker \varphi$ if $\dim V < \infty$ (rank-nullity)

8. $\mathrm{rk}(\varphi) := \dim \varphi(V)$ **rank of** φ

3.2.4 $\mathrm{Hom}_K(V, W)$

1. $\mathrm{Hom}_K(V, W)$ is a K-vector space

2. A K-vector space $(A, +, \cdot)$ together with a map $\circ : A \times A \to A$ is a K**-algebra** if

   - $(A, +, \circ)$ is a ring with identity
   - $a \in K, v, w, \in A \implies a \cdot (v \circ w) = (a \cdot v) \circ w$

   $\varphi : A \to B$ is an **algebra homomorphism** φ is a vector space homomorphism and $\varphi(a \circ b) = \varphi(a) \circ \varphi(b), \quad \varphi(1_A) = 1_B$ for all $a, b \in A$

   Example 1.
   - $\mathrm{End}_K(V) := \mathrm{Hom}_K(V, V)$ set of **endomorphisms** is a K-algebra and $\circ$-multiplication is composition
   - $K[x_1, \ldots, x_n]$ is a K-algebra

3. $\mathrm{GL}(V) = \mathrm{Isom}_K(V, V) = \{\varphi \in \mathrm{End}_K(V) \mid \varphi \text{ isomorphism } \}$ **general linear group**

3.3 Coordinates and matrices

$\dim_K V < \infty$. New convention: $S = (s_1, \ldots, s_n)$ **ordered basis of V**.

1. $v \in V, v = \sum_{i=1}^n a_i s_i \implies v_s = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$ **coordinate vector with respect to S**

   $a_i =$ **coordinate of v** with respect to S

   $\gamma_S : V \to K^n, \quad v \mapsto v_s$ **coordinate representation of V with respect to S** (this is an isomorphism by the isomorphism theorem)

2. $S = (s_1, \ldots, s_n), T = (t_1, \ldots, t_n)$

$$s_j = \sum_{i=1}^n c_{ij} t_i, \quad j = 1, \ldots, n$$

$$C_T^S := \begin{pmatrix} c_{11} \ldots c_{1n} \\ \ldots \\ c_{n1} \ldots c_{nn} \end{pmatrix} \in \mathrm{Mat}_n(K)$$

   base change matrix from S **to** T

   Then $V_T = C_T^S V_S, \quad C_T^S C_S^T = C_S^T C_T^S = I$

3. $\varphi : V \to W$ linear

   $S = (s_1, \ldots, s_n)$ basis of V

   $T = (t_1, \ldots, t_m)$ basis of W

   $\varphi(s_j) = \sum_{i=1}^{m} d_{ij} t_i, \quad j = 1, \ldots, n$

   $D_T^S := (d_{ij})$ representation matrix of φ with respect to S and T

4. $$\begin{array}{ccc} V & \xrightarrow{\varphi} & W \\ {\scriptstyle \gamma_V}\downarrow & & \downarrow{\scriptstyle \gamma_T} \\ K^n & \xrightarrow{D_T^S(\varphi)} & K^m \end{array}$$

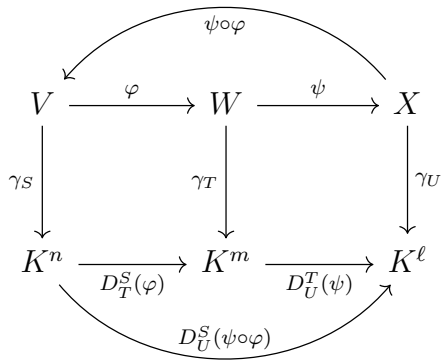
   $\varphi(v)_T = D_T^S(\varphi) v_S$

5. $D_T^S : \mathrm{Hom}_K(V, W) \to K^{m \times n}, \quad \varphi \mapsto D_T^S(\varphi)$ is an isomorphism of K-vector spaces

6. V, W, X vector spaces with bases S, T, U. $\varphi \in \mathrm{Hom}_K(V, W), \psi \in \mathrm{Hom}_K(W, X)$. Then

$$D_U^S(\psi \circ \varphi) = D_U^T(\psi) D_T^S(\varphi)$$

Note 1. If V is a vector space over K, $\dim V = n < \infty \iff V \cong K^n$



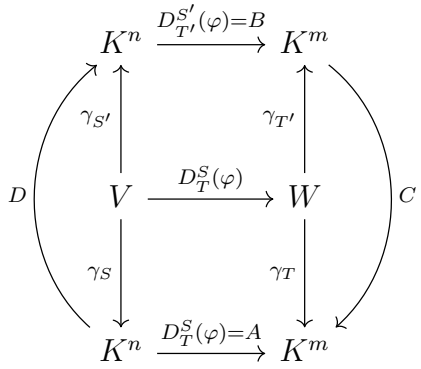
Vector space isomorphism $D_S^T : \mathrm{Hom}_K(V, W) \to K^{m \times n}$
K-algebra isomorphism $D_S = D_S^S : \mathrm{End}_K(V) \to K^{n \times n}$
Group isomorphism $D_S|_{\mathrm{GL}(V)} : \mathrm{GL}(V) \to \mathrm{GL}_n(K)$

3.3.1 Equivalence and rank of matrices

1. $A, B \in K^{m \times n}$. We say A is **equivalent to** B if there exists $C \in \mathrm{GL}_m(K), D \in \mathrm{GL}_n(K)$ so that $B = CAD$. Equivalent matrices represent the same linear map with respect to different (pairs of) bases.

2. $A, B \in K^{n \times n}$. We say A is **similar to** B if there exists $C \in \mathrm{GL}_n(K)$ so that $CAC^{-1} = B$. Similar matrices represent the same endomorphisms with respect to different bases.

   Both of these define equivalence relations.



For (2), in the diagram above take $D = C^{-1}$.

3. $\mathrm{rk}A = \mathrm{rk}\varphi$ if $A = D_T^S(\varphi)$ = number of linearly independent columns

   $A \in K^{m\times n} \implies A$ is equivalent to a unique matrix of the form

$$\left(\begin{array}{ccc|c} 1 & & 0 & \\ & \ddots & & \\ 0 & & 1 & \\ \hline & & & \end{array}\right)$$

   (rank normal form)

   Proof idea: $A = D_T^S(\varphi), \quad \varphi : V \to W$. Pick a complement X to $\ker\varphi$ in V. $V = X \oplus \ker\varphi$. Take ordered basis S of V. Then $\dim X = \mathrm{rk}(\varphi)$.

   $T = (\varphi(s_1), \ldots, \varphi(S_{\mathrm{rk}\varphi}), \cdots$ basis for complement of $\varphi(V))$.

   $V = X \oplus \ker\varphi \to W = \varphi(V) \oplus U, \quad S \to U$

   So $K^{m\times n}$ has $\min\{m, n\} + 1$ equivalence classes.

4. Elementary operations with elementary matrices

$$V_{ij} = \begin{pmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 0 & \cdots & 1 & & \\ & & & & & & \\ & & \vdots & \ddots & \vdots & & \\ & & 1 & \cdots & 0 & & \\ & & & & & & \\ & & & & & \ddots & \end{pmatrix} \begin{matrix} \\ \\ \\ \\ \\ \\ \\ \\ \end{matrix} \begin{matrix} \\ \\ \\ \\ \\ \\ \\ \\ \end{matrix}$$

   Multiplying with V_{ij} from the left switches the i-th and j-th row

   Multiplying with V_{ij} from the right switches the i-th and j-th column

$$A_{ij}(\alpha) = \begin{pmatrix} 1 & & & & & & \\ & \ddots & & & & & \\ & & 1 & \cdots & \underset{(i,j)}{\alpha} & & \\ & & & \ddots & & & \\ & & & & & 1 & \end{pmatrix}, \quad \alpha \in K$$

Multiplying with $A_{ij}(\alpha)$ on the left adds the i-th to α times the j-th row

$$\mu_i(\beta) = \begin{pmatrix} 1 & & & & \\ & \ddots & & & \\ & & \underset{(i,i)}{\beta} & & \\ & & & \ddots & \\ & & & & 1 \end{pmatrix}, \quad \beta \in K \setminus \{0\}$$

Multiplying with $\mu_i(\beta)$ on the left multiplies the i-th row by β

- $\operatorname{rk}(A)$ is invariant under elementary operations
- A finite number of elementary operations transforms A into rank normal form
- Gauss algorithm: only left operations
- Every invertible matrix is a product of elementary matrices

3.4 Linear operators

3.4.1 Linear forms and dual spaces

$V^* = \operatorname{Hom}_K(V, K)$ is called the **dual space** of a vector space V
$\dim V = \dim V^*$ if $\dim V < \infty$
$\dim V^* = \dim(\operatorname{Hom}_K(V, K)) = \dim V \dim K = \dim V$
$S = (s_1, \ldots, s_n)$ basis

Dual basis: $s_i^*(s_j) = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & \text{otherwise} \end{cases}$
$V \cong V^*$ if $\dim V < \infty$

Example 1. $V = \mathbb{R}^2, S = (e_1, e_2)$ standard basis, $T = (v_1, v_2)$ where $v_1 = e_1, v_2 = e_1 + e_2$.
To express v_1^*, v_2^* in terms of e_1^*, e_2^* we observe that

- $e_1^*(e_1) = 1, \quad e_1^*(e_2) = 0$
- $e_2^*(e_1) = 0, \quad e_2^*(e_2) = 1$
- $v_1^*(e_1) = 1, \quad v_2^*(e_1 + e_2) = 1$

So $v_1^* = e_1^* - e_2^*$ and $v_2^* = e_2^*$

Note 2. 1. $S^* = \{s^* \mid s \in S\}$ (where S is a basis of V) is linearly independent even if $\dim V = \infty$

2. If $\dim V = \infty$ then $V \ncong V^*$ (in fact $\dim V^* > \dim V$)

   We will show something weaker: if S is a basis of V then S^* does not span V^*

   Say $S = (s_i \mid i \in I)$. Define $\varphi : V \to K$ linear, so that $\varphi(s_i) = 1 \forall i \in I$. As in the isomorphism theorem, this defines φ uniquely.

   Suppose $\varphi = \sum_{j \in J} a_j s_j^*$ for some $J \subseteq I$ finite.

   $i \notin J \implies \varphi(s_i) = \sum_{j \in J} a_j s_j^*(s_i) = 0 \to\leftarrow$ contradicts choice of map.

Example 3. Let $U \leq \mathbb{R}^{\mathbb{N}} = \text{Maps}(\mathbb{N}, \mathbb{R}) = \{\text{sequences in } \mathbb{R}\}$ be a subspace with finitely many entries nonzero. This is countably infinite dimensional. But $U^* \cong \mathbb{R}^{\mathbb{N}}$ is uncountably infinite dimensional.

3.4.2 Duality theorems

Theorem 4. *If* $\dim V < \infty$, *then there exists a natural isomorphism*

$$\theta : V \to (V^*)^* =: V^{**} = \text{Hom}_K(V^*, K)$$

$$v \mapsto \theta(v)$$

where for $f \in V^*, \theta(v)(f) = f(v)$

Note: When $\dim V = \infty$, we can still define θ and it is injective.

Proof. Check that:

1. $\theta(v) \in \text{Hom}_K(V^*, K)$ i.e. is linear for all $v \in V$
2. θ is linear
3. θ is injective: Suppose $0 \neq v \in \ker \theta$. Take S a basis so that $v \in S.S^* = (s^* \mid s \in S)$
   $1 = v^*(v) = \theta(v)(v^*) = 0 \to\leftarrow$

Since the dimensions are the same, if $\dim V < \infty$ this is also surjective, hence an isomorphism. □

Notation: $f \in V^*, v \in V, f(v) =: \langle f, v \rangle \in K$.
So $\langle \cdot, \cdot \rangle : V^* \times V \to K$.
Then (if $f, g \in V^*, v, w \in V, a \in K$)

- $\langle f + g, v \rangle = \langle f, v \rangle + \langle g, v \rangle$
- $\langle f, v + w \rangle = \langle f, v \rangle + \langle f, w \rangle$
- $\langle af, v \rangle = a\langle f, v \rangle = \langle f, av \rangle$
- $\langle f, v \rangle = 0 \forall v \in V \implies f = 0$
- $\langle f, v \rangle = 0 \forall f \in V^* \implies v = 0$

3.4.3 Orthogonality

1. V K-vector space, V^* dual space: $v \in V, f \in V^*$ are **orthogonal** if $\langle f, v\rangle = 0$. Notation: $f \perp v$.

   $S \subseteq V \implies S^\perp = \{f \in V^* \mid \langle f, s\rangle = 0 \text{ for all } s \in S\}$
   orthogonal complement of S in V^*

   $T \subseteq V^* \implies T^\perp = \{v \in V \mid \langle f, v\rangle = 0 \text{ for all } f \in T\}$
   orthogonal complement of T in V

   Note 5. $S \subseteq V, S^\perp \leq V^*$ (only depends on $\operatorname{Span} S$)

   $T \subseteq V^*, T^\perp \leq V$

2. $\dim V = n < \infty$
   $S \subseteq \tilde{S} \subseteq V \implies \tilde{S}^\perp \leq S^\perp \leq V, \quad \langle S\rangle^\perp = S^\perp$

   $W \leq V \implies \dim W^\perp = \dim V - \dim W$

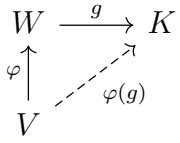
   $(W^\perp)^\perp = W$

   $W_1, W_2 \leq V \implies (W_1 \cap W_2)^\perp = W_1^\perp + W_2^\perp$ and $(W_1 + W_2)^\perp = W_1^\perp \cap W_2^\perp$

3.4.4 The dual of a linear map

Theorem 6. *If K is a field, V, W are K-vector spaces, and $\varphi \in \operatorname{Hom}_K(V, W)$, then there exists a unique $\varphi^* : W^* \to V^*$ satisfying $\langle \varphi^*(g), v\rangle = \langle g, \varphi(v)\rangle$ for all $g \in W^*, v \in V$. In particular, $*$ is a contravariant functor $\operatorname{Vect}_K \to \operatorname{Vect}_K$.*

Proof. $\varphi^*(g) = g \circ \varphi$
$g : W \to K, \quad \varphi^*(g) : V \to K$



- Linear
- Check: uniqueness

□

Theorem 7. *Let $\varphi \in \operatorname{Hom}_K(V, W), \varphi^* \in \operatorname{Hom}_K(W^*, V^*)$.*

1. $\ker \varphi^* = \operatorname{Im}(\varphi)^\perp$. *Hence φ^* is injective $\iff$ φ is surjective.*
   And $\ker \varphi = \operatorname{Im}(\varphi^*)^\perp$, *hence φ^* is surjective $\iff$ φ is injective.*

2. *If* $\dim V, \dim W < \infty, \operatorname{rk}(\varphi) = \operatorname{rk}(\varphi^*)$

Theorem 8. *Suppose V, W are finite-dimensional K-vector spaces with bases S, T and dual bases S^*, T^*. If $\varphi \in \operatorname{Hom}_K(V, W)$ and $\varphi^* \in \operatorname{Hom}_K(W^*, V^*)$,*

$$D_{S^*}^{T^*}(\varphi^*) = (D_T^S(\varphi))^{tr}$$

Observe that this forces $\operatorname{rk}\varphi = \operatorname{rk}(D_T^S(\varphi))$ = number of linearly independent columns to equal $\operatorname{rk}(\varphi^*)$ = number of linearly independent rows.

3.5 Tensor products

3.5.1 Universal property and existence

Definition 1. Let V, W, X be K-vector spaces. A map $\varphi : V \times W \to X$ is **K-bilinear** if it is linear in each component.

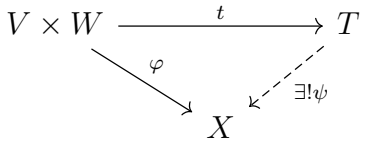
Example 2. 1. Dual pairing $V \times V^* \to K$

2. $\varphi : K^2 \times K^2 \to K, \quad \left(\begin{pmatrix} r \\ s \end{pmatrix}, \begin{pmatrix} t \\ u \end{pmatrix} \right) \mapsto ru - st$

Definition 3. Let V, W be K-vector spaces:

A **tensor product** of V, W is a K-vector space T together with a bilinear $t : V \times W \to T$ satisfying the UMP:

If $\varphi : V \times W \to X$ is bilinear for some K-vector space X, there is a unique $\psi \in \mathrm{Hom}_K(T, X)$ such that the following diagram commutes:



Theorem 4. *Let V, W be K-vector spaces. The tensor product of V and W exists and is unique up to isomorphism. (Notation: $V \otimes W, V \otimes_K W$).*

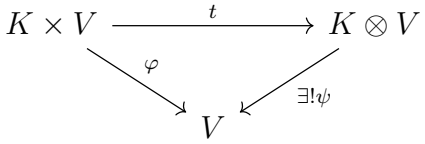
We call $t(v, w) =: v \otimes w$ a **pure tensor.**

Corollary 5. *In $V \otimes W$, for all $v_i, v \in V, w_i, w \in W, a \in K$:*

1. $(v_1 + v_2) \otimes w = v_1 \otimes w + v_2 \otimes w$
2. $v \otimes (w_1 + w_2) = v \otimes w_1 + v \otimes w_2$
3. $a(v \otimes w) = av \otimes w = v \otimes aw$
4. $0 \otimes w = 0 = v \otimes 0$

Proposition 6. *$K \otimes_K V \cong V$. More generally, $K^n \otimes_K V \cong V^n$ if $n < \infty$ (where V is a K-vector space).*

Proof.



$\psi t = \varphi \implies \psi(a \otimes v) = av$.
$\varphi(a, v) \mapsto a \cdot v$ is bilinear by V being a K-vector space.
ψ is surjective: $\psi(1 \otimes v) = v$ for all $v \in V$.
If $\sum_{i=1}^n a_i x_i = 0$ then $\sum_{i=1}^n a_i \otimes x_i = 1 \otimes \sum_{i=0}^n a_i x_i = 0$ so ψ is injective.
Hence ψ is an isomorphism. □

Proof. (Of Theorem 1)

1. Uniqueness: as usual
2. Existence: $I := V \times W$

   $\hat{T} := \bigotimes_I K$ vector space with basis elements $\{e_{(v,w)} \mid (v,w) \in V \times W\}$

   $U := \langle e_{(v_1+v_2,w)} - e_{(v_1w)} - e_{(v_2w)}, \dots \rangle$

   Define $V \otimes W := \hat{T}/U$.

   Check: this works.

□

Corollary 7. *For all K-vector spaces V, W, X:*

1. $V \otimes W \cong W \otimes V$.
2. $V \otimes (W \otimes X) \cong (V \otimes W) \otimes X$.

Proof.

1. $$\begin{array}{ccc} & & V \times W \xrightarrow{t} V \otimes W \\ & & \downarrow{\scriptstyle \varphi} \quad \nearrow_{\exists! \psi} \\ W \times V & \xrightarrow{t'} & W \otimes V \end{array}$$

   $\varphi(v,w) = w \otimes v$ is bilinear

   $\psi t = \varphi$ i.e. $\psi(v \otimes w) = w \otimes v$

   By symmetry, we get $\psi' : W \otimes V \to V \otimes W$. Can check: $\psi\psi' = \mathrm{id}, \quad \psi'\psi = \mathrm{id}$.

   Only have to check that they're inverses on pure tensors.

2. In notes.

□

Note 8. $\mathrm{Hom}(V \otimes W \otimes X, Y)$ corresponds to multilinear $V \times W \times X \to Y$.

Theorem 9. *If V, W, X are K-vector spaces:*

$$(V \oplus W) \otimes X \cong (V \otimes X) \oplus (W \otimes X)$$

Proof. In notes. □

Corollary 10. $\dim(V \otimes W) = \dim V \dim W$ *if* $\dim V, \dim W < \infty$.

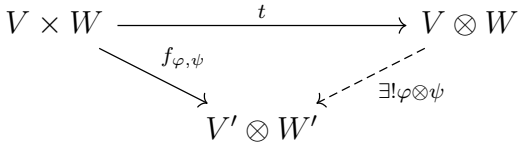
3.5.2 Tensor product of homomorphisms

Theorem 11. *Let V, W, V', W' be K-vector spaces. Let $\varphi \in \mathrm{Hom}(V, V'), \psi \in \mathrm{Hom}(W, W')$. Then there exists unique $\varphi \otimes \psi \in \mathrm{Hom}(V \otimes W, V' \otimes W')$ so that*

$$(\varphi \otimes \psi)(v \otimes w) = \varphi(v) \otimes \psi(w)$$

for all $(v, w) \in V \times W$.

Proof.



$f_{\varphi,\psi}(v, w) = \varphi(v) \otimes \psi(w)$ □

Theorem 12. *If V, V', V'', W, W', W'' are K-vector spaces, $\varphi \in \mathrm{Hom}(V, V'), \varphi' \in \mathrm{Hom}(V', V''), \psi \in \mathrm{Hom}(W, W'), \psi' \in \mathrm{Hom}(W', W''),$ then:*

$$(\varphi' \otimes \psi')(\varphi \otimes \psi) = (\varphi' \circ \varphi) \otimes (\psi' \circ \psi) \in \mathrm{Hom}(V \otimes W, V' \otimes W')$$

That is, the assignment $(V, W) \mapsto V \otimes W$ and $(f, g) \mapsto f \otimes g$ makes $\otimes$ into a functor

$$\otimes : \mathbf{Vect}_k \times \mathbf{Vect}_k \to \mathbf{Vect}_k.$$

Example 13. Suppose $V = K^2, W = K^2, V' = V, W' = W$. Say $D_S(\varphi) = A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ ($\varphi \in \mathrm{End}(V)$) and $D_T(\psi) = B = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$ ($\psi \in \mathrm{End}(W)$). Then what is the matrix representation of $\varphi \otimes \psi$?

Say $S = (s_1, s_2)$ and $T = (t_1, t_2)$. Then
$\varphi(s_1) = s_2$
$\varphi(s_2) = s_1$
$\psi(t_1) = t_1$
$\psi(t_2) = 2t_1 + 3t_2$
Then e.g. $(\varphi \otimes \psi)(s_1 \otimes t_1) = \varphi(s_1) \otimes \psi(t_1) = s_2 \otimes t_1$ gives first column of the matrix.

$$D_{S \otimes T}(\varphi \otimes \psi) = \begin{pmatrix} & & 1 & 2 \\ & & 0 & 3 \\ 1 & 2 & & \\ 0 & 3 & & \end{pmatrix}$$

In general, let V, V', W, W' be finite-dimensional with bases S, S', T, T'. Let $\varphi \in \mathrm{Hom}(V, V')$ and $\psi \in \mathrm{Hom}(W, W')$. If $(a_{ij}) = A = D^S_{S'}(\varphi) \in K^{k \times l}, (b_{rs}) = B = D^T_{T'}(\psi) \in K^{m \times n}$, the matrix $D^{S \otimes T}_{S' \otimes T'}(\varphi \otimes \psi)$ is the Kronecker product:

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & \dots \\ a_{21}B & a_{22}B & \dots \\ \vdots & \vdots & \ddots \end{pmatrix}.$$

That is, $A \otimes B$ represents $\varphi \circ \psi$ with respect to $\{s_i \otimes t_j\}, \{s'_i \otimes t'_j\}$.

3.5.3 Hom and Tensor

Theorem 14. *Let V, W, X be K-vector spaces:*

$$\mathrm{Hom}(V \otimes W, X) \cong \mathrm{Hom}(V, \mathrm{Hom}(W, X))$$

Proof. **Exercise**. □

Conclude: Hom and $\otimes$ are adjoint functors.

3.5.4 Exactness of $\otimes$

Theorem 15. *Let V be a K-vector space and $0 \to V_1 \to V_2 \to V_3 \to 0$ a short exact sequence. Then*

1. $$0 \to V \otimes_K V_1 \to V \otimes_K V_2 \to V \otimes_K V_3 \to 0$$
   is exact
2. $$0 \to \mathrm{Hom}(V_3, W) \to \mathrm{Hom}(V_2, W) \to \mathrm{Hom}(V_1, W) \to 0$$
   is exact.
3. $$0 \to V_3^* \to V_2^* \to V_1^* \to 0$$
   is exact

Note 16. If V, W are K-vector spaces, then $\mathrm{Hom}_K(V, W)$ is a K-vector space, hence an abelian group. This is en example of an "abelian category."

Note 17. **Additive functors** are functors that induce homomorphisms on Hom groups;
An additive functor $F : \mathcal{C} \to \mathcal{D}$ is **left exact** if for any exact sequence $0 \to X \to Y \to Z$, the sequence $0 \to F(X) \to F(Y) \to F(Z)$ is also exact and **right exact** if for any exact sequence $X \to Y \to Z \to 0$, $F(X) \to F(Y) \to F(Z) \to 0$ is also exact.

F is **exact** if F is left and right exact.

Thus Theorem 1 $\implies$ Hom and $\otimes$ are exact functors.

3.6 Tensor products and the tensor algebra

Definition 1. If V is a K-vector space and $m \in \mathbb{N}$,

$$T^m(V) := V \otimes \cdots \otimes V$$

(m times) is called the **m-th tensor power.** Other notation: $V^{\otimes m}$. ($T^0(V) := K$).

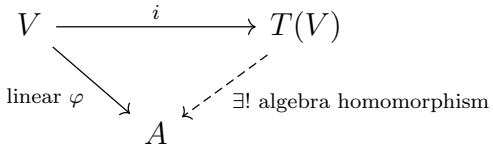
The **tensor algebra** of V is

$$T(V) = \bigoplus_{m \in \mathbb{N}} T^m(V) = K \oplus V \oplus (V \otimes V) \oplus \cdots$$

with multiplication induced by the tensor product.

Facts:

- $T(V)$ is a noncommutative graded k-algebra with multiplication $T^k(V) \times T^l(V) \to T^{k+1l}(V)$ induced by tensor product
- $T(V)$ has the following UMP:



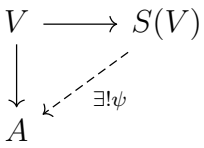
(A a K-algebra)

That is, every other K-algebra that admits a linear map from V is a quotient of $T(V)$

Important quotients of $T(V)$:

1. $T(V)/(v \otimes w - w \otimes v \mid v, w \in W) =: S(V)$ **symmetric algebra on V**

   UMP:



   (A a commutative K-algebra)

   That ism every other K-algebra that admits a linear map from V is a quotient of $S(V)$

$$S(V) = K \oplus V \oplus S^2(V) \oplus \cdots$$

   If $\dim V = n < \infty, S(V^*) = K[x_1, \ldots, x_n]$.

   $\dim(S^k(V)) = \binom{n+k-1}{k}$

2. $T(V)/(v \otimes v \mid v \in V) = \Lambda(V)$ **exterior algebra**

   The image of $a \otimes b$ in $\Lambda(V)$ is written $a \wedge b$.

   $\wedge$ is **alternating** ($x \wedge x = 0 \forall x \in V \cong \Lambda^1(V)$) and **anticommutative** ($x \wedge y = -y \wedge x \forall x, y \in V$).

   $\Lambda(V) = K \oplus V \oplus \Lambda^2(V) \oplus \cdots$ (this ends at $\Lambda^n(V)$ if $n = \dim V < \infty$).

   If $\dim V = n < \infty, \dim \Lambda^k(V) = \binom{n}{k}$. So $\Lambda^k(V) = 0$ if $k > n$..